Trading Literature Atlas

For Semi-Automated Systematic Trading

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Abstract

Research that matters for systematic trading is scattered across communities that rarely read one another. Time-series classification, deep learning, empirical finance, market microstructure, derivatives, rough-path mathematics, statistics, decision theory, language models and practitioner writing each study objects that a trading desk needs, such as regimes, labels, liquidity, cross-sections and search over signals, but they name them differently and often study them in isolation. This manuscript is a contribution in bridge-building. It builds a common vocabulary so that traders can find better alphas and so that mathematicians and machine-learning researchers can see where their skills apply in a hard, noisy, adversarial domain. The setting throughout is a semi-automated systematic fund.

The evidence base is a Zotero collection of 1065 papers. An automated classification of titles and abstracts assigned each paper to one of 11 research silos, to one or more of 11 stages of a fund's pipeline, and to concept tags. Fourteen topical chapters, one per cluster of the collection, review the papers on their own terms and close with bridge notes that translate jargon between silos. Desk notes turn ideas into inputs, outputs and failure modes for a trading desk. A shared vocabulary, a set of cross-silo bridges, a blueprint of the fund and a list of open gaps tie the chapters together.

The reader gets a map of the literature indexed by the stages of a real pipeline, a glossary that translates between communities, a set of concrete cross-silo proposals, and an honest account of what the method can and cannot claim. Silo and stage labels rest on abstracts, not full texts, and they are produced by an automated classifier. Synthesis and proposals are ours, not the cited authors'.

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Part I

Orientation

Chapter 1

Introduction: A Library in Shards

1.1 A library in shards

A systematic fund is a pipeline. Data come in, features are built, signals and forecasts are produced, the market regime and the risk budget are assessed, a portfolio is sized, orders are executed, and the whole chain is validated and, increasingly, automated. Almost every link in that chain has a research literature. The difficulty is that the literatures do not form one body of knowledge. They form shards, each held together by a shared training, a shared set of benchmark data and a shared dialect.

The collection behind this manuscript makes the point plainly. It holds 1065 papers on trading, gathered over time in a Zotero collection called `trading`. They span 1975 to 2027, but the mass is recent: 597 carry a year of 2024 or later and 500 a year of 2025 or later (counted from the year field of our records, whose latest value is 2027). When the papers are sorted by the research community that they appear to belong to, eleven silos emerge, from time-series machine learning (180 papers) and deep learning (176) through empirical finance and econometrics (136), market microstructure (115), decision theory and reinforcement learning (104), statistics (95), language models and agents (88), practitioner writing (55), rough-path and topological mathematics (49), derivatives (43) and work imported from outside finance (24). Chapter 2 defines these silos. For now the relevant fact is that the silos differ in what they publish, how they validate and which words they use, and that a desk needs the output of all of them.

1.1.1 One object, many names

The clearest symptom of a silo is a single underlying object that carries a different name, a different mathematical form and a different evaluation protocol in each community. Five examples from the collection show the pattern. In each we point to representative papers. We did not trace citation links between them, so we claim only what the titles and abstracts show: that each community describes the object in its own terms.

The market that changes. Statisticians treat the problem as change-point detection. Habibi (2022) introduce Bayesian online change-point detection to finance, with its focus on abrupt changes in the generative parameters of a series. Market-microstructure researchers speak of structural breaks and regimes in the order flow, and ask how to identify them in real time for trading and risk systems (Jebali, 2026). Time-series data miners speak of discovering multiple phases of dynamics in multivariate series (Wang and Hsieh, 2021). A mathematics-side paper embeds persistent homology in a Markov regime-switching filter to detect volatility regimes earlier (Faris, 2026). Four silos, four vocabularies (change point, structural break, phase, regime), one operational question: when should the model, the risk limit or the execution schedule be re-estimated?

The target. Supervised learning needs a label, and the label is rarely a given. A time-series group shows that, holding the model fixed, the choice of target changes the results of stock selection (Guida and Coqueret, 2026). A derivatives group labels intraday bars by the repriced profit of an option position, in place of price direction (Vaghela, 2026). A practitioner study documents a failure mode in which success at predicting triple-barrier labels does not carry over to portfolio performance (Lima, 2026). Decision theorists have a name for the underlying issue, which is to train the predictor for the decision that it feeds (Caunhye et al., 2026; Zhang and Li, 2026). These are one problem under four headings: label design, target engineering, label-to-P&L gap, and decision-focused learning.

Who moves with whom. An econometrician tests whether one sectoral index Granger-causes another (Sahoo and Kumar, 2025), in the line of the classic collected econometrics papers (Granger et al., 2001). A time-series data-mining paper builds lead–lag relationships into the selection of pairs (Gupta and Chatterjee, 2020). A deep-learning paper models delayed dependence between stocks with a graph neural stochastic differential equation (You et al., 2026). A microstructure study finds that the aggressive trades of high-frequency traders lead those of other investors (Hirschey, 2021). Lead–lag, Granger causality, delayed graph dependence and order anticipation are four views of directed information flow.

The correlation graph. A time-series paper finds that a graph built from stock correlations carries an economic taxonomy (Mantegna, 1998). Another asks how well correlation-filtering methods perform on noisy estimates (Millington and Sánchez Granero, 2022). An empirical-finance paper examines partial correlation as an alternative to the Pearson coefficient for equity co-movement (Kenett et al., 2014). A deep-learning paper learns the financial network instead of taking it as given, in an asset-pricing model (Uddin et al., 2021). A mathematical paper allocates a portfolio from a network built from path signatures (Gregnanin et al., 2024). A desk could use any of these, and could not easily tell that they are answers to the same question.

The search over signals. Large-language-model agents explore a space of formulaic alphas with memory of the search history (Yu et al., 2026e). Evolutionary algorithms do the same without language models (Zhang et al., 2020). Reinforcement-learning researchers recast alpha discovery as sequential decision-making (Zhu and Zhu, 2025). Scientific-computing researchers call the procedure symbolic regression (Cranmer, 2023). A statistician bounds the false discovery rate among published cross-sectional predictors (Chen, 2022). Every one of these is a search over a combinatorial space with noisy feedback. Only one of them sets the multiplicity of the search at the centre of its measure of success.

Bridge. Why this is a bridge problem and not a survey problem. A conventional review would take each of these five objects, pick the community that looks most natural, and review it. The shards would survive. Our aim is to place the five stories next to one another, to give each object one shared term, and to say for each community what it could take from the others.

1.2 Who this is for

Two readers motivate this manuscript. They need different things and should enter at different places.

1.2.1 Portfolio managers and quantitative researchers

You know the pipeline and the cost of a mistake in it. You may not know that a method you need exists under an unfamiliar name, in a venue you do not read. Use the manuscript as a lookup.

Start from the stage of your pipeline that is giving you trouble, find it in the silo-by-stage table of Chapter 2, read the topical chapter that covers it, and look at the desk notes. Each desk note gives a stage, an input, an output and a list of things that can go wrong, so that the idea can be tried in a notebook before anyone is convinced by it. Then read the bridge notes at the end of the chapter, which say which other chapter may hold a tool you are missing. If you manage a team, Chapter 18 assembles the pieces into a blueprint, and Chapter 19 lists what the literature does not yet cover.

1.2.2 Mathematicians and machine-learning researchers

You know the tools: signatures, topology, change-point theory, conformal prediction, foundation models, decision-focused learning. You may not know which problem in trading is shaped to fit them, or which of the problem's difficulties (non-stationarity, tiny signal-to-noise, adversarial counterparties, selection bias over many trials) will break a method that works well on a standard benchmark. Start with the vocabulary chapter and its translation table, which maps your terms onto the trader's. Then read the chapters of the parts on Representation and Pattern and on Research Operations, which are written for the reader who wants to know what a correct evaluation looks like. The key lesson of the Research Operations part is that a result in this domain is only as good as its validation: look first at the backtesting and leakage material in Chapter 14 before you invest in an architecture.

1.3 How the atlas was built

The method has four steps, and each of them has a limit that we state here and will repeat when it matters.

1. **Collection.** The starting point is a Zotero collection called `trading`, with sub-collections that group papers by topic. It holds 1065 papers. The collection is a convenience sample of what the owner found or was sent. It is not a systematic search, and its coverage is uneven. We make no claim to completeness, and the counts in this manuscript describe the collection, not the field.

2. **Automated classification.** Every paper was read by a language-model classifier, in batches, from its title and abstract. For each paper the classifier assigned one of 11 silos, one or more of 11 pipeline stages (775 papers received one stage, 290 received two or three), a set of concept tags (122 distinct tags in all), a relevance score, a one-line desk use and a one-line cross-silo bridge. The classifier read abstracts only. Seven abstracts in the collection are shorter than 200 characters, six of them empty, so the labels for those papers lean on the title. A paper's silo is a judgement about its dialect, and borderline papers could be assigned either way. We did not manually audit every label.

3. **Chapters.** Papers were grouped by topic, using the sub-collections of the library as a guide, into 14 topical chapters, from 29 papers (market structure) to 134 (microstructure). Each chapter was written from the abstracts of its papers. Within a chapter, we report what a paper's abstract says and attribute only that; the synthesis, the proposals and the gap-filling are our own and are worded as such.

4. **Synthesis.** The final part draws the chapters together: a catalogue of cross-silo bridges, a blueprint for a semi-automated fund and an inventory of gaps.

Desk note. Limits of the method. (1) Grounded in titles and abstracts, not full texts: a method that is described briefly in an abstract may be treated more generously or more harshly than the paper deserves, and details such as costs, universes and test periods are often missing from abstracts. (2) The classification is automated and the labels are noisy, so a count by silo or stage is an estimate and not a measurement. (3) Authors are recorded by surname only, and the reference list carries surnames only, so we make no claim about who works in which silo, and we do not trace author networks or citation links. (4) Many abstracts report in-sample or gross results. We repeat what the abstract states and flag when the validation is unknown. (5) Sections that propose a design are proposals. They are not findings of the cited papers.

1.4 A roadmap

The manuscript has six parts. The structure follows the pipeline of a fund, from its inputs to its governance, and ends with a synthesis.

Orientation. This chapter and Chapter 2, which defines the silos and stages, tabulates where the papers sit, and offers a translation table of shared concepts across silos.

Data and Substrate. Before any signal there is data, and before a market exists in the lab there is a simulator. Chapter 3 reviews synthetic data, generative models and agent-based simulators as a training and testing ground. Chapter 4 describes the participants, rules and reforms that produce the data in the first place, and how they create natural experiments.

Representation and Pattern. Chapter 5 covers time-series forecasting, classification and mining, with similarity search and motifs as a shared engine. Chapter 6 covers the representations (hand-built features, signatures, topology, self-supervised encoders and foundation models) that turn a path into something a model can use. Chapter 7 covers regimes, change points, networks and systemic risk.

Alpha. Four chapters deal with the sources of expected return. Chapter 8 reviews return forecasting with deep sequence and graph models. Chapter 9 reviews factors, anomalies and relative value, including the mining of formulaic alphas. Chapter 10 reviews trading strategies and signals, from technical rules to reinforcement-learning traders. Chapter 11 reviews volatility and derivatives, where option-implied information feeds back into equity signals.

Implementation. Chapter 12 deals with order books, market impact and execution, and Chapter 13 with portfolio construction and decisions under uncertainty. This is where a forecast becomes a position and a position becomes a fill.

Research Operations. Chapter 14 deals with labels, backtesting, overfitting and the statistics of selection. Chapter 15 deals with language models, text and agentic research. Chapter 16 deals with general methods imported from other domains: anomaly detection, change-point methods, symbolic regression, validation tools and the automation of the research loop.

Synthesis. Chapter 17 collects the cross-silo bridges that the topical chapters propose and develops twelve of them into research proposals. Chapter 18 assembles them into a blueprint of a semi-automated systematic fund. Chapter 19 lists the gaps and open problems that the collection leaves, and concludes.

Two conventions run through the topical chapters. Each ends with a section on bridges out of the chapter, and each contains desk notes. Chapter 2 explains the boxes before the reader

meets them. Citations use author–year form, and every attributed statement is limited to what the cited abstract supports.

Chapter 2

A Shared Vocabulary

A bridge needs two banks. This chapter fixes the shared terms that every later chapter uses: the eleven research communities (silos) that the collection falls into, the eleven stages of a semi-automated systematic fund against which each paper is placed, and a translation table, a Rosetta stone, between the dialects in which the same object is described. It also explains the two kinds of box that run through the topical chapters.

2.1 Eleven silos

A silo, in this manuscript, is a research dialect: a community with its own training, benchmark problems, venues and words. It is not a verdict on quality. The eleven labels below were assigned to each paper by an automated classification of its title and abstract (one label per paper), so each is a judgement about which dialect the paper speaks. Counts in parentheses are the number of papers in our collection of 1065.

tsml (180): time-series machine learning and data mining.
Similarity search, motifs and discords, shapelets, clustering and classification of series, labelling schemes and feature libraries. The default question is how to find structure in a sequence.

dl (176): deep learning and foundation models.
Sequence, attention and graph networks, self-supervised encoders, generative models, pre-trained forecasters. The default question is how to learn the mapping from history to a target.

fin-econ (136): empirical finance and econometrics.
Anomalies, factor models, event studies, volatility models, flows and behavioural effects, often on a single national market. The default question is whether an effect is priced and why.

microstructure (115): market microstructure, order books and execution.
Spreads, depth, order flow, price discovery, impact, auctions and the behaviour of high-frequency participants. The default question is how trades form prices.

decision (104): decision theory, control and reinforcement learning.
Optimal execution, stochastic control, portfolio allocation, predict-then-optimise, reinforcement-learning traders. The default question is what to do given a belief.

stats (95): statistical testing, change-point and Bayesian methods.
Hypothesis tests, forecast comparison, change-point detection, uncertainty sets and point processes. The default question is how sure we are.

nlp-llm (88): text, language models and agents.
News and filings as features, language-model reasoning, alpha-mining agents, multi-agent

trading systems, security of agents. The default question is what a language model can read or do.

practice (55): practitioner and research-methodology writing.
Backtesting practice, platforms, walk-forward protocols, overfitting diagnostics, discretionary-rule write-ups. The default question is what survives contact with a real desk.

math (49): rough paths, topology, algebra and geometry.
Path signatures, persistent homology, geometric and information-theoretic constructions. The default question is what the right invariant of a path is.

quant-deriv (43): derivatives, volatility modelling and mathematical finance.
Implied-volatility surfaces, risk premia, option pricing and hedging, scenario P&L. The default question is what the option market implies.

other-domain (24): work imported from outside finance.
Root-cause analysis in software systems, anomaly detection in medicine, control charts, symbolic regression in physics. The default question is whether a tool built elsewhere transfers.

2.2 Eleven pipeline stages

The second axis follows a semi-automated systematic fund from its inputs to its governance. Each paper was assigned one or more stages (a paper may serve two, for instance a labelling scheme that is both a data and a forecast tool). The assignment is again automated and rests on the abstract.

data.
Acquisition, cleaning, sampling and labelling of market data, including simulated and synthetic data.

features.
Transformations and representations that turn raw series into model inputs: indicators, embeddings, signatures, topological summaries.

signal.
A rule or score that proposes a trade or ranks assets: technical rules, factor scores, mined formulaic alphas, event signals.

forecast.
A model that predicts a return, a direction, a volatility or a distribution.

regime.
Detecting that the market state has changed: change points, latent states, early-warning indicators.

portfolio.
Turning forecasts and signals into positions: allocation, sizing, constraints, decision-focused learning.

execution.
Turning positions into fills: order placement, schedules, market impact and transaction-cost analysis.

risk.
Measuring and limiting loss: volatility, tails, drawdown, liquidity stress and systemic contagion.

silo	data	features	signal	forecast	regime	portfolio	execution	risk	validation	automation	theory
tsml	66	33	53	37	14	6	1	11	23	7	3
dl	26	19	32	113	6	15	1	4	12	2	5
fin-econ	1	6	36	19	13	5		38	20	1	19
microstructure	9	7	28	2	1		39	24	11	3	17
quant-deriv	2		14	5	1			16	8		6
math	1	6	4	5	5	6	3	6	1	1	22
stats	8	3	12	8	14	7	2	17	39	1	10
decision			43	2	4	22	26	3	6	4	25
nlp-llm	4	1	47	10		3		7	14	14	2
practice	1		17	1		3	4	5	30	10	2
other-domain	4	4	2	4	1	1			3	3	4

Table 2.1: Silo by pipeline stage: number of papers (a paper with several stages is counted once per stage). Silo and stage labels come from an automated classification of titles and abstracts. A blank cell is zero.

validation.
Testing whether a result is real: backtests, labels, out-of-sample protocols, multiple-testing and leakage controls.

automation.
Making the research loop run itself: agents, platforms, harnesses and tooling.

theory.
Mathematical or conceptual foundations that do not yet sit in one operational stage.

The topical chapters often abbreviate this list to data, features, signal, regime and risk, portfolio, execution, validation and research automation. That shorthand folds forecast into signal, pairs regime with risk and leaves theory implicit; it is the same pipeline. Chapter 18 uses a third, operational cut of ten stages, which it maps back onto these.

2.3 Where the literature sits

Table 2.1 cross-tabulates the two axes. Because a paper can carry up to three stages, the cells add to 1357 stage assignments over 1065 papers (775 papers have one stage, 290 have two or three).

Where the mass sits. The signal stage is the largest (288 assignments), followed by forecast (206) and validation (167); portfolio (68), regime (59) and automation (46) are the thinnest, with execution (76) close behind. The silos pile up in different places. Deep learning concentrates on forecasting: 113 of its 235 stage assignments, and 113 of the 206 forecast assignments overall, are its own. Time-series machine learning is the main supplier of data-stage work (66 assignments, much of it labelling and sampling) and also contributes to signal and feature construction. Microstructure is the home of execution (39) and a major supplier of risk work (24). Statistics and practitioner writing hold the validation stage (39 and 30). Language-model papers cluster in signal (47), where alpha-mining agents sit. Mathematics is the one silo whose largest cell is theory (22 of 60).

Where it is empty, and why that matters. Fourteen of the 121 cells are zero, and 23 more hold one or two papers. The zeros are informative because each marks a hand-off that no one

in the collection is working on, at least in a way that the classifier could see. Some examples: no derivatives paper is placed in execution, portfolio, features or automation, so option-implied information is largely not carried through to sizing and order placement; no microstructure paper is placed in portfolio, so cost curves reach allocators through other silos if at all; no language-model paper is placed in regime or execution, so agents that trade are rarely paired with change-point detectors; decision-theory papers have no data or feature stage, so control formulations assume their inputs; and practitioner writing has no feature or regime stage. Only two silos (microstructure and decision) place five or more papers in execution. Some of these gaps may be artefacts of the classifier, which gave each paper a single dominant silo and at most three stages. We treat the table as a prompt for the bridge-building of Chapter 17 and for the gap analysis of Chapter 19, not as a measurement of the field.

2.4 A Rosetta stone

The table that follows is the core of this chapter. Each row is a shared concept, taken from the concept index of the collection (122 distinct tags, assigned by the same automated classifier; the table keeps those that span silos or recur most). For each we give the term we will use, how the relevant silos name or frame it, the chapters in which it is most prominent, and one to three representative papers. The silo framings are our translations. They are meant to help a reader recognise an object in an unfamiliar dialect. They are not attributions to the cited papers, whose abstracts we checked only for fit with the row. Cited papers were chosen from the concept index and checked against their abstracts; they are examples, not a complete census.

Table 2.2: Rosetta stone of shared concepts.

Shared term	How the silos frame it	Chapters	Examples
Regime shift or change point	Statistics: change point, online detection, hazard rate. Econometrics: structural break. ML: concept drift, distribution shift. Time-series mining: segmentation, phase; a discord is a transient cousin. Microstructure: liquidity or intensity regime. Trader: market-state change.	7, 5, 10, 8, 16	Habibi (2022); Jebali (2026); Wang and Hsieh (2021)
Early warning or critical transition	Topology: persistent-homology landscapes. Physics: log-periodic singularity, criticality. Trader: bubble or crash precursor. Statistics: a change-point test with a short delay.	6, 7	Gidea and Katz (2017); Akingbade et al. (2023)
Forecast evaluation	Statistics: predictive-ability test, pairwise comparison of forecast errors. ML: held-out metric against a benchmark. Finance: out-of-sample R^2. Trader: live track record net of costs. For cross-sectional returns the stringent benchmark is the panel historical mean, not the zero forecast; for volatility it is GARCH(1,1), which the cited comparison finds beaten mainly by models with a leverage effect.	8, 5, 14, 16, 15	Reinhard Hansen and Lunde (2003); Bandi et al. (2026); Diao (2026)
Label design	Time-series mining: labelling method, target engineering. ML: ground truth, supervised target. Derivatives: scenario-P&L label. Decision theory: surrogate for the decision loss. Trader: what a trade is meant to capture.	14, 8, 10, 16	Guida and Coqueret (2026); Vaghela (2026); Lima (2026)
Event labels and the side/size split	Practitioners: triple barrier, meta-labelling, information-driven bars. Probability: first-passage or stopping time. Microstructure: event-time clock. Control: a decision rule split into direction and size.	14, 8, 10	Grądzki et al. (2025); Meyer et al. (2022); Zielkowski (2026)

Shared term	How the silos frame it	Chapters	Examples
Representation learning	Deep learning: embedding, self-supervised or contrastive encoder. Time-series mining: feature library, transform. Mathematics: signature, topological summary. Trader: indicator set.	5, 6, 8, 10	Sun et al. (2022a); Demirhat et al. (2026); Hwang et al. (2024)
Foundation model	Deep learning: pretrained time-series model, tokeniser. Language models: a series read as a token stream. Trader: a generic forecaster not fitted to the desk's universe.	6, 5, 16, 3	Shi et al. (2025b); Rahimikia et al. (2025a); Zhu et al. (2025a)
Sequence model	Deep learning: LSTM, transformer, state-space model. Econometrics: ARMA or GARCH hybrid. Trader: a forecaster with memory.	8, 5, 10	Coelho E Silva et al. (2024); You et al. (2026)
Similarity search and motif	Time-series mining: nearest neighbour, matrix profile, motif, shapelet. Technical analysis: chart pattern. Mathematics: shape of a path. Deep learning: retrieval.	5, 10, 15, 6	Cremona et al. (2023); Rodrigues et al. (2019); Jang et al. (2026)
Trend, momentum and technical rule	Trader: technical rule, breakout, moving-average cross. Finance: time-series or cross-sectional momentum, timing strategy. ML: handcrafted features. Evolutionary computation: a rule space to be searched.	10, 8, 14	Basanisi and Torresetti (2023); Han et al. (2013); Baltas and Kosowski (2013)
Liquidity	Microstructure: spread, depth, resilience. Finance: illiquidity premium, liquidity risk. Trader: the cost of size. Regulators: market quality.	12, 11, 10, 3, 4	Shaik and Bairaria (2026); Nawn and Banerjee (2019)
Order flow	Microstructure: signed volume, imbalance. Statistics: point-process intensity. Finance: investor flow, herding. Deep learning: trade sequence. Trader: the tape.	12, 7, 11	Hirschey (2021); Jebali (2026); Garg and Mitra (2015)
Limit order book	Microstructure: book state, depth profile. Deep learning: spatial–temporal patch of an image. Control: state of an execution problem. Simulation: environment.	12, 3	Noble et al. (2026); Panayi et al. (2015)
Market impact	Microstructure: temporary and permanent impact, price pressure. Finance: short-term reversal, liquidity-provision premium. Portfolio: capacity. Trader: slippage.	12, 3, 4, 14	Teodorovic (2011); Wu and Jegadeesh (2020)
Execution and transaction cost	Control: optimal liquidation, stochastic control. Statistics: Bayesian transaction-cost analysis. Practitioners: implementation shortfall. ML: execution agent.	12, 10, 15, 13	Graf and Mastrolia (2026); Markov (2019)
Anomaly (two senses)	Finance: a cross-sectional return pattern not explained by risk. Time-series mining and other-domain work: an outlier or abnormal subsequence. Regulators: manipulation signature.	9, 5, 12, 4, 16	Bali et al. (2008); Li et al. (2020); Chen and Hybinette (2026)
Factor model	Finance: priced risk factor, spanning. ML: latent factor, learned loading. Statistics: low-rank decomposition. Practitioners: alpha factor.	9, 8, 15, 13	Holcblat et al. (2022); Gopal (2024); Uddin et al. (2021)
Alpha mining	Language-model agents: search-space exploration with memory. Evolutionary computation: genetic programming. Reinforcement learning: sequential decision-making. Science: symbolic regression. Statistics: multiple testing.	15, 9, 14, 10, 16	Yu et al. (2026e); Zhang et al. (2020); Zhu and Zhu (2025)
Backtest overfitting	Practitioners: overfitting, curve fitting. Statistics: selection bias. ML: validation leakage. Agents: reward hacking of the evaluator.	14, 15, 10, 8	Inglese (2026); Sheppert (2026)
Multiple testing	Statistics: false discovery rate, deflated Sharpe ratio. Finance: factor zoo, data mining. Agents: number of trials.	14, 15, 5, 10	Chen (2022); Reinhard Hansen and Lunde (2003)

Shared term	How the silos frame it	Chapters	Examples
Data leakage	ML: train–test contamination, memorisation. Practitioners: look-ahead bias, point-in-time data. Backtesting: overlapping labels.	15, 14, 8	Roy and Roy (2026); Cai et al. (2026); Diao (2026)
Walk-forward validation	Practitioners: walk-forward, rolling origin. ML: temporal cross-validation. Statistics: sequential out-of-sample testing.	14, 10, 11	Deep et al. (2025a); Sheppert (2026)
Realised volatility	Finance: GARCH-type conditional volatility. Microstructure: intraday estimator from ticks. Probability: barrier-hitting time scales with volatility. Trader: risk scaling.	11, 12, 8	Dai et al. (2025a); Zielkowski (2026); Ahluwalia et al. (2023)
Implied volatility and surface	Derivatives: smile, skew, term structure, risk-neutral density. Finance: option-implied predictor of stock returns. Deep learning: generated surface. Trader: fear gauge.	11, 9, 8, 12	Shaikh and Padhi (2014); Eksi and Roy (2025); Han et al. (2026)
Network and contagion	Finance: contagion, spillover. Statistics: graphical model, precision matrix. Deep learning: learned graph. Mathematics: filtered correlation graph. Agents: adversarial signal propagation.	7, 8, 11, 15	Banerjee and Guhathakurta (2020); Uddin et al. (2021); Sua et al. (2026)
Portfolio optimisation	Decision theory: mean–variance, robust optimisation. Statistics: inference on the efficient frontier. Mathematics: path-dependent allocation. Trader: sizing.	13, 10, 15	Bodnar et al. (2009); Mendes and Lavrado (2017); Gregnanin et al. (2024)
Decision-focused learning	Operations research: predict-then-optimise. ML: end-to-end training on the decision loss. Trader: train on P&L, not on error.	13, 5	Caunhye et al. (2026); Zhang and Li (2026)
Uncertainty and abstention	Statistics: prediction set, coverage guarantee. ML: calibration, selective prediction. Trader: confidence that scales a position or vetoes a trade.	13, 11, 8	Taheri Hosseinkhani (2026); Singh et al. (2026)
Reinforcement learning	Control: Markov decision process. ML: agent and reward. Practitioners: adaptive strategy. Execution: liquidation policy.	10, 13, 15, 14	Zhu and Zhu (2025); Graf and Mastrolia (2026); Sun et al. (2023a)
Path signature	Mathematics: iterated-integral feature, rough path. ML: universal feature map, kernel. Trader: an order-aware summary of a path.	6, 13, 7	Bayer et al. (2026a); Fermanian (2019); Gregnanin et al. (2024)
Topology	Mathematics: persistent homology, landscape. Time-series mining: shape statistics. Trader: crash or bubble gauge.	6, 7	Gidea and Katz (2017); Akingbade et al. (2023)
Synthetic data	Deep learning: GAN, diffusion, flow-matching generator. Microstructure: simulator. Derivatives: generated surfaces. Practitioners: placebo data for testing.	3, 16, 14	Zhang et al. (2026d); Caulfield and Gleeson (2024); Han et al. (2026)
Agent-based model	Microstructure: interacting low- and high-frequency traders. Economics: heterogeneous agents. Language models: role-playing agents.	3, 4, 12	Leal et al. (2014); Gao et al. (2024)
Self-exciting flow	Statistics: Hawkes process, branching ratio. Microstructure: order-arrival clustering. Physics: reflexivity, endogeneity.	12, 5	Bacry et al. (2015); Filimonov and Sornette (2012)
Pairs and mean reversion	Finance: cointegration, distance method. Statistics: copula dependence. Time-series mining: lead–lag clustering. Microstructure: price error around a latent efficient value.	9, 12, 10	Wilkens (2025); Stander et al. (2013); Amaral (2026)
Causality and lead–lag	Econometrics: Granger causality. Microstructure: natural experiment with an exogenous instrument. Time-series mining: lead–lag in pair selection. Deep learning: delayed graph dependence.	12, 9, 5, 16	Granger et al. (2001); Aggarwal and Thomas (2014); Gupta and Chatterjee (2020)
Sentiment and text	Language models: polarity and event scores. Finance: investor attention and sentiment. Trader: news flow.	15, 8, 10, 4	Zhang et al. (2025d)

Shared term	How the silos frame it	Chapters	Examples
LLM agent	Language models: role-based multi-agent system, memory, harness. Practitioners: production use. Security: attack surface.	15, 16, 13	Xiao et al. (2025b); Sua et al. (2026)
Behavioural bias	Finance: lottery preference, herding, retail trading. Microstructure: attention and price pressure. Trader: crowding.	12, 10, 4	Bali et al. (2008); Agarwal et al. (2025)

Two cautions apply to the table. First, the shared term is a convenience, and in places the silos mean different things by the same word: "anomaly" is the sharpest case, and "regime" is a close second. Second, a row lists only the silos that appear in the concept index for that tag, so an empty framing does not mean that a community lacks the idea. It means that we saw no paper in the collection that frames it.

2.5 How to read the boxes

Two kinds of box recur in every topical chapter. They have different purposes and different standards of evidence.

Desk notes.
A desk note converts a cluster of papers into a design that a desk could try. It states a pipeline stage, an input, an output and what can go wrong, and it is written as a proposal: "a natural use is" or "a possible procedure" marks our design, not a result of the cited work. The cited papers support only what their abstracts state, such as a method, a data set or a reported finding. The "what can go wrong" list matters as much as the design, since many of the failure modes (look-ahead in labels, costs omitted from the reported edge, a detector that fires after the loss, thresholds fitted on the window that the strategy research already touched) are those that the validation chapters warn about. Treat a desk note as a hypothesis to be tested with the protocol of Chapter 14, never as a recommendation to deploy.

Bridge notes.
A bridge note sits in the closing section of a chapter, "Bridges out of this chapter". It takes an idea from one silo and says, in both directions, what another silo calls it and what each side could borrow. A typical sentence is of the form "what microstructure calls X is what time-series mining calls Y". These translations are our synthesis, informed by the abstracts, and several are also candidates for the catalogue of Chapter 17. A bridge note is a claim about vocabulary and a suggestion for work, not a finding that the two communities have shown the equivalence.

Across the 14 topical chapters there are 66 desk notes and 56 bridge notes. A reader short of time can read the opening of a chapter for its dialects, the bridge section for the connections, and the desk notes for the stage of the pipeline that interests them. The Rosetta stone above is the index to all of this: find the shared term, read the framings, and follow the chapters column to the place where the papers are discussed.

Part II

Data and Substrate

Chapter 3

Synthetic Data, Simulation and Market Generators

Three communities meet in this chapter and rarely read one another. Deep-learning researchers speak of *generative models*, *foundation models*, *tokenisation* and *diffusion*; they want realistic samples. Market-microstructure and agent-based-modelling researchers speak of *simulators*, *zero-intelligence agents*, *calibration* and *stylised facts*; they want a mechanism that explains why prices behave as they do. A third, smaller group (financial mathematics, time-series mining, behavioural finance) contributes scoring rules, anonymisation and equilibrium arguments. In the shared vocabulary of the atlas these are all instances of one object: a *synthetic-data* engine that produces counterfactual paths, order flow or scenarios, because history gives only one realised trajectory.

The translation is mostly a matter of purpose. What the deep-learning silo calls a "sample" or a "rollout" is a *scenario*; what the agent-based silo calls "calibration" is *fitting*; "stylised facts" (heavy tails, volatility clustering, absent return autocorrelation, square-root impact) are the common *validation* checklist. The two main families differ in what they give a desk. Data-driven generators reproduce statistics without explaining them. Agent-based simulators offer causal levers (add a seller, remove a market maker) but fit statistics less tightly.

In the fund pipeline the chapter sits at the *data* stage, as augmentation and scenario generation, and at the *validation* stage, as the sandbox against which execution code, risk limits and learned policies are tested before capital is committed. It feeds *risk* and *execution* indirectly, through stress scenarios and cost assumptions. Cross-references: Chapter 12 for real order-book data, Chapter 14 for validation practice.

3.1 Generative models for return paths

The oldest question is whether a neural network can emit price paths that look like markets. Wiese et al. (2020) build Quant GANs from temporal convolutional networks, with a generator constructed so that a transition to a risk-neutral distribution is possible; they report volatility clusters, leverage effects and serial autocorrelations. Kwon and Lee (2024) ask the sceptical version: can GANs learn random-walk, mean-reverting and jump behaviour in univariate and multivariate series? Caulfield and Gleeson (2024) compare deep generative models with parametric alternatives on multivariate series using distribution-moment distances, then test the best on an implied-volatility trading task. Their message, together with the previous two, is that the generator must be benchmarked against classical models, not assumed superior.

Two papers attack the failure mode that matters to traders. Zhang et al. (2026d) argue that GAN output can look realistic yet collapse under trading backtests, and add differentiable stylised-fact constraints for asymmetry and tail events. Cliff (2006) is unrelated in method but relevant in spirit: a 60-parameter search finds parameters that beat the original, and the paper

casts doubt on earlier conclusions. Any simulator-tuned result inherits this sensitivity.

Extensions add structure. Gregnanin et al. (2026) combine path signatures, LSTMs and graph networks on visibility graphs; Masi et al. (2026) pair a conditional GAN for correlated mid-price and volume with a diffusion model whose critic guides sampling; Xia et al. (2026) condition a GAN on macroeconomic indicators and train robust policies against it. Crisan et al. (2026) address small data with a rough-path perspective and a parsimonious variational autoencoder. A fuller account of representations such as signatures is in Chapter 6.

Desk note. Stage: validation, with data augmentation. Input: daily or intraday returns for a universe. Output: a generator-gated stress set. A natural use is to train a generator on the history, sample many paths, and run the *unchanged* strategy code on them, reading the dispersion of drawdown and turnover rather than any single P&L. Require that a stylised-fact checklist passes before a path set is used. Risks: the generator is fitted on the same history as the strategy, so leakage and look-ahead are easy; GAN mode collapse produces implausible extremes; and synthetic paths inherit the regime mix of the training window, so they cannot supply a regime the data never showed.

Between the two sections sits a practical point about data volume. Daily return histories are short, so a generator trained on one asset memorises more than it generalises; the tick-level models below are trained on far larger event sets (billions of trade events in the case of Kawawa-Beaudan et al. (2026)), which is why cross-asset transfer is plausible there and doubtful here.

3.2 Diffusion, wavelets and foundation models

Diffusion has become the default generative family. Wang and Ventre (2026) survey it by data type (time series, order books, tabular), and Bayer et al. (2026b) survey neural market generators more broadly, including GPT-style models. Takahashi and Mizuno (2025) convert multivariate price, volume and spread series to wavelet images and denoise them with a DDPM; Wang and Milenkovic (2026) instead diffuse directly on wavelet coefficients with level-specific energy constraints. Tang et al. (2026a) extend Tweedie's formula to geometric Brownian motion, squared Bessel and Cox–Ingersoll–Ross processes, so state-dependent volatility enters the denoiser. Waller et al. (2026) replace transformer feed-forward blocks with quantum layers, evaluated on two stocks. Choi et al. (2026a) move the idea to firm-level annual statements in latent space. Read together, they also show the field's trade-off: more expressive denoisers fit multi-scale structure, while the parametric route of Tang et al. (2026a) keeps interpretable volatility dynamics that a risk team can audit. Across these, the idea is the same: match multi-scale structure, not just marginal distributions. Their claims are made on their own datasets, and nothing in the abstracts shows out-of-sample trading gains.

3.3 Limit-order-book and order-flow generators

At tick level the object is an event stream. Coletta et al. (2022) learn a single "world agent" from data, avoiding hand-set agent populations. Berti et al. (2025) generate order flow conditioned on market state with a transformer diffusion model, and note that metrics for such simulators are scarce, so they adapt a predictive score. Wang et al. (2026a) use conditional flow matching for LOB trajectories across Hong Kong symbols, and report that it needs far fewer solver steps than diffusion. Zhang et al. (2026e) model order events and book state jointly as a foundation model, motivated by counterfactual questions. Kawawa-Beaudan et al. (2026) tokenise trade events with scale-invariant features for a 524M-parameter transformer that works across thousands of equities. Noble et al. (2026) are the practical counterweight: a simulator for large-tick stocks projected on spread and volume imbalance, with a power-law impact kernel, aimed at realistic fills and costs.

Desk note. Stage: execution and validation. Input: message-level or top-of-book data for the instruments traded. Output: a slippage and fill-rate estimate with its dispersion for an execution algorithm. A natural use is to replay the same parent order through many generated books, conditioned on a spread or volatility scenario, and compare schedules. Risks: generators trained on passive history do not react to our own orders, so impact is understated unless a feedback kernel is added; costs and queue position are often the first things a generator gets wrong; and capacity conclusions do not transfer from large-tick to small-tick names.

3.4 Agent-based simulators

The agent-based line reproduces mechanisms. Byrd et al. (2020) provide ABIDES, an open-source message-level simulator with configurable latency, used for market-impact experiments. Gao et al. (2024) calibrate a high-frequency model with machine-learning surrogates and recreate the 2010 flash crash by adding an institutional seller; Leal et al. (2014) reach a related conclusion in a model of low- and high-frequency traders, linking flash crashes to spreads and sell-side synchronisation. Vytelingum et al. (2025) model liquidation cost by letting traders update beliefs about fundamental value from order flow. Lussange et al. (2019) replace zero-intelligence agents with reinforcement-learning agents calibrated to London data. Nakagawa et al. (2025) derive microfoundations for GARCH from noise, fundamental and AI traders. Weng (2026) generate momentum and reversal from herding and delayed information diffusion. Zeng et al. (2012) supply a reputation game between institutions and individuals, and Davidovic et al. (2019) a network model where uncertainty about exposures induces liquidity hoarding.

Several papers study instability. Cliff (2022) shows that competing adaptive traders converge on suboptimal outcomes; Rocha et al. (2021) shows that reaction speed can reverse rankings of strategies; Zhou et al. (2026b) find that square-root impact is necessary for self-sustained manipulation cycles in a learning-agent market; Novotny (2026) treat fundamental anchoring as the stabiliser of liquidity, with no stress transmission between coupled books. Maitrier et al. (2025) generate synthetic flow from superposed metaorders to test the square-root law. Calvez and Cliff (2018) clone a trader's behaviour from quotes with deep networks, and Inverso et al. (2026) outline how formal methods might check emergent outcomes.

Desk note. Stage: risk and portfolio limits. Input: the fund's position and liquidity profile. Output: a crowding or fire-sale limit. A proposed use is to run an agent-based market with a large seller whose size equals a plausible unwind, vary the strength of the fundamental anchor, and set position caps at the size where one-sided book stress becomes self-sustaining. Risks: results depend on hand-picked agent populations; mechanisms found in a toy model are hypotheses, not estimates; and regime dependence is unavoidable, since the anchor strength that matters is exactly what is unobserved.

3.5 Evaluation, benchmarks and infrastructure

Sun et al. (2026) supply thirty controlled environments across six mechanism families, so forecast failures can be attributed to volatility clustering, heavy tails, regime switching or jumps. Galas (2014) describe a simulation environment for testing algorithm stability and parameter sensitivity. Der et al. (2023) synthesise series that preserve matrix-profile similarity joins while masking raw values, a route to sharing data with outsiders.

A common thread is that evaluation of generators lags their construction. Berti et al. (2025) and Kwon and Lee (2024) both note the lack of agreed metrics, which suggests a desk should score a generator by what it will be used for: a predictive score for forecasting models, fill and cost error for execution, tail coverage for risk. The same logic explains why a harness such as

that of Galas (2014) matters more than any one generator, since it fixes the protocol under which every generator is compared.

3.6 Bridges out of this chapter

Bridge. Microstructure and deep learning. What the agent-based literature calls a "calibrated population" is what the generative literature calls a "learned world agent" (Coletta et al., 2022) or a "foundation model" (Zhang et al., 2026e); both answer "what would the book have done?" Execution teams in Chapter 12 can use the same stylised-fact checklist to judge either.

Bridge. Time-series mining and backtesting. The mechanism-labelled environments of Sun et al. (2026) give TS-mining groups a labelled benchmark in which what they call a "regime change" or "discord" is known by construction. Chapters 5 and 14 can use that as a positive control for detectors and for overfitting tests.

Bridge. Market structure and regimes. Herding, anchoring and liquidity-hoarding models turn the behavioural vocabulary of Chapter 4 ("herding", "confidence") into simulator parameters, and offer regime-switch triggers for the models of Chapter 7.

Chapter 4

Market Structure and Participants

Four communities meet in this chapter and rarely read one another. Financial economists write about *participants*, *flows*, *anomalies* and *behavioural biases*. Microstructure researchers write about *venues*, *auctions*, *liquidity* and *market impact*. Time-series and deep-learning researchers, who reach this topic through surveillance, write about *anomaly detection* and *imbalanced classification*. A fourth group, drawn from law, science and technology studies, control theory and ecology, writes about *regulation*, *stacks*, *feedback* and *fragility*.

In the shared vocabulary of the atlas these become a small set of terms. A regulatory reform is a candidate *regime-shift*. A fund flow or crowd of retail traders is an *order-flow* or *sentiment* feature. A manipulation alert is an *anomaly* score. A "fragile" market is one whose *market-impact* and *network-contagion* behaviour changes sharply in the tails. The chapter is therefore less about alpha than about the ground the alpha stands on: who trades, under which rules, and how the rules break.

In the fund pipeline the chapter sits mostly at the regime and risk stages, with side entrances into data (which volume can be trusted), signal (flow and attention features), execution (auctions, expiry days) and validation (simulation, falsification). Most examples come from India, the market this atlas keeps returning to.

4.1 Rules as regimes: reforms and natural experiments

A backtest that spans a rule change spans two markets. Potharla (2025) chart the 2010–2025 growth and reform of Indian capital markets, and Gunnam et al. (2024) review SEBI's derivatives measures: larger lot sizes, fewer weekly expiries, higher expiry-day margins, upfront premium collection and tighter intraday monitoring. A natural use is a dated table of regime flags, joined to every Indian backtest as structural-break dummies.

Three papers show how to measure what a rule change did. Wang et al. (2026b) treat China's 2024 program-trading provisions as a quasi-natural experiment. They build a stock-level HFT-intensity index and use difference-in-differences, finding larger spread reductions where prior HFT exposure was higher, and larger effects after negative-return periods. Hussain and Olekar (2026) exploit the 2025 swap of weekly expiry days between two Indian exchanges and subject each estimate to a four-part falsification battery, because only two exchanges are involved. The pairing is useful: one paper supplies the exposure-based design, the other a discipline for distrusting it. Khatwani (2026) describe the Closing Auction Session that replaced VWAP-based closing prices for F&O stocks, and report early sessions only; their observations are anecdotal until more data accumulate. Older and cross-market evidence enters through Hillion and Suominen (2004), whose agency model of closing-price manipulation explains why the last minute at the Paris Bourse was so active before a closing call auction, so the closing window deserves both execution care and surveillance. Babu and Dsouza (2021) fit the same event-study mould for 52 Indian IPOs in 2018–2020, examining abnormal returns and their link to over-subscription,

profit after tax, promoter holdings and issue price.

Desk note. Regime calendar for Indian backtests (regime stage; validation). Input: dated SEBI circulars, lot-size, margin and expiry changes, auction-regime start dates (Gunnam et al., 2024; Potharla, 2025; Khatwani, 2026). Output: a categorical regime feature plus a hard rule that no model is fit across a flagged break without a break dummy, and that cost models are re-estimated per era. Failure modes: look-ahead if announcement and effective dates are confused; too many flags for too few observations; and, as Hussain and Olekar (2026) warn, assuming a rule change moves volatility before placebo tests agree.

4.2 Participants and flows

Several papers describe who is on the other side of a trade. Agarwal et al. (2025) study Indian retail options traders, who concentrate in short-maturity index options, hold briefly and lose money; supply-side interventions raised trading, and participation costs pushed investors toward cheaper, shorter-dated out-of-the-money options. Saw and Santosh (2026) survey 320 retail investors and report that herd behaviour shapes perception and participation, partly mediated by perception and moderated by financial literacy. Questionnaire evidence is weaker than trade data, and a desk should treat it as hypothesis generation. Bunkar and Ramaiah (2024) use a similar survey design for intention to use algorithmic platforms, a rough guide to how much retail flow may become automated.

Flows can also be measured. Madhumathi et al. (2013) find that Indian equity-fund flows respond to lagged returns and that outflows follow negative market returns without a spiralling effect. Sicherman et al. (2016) use daily account logins: logins fall by 9.5% after market declines and when the VIX is high. Gabaix and Koijen (2021) close the loop with the inelastic markets hypothesis: because institutions are constrained, flows move aggregate prices, and a dollar invested raises market value by about five dollars in their estimates.

Desk note. Flow and attention overlay (signal and risk stages). Inputs: mutual-fund flow data, broker login or search-activity indices, and short-dated retail option volume (Madhumathi et al., 2013; Sicherman et al., 2016; Agarwal et al., 2025). Output: a slow crowding or flow-pressure feature, or an exposure cap in short-dated index products, and a candidate contrarian overlay (Saw and Santosh, 2026). Risks: flow data arrive with a lag, so the feature may be stale on arrival; attention proxies are a consequence of returns as much as a cause; and elasticity estimates (Gabaix and Koijen, 2021) are macro-level and not a capacity model for one book.

Taken together, these papers suggest a layered picture: slow, measurable flows set the background pressure on prices, while attention and herding explain why retail participation clusters in particular products and moments. None is a trading rule by itself. Each could serve as a conditioning variable, telling a model when its usual relationships are likely to be distorted by who is active.

4.3 Manipulation and surveillance

Surveillance is where classification methods meet market rules. Zulkifley et al. (2023) survey AI detectors of stock manipulation and give a baseline taxonomy. The newer papers differ by data and idea. Chen and Hybinette (2026) detect pump-and-crash patterns in the velocity of option Delta and price at minute frequency on BANKNIFTY, with a time-partitioned test and SHAP attribution; conditioning on hidden-Markov regimes trades recall for precision, a useful negative result. Abdulrazaq and Musa (2026) address class imbalance in cryptocurrency pump-and-dump data with SmoteGAN augmentation and a TabNet classifier, reporting precision and recall on

25-second intervals. Zwydak et al. (2026) instead use complexity measures (multifractality, entropy, detrended cross-correlation) to flag exchanges whose trade counts rise without matching volume. Y et al. (2026) fuse disclosure text, online sentiment and trading features at firm-month level against CSRC sanctions. Tsaknaki et al. (2026) flip the viewpoint: in an Almgren–Chriss setting an RL agent learns manipulative strategies, a red-team generator for detectors.

4.4 Algorithmic fragility, opacity and oversight

The last group asks what happens when many fast agents interact. Jain et al. (2016) study Tokyo's Arrowhead platform and argue for CoVaR-style correlation measures in tail states. Ng (2026) classify AI-driven flash crashes as herding, model-error cascades or GenAI disinformation. Dou et al. (2025) model coordinated AI speculators who create and then unwind bubbles. Bajpai (2025a) recast markets as adversarial control systems with a proposed Control Fragility Index, which is unvalidated. Wallace and Wallace (2025) offer a biological analogy for limits on HFT-AI systems, and Farmer and Skouras (2013), known to us by title only, take an ecological view of computer trading. Aldridge (2026) surveys how trading rules evolved through electronic books, auctions and AI agents.

On the oversight side, Raschner (2021) analyses MiFID II's testing and simulation requirements, Gulhane (2026) argues that undisclosed strategy patterns across broker APIs form functional dark liquidity in India, and Hansen (2025) draw on 182 practitioner interviews to argue that analysis should shift from algorithm logic to the layered stack and its dependencies.

Desk note. Pre-deployment simulation and stress set (validation and risk). Input: a simulated market or replayed tick data, scenarios from the flash-crash taxonomy (Ng, 2026; Dou et al., 2025), and a stack dependency map (Hansen, 2025). Output: documented pass criteria, in the spirit of Raschner (2021), and tail-state position limits (Jain et al., 2016). Risks: simulators embed their builders' assumptions, so passing proves little about unseen regimes; and scenario sets tend to fit last year's crash.

Read together, the group offers a vocabulary of failure modes rather than forecasts. A fund can use it to ask which of its own strategies would be crowded, correlated with others in a stress, or exposed to a venue or broker dependency, and to write those questions into its risk review before an incident does.

4.5 Bridges out of this chapter

Bridge. To microstructure and execution (Chapter 12). What this chapter calls a "closing-window manipulation signature" (Hillion and Suominen, 2004) is what order-book work calls an end-of-day liquidity and impact regime. Expiry and auction reforms are natural experiments for the impact and cost models there.

Bridge. To time-series mining and regimes (Chapters 5 and 7). A manipulation episode is, in mining language, a discord or anomalous subsequence; velocity features (Chen and Hybinette, 2026) and complexity diagnostics (Zwydak et al., 2026) are representations that anomaly search could reuse. Regulatory dates are the change-point candidates that regime methods usually have to discover blindly.

Bridge. To synthetic data and validation (Chapters 3 and 14). Simulation-based certification and rare-event augmentation are the same need seen from law and from deep learning. The falsification battery of Hussain and Olekar (2026) is a multiple-testing safeguard that backtesting chapters can borrow.

Part III

Representation and Pattern

Chapter 5

Time-Series Methods: Forecasting, Classification and Mining

Time-series mining is the least "financial" chapter of this atlas, and that is exactly why it is here. Most of the papers below were written by people who look at water-treatment plants, accelerometers, servers, hospital wards and seismometers, and who have never been asked to produce a Sharpe ratio. Three communities meet in this chapter. The data-mining school (the *tsml* silo, centred on the matrix-profile programme) speaks of *primitives*, *motifs*, *discords*, *shapelets* and *similarity joins*. The statistics school speaks of dynamics, long memory, spectra and system identification. The finance and econometrics school speaks of clusters of stocks, volatility regimes and "mechanisms". A small group of deep-learning and LLM papers appears at the edges.

The atlas's shared vocabulary lets us translate. A *motif* is a recurring pattern, which a trader would call a setup or a template. A *discord* is the subsequence least like anything else, which is an anomaly, an outlier bar sequence or a dislocation. *Similarity search* is nearest-neighbour matching of a query shape against history, the ancestor of every "find days like today" study. *Clustering* is grouping series by shape, dynamics or features, and the trader's version is a peer group, a sector proxy or a hedge basket. *Representation* and *feature extraction* turn a window of raw prices into a vector that a model can consume. The recurring concepts here are similarity-search, motif, anomaly, representation-learning, regime-shift and cross-section.

In the fund pipeline (data → features → signal → regime/risk → portfolio → execution → validation → research automation) this chapter sits mostly at the first two stages. It supplies distance functions, indexes, descriptors and unsupervised groupings that later stages consume. It supplies few finished alphas. This half of the chapter covers the mining primitives, clustering and feature libraries. The second half turns to forecasting and classification models proper. Chapter 6 covers learned representations in more depth, and Chapter 7 picks up the regime and change-point uses of the same tools.

5.1 Similarity and the matrix profile as a shared engine

Nearly everything in the mining silo reduces to one operation: compare a subsequence with other subsequences. Berndt and Clifford (1994) is the historical anchor, an early experiment applying dynamic time warping (DTW) to find patterns in time series, with stock prices named among the example data. DTW tolerates stretching along the time axis, so two similar shapes that unfold at different speeds still match. Rakthanmanon et al. (2012) then attack the practical objection, that similarity search is the bottleneck of mining algorithms, and show how to search and mine trillions of subsequences under DTW. For a desk this is the difference between a research notebook and a screen that runs across a whole universe.

The *matrix profile* generalises the idea. Yeh et al. (2016b) frame all-pairs similarity joins

as a unifying view that includes motifs, discords and shapelets: for each subsequence, store the distance to its nearest neighbour elsewhere in the series. Low values mark motifs and high values mark discords. The follow-ups are mostly about scale and convenience. Zhu et al. (2016) push exact motifs and joins past one hundred million points using GPUs, Zhu et al. (2018) give an interactive-speed algorithm (SCRIMP++), and Zimmerman et al. (2019) scale GPU computation to a quintillion pairwise comparisons a day. Yeh et al. (2016a) use the profile to visualise salient subsequences in massive series. Shahcheraghi et al. (2024) bring the same scaling ideas to recurrence plots and similarity matrices, from which features can be derived for downstream analytics.

Several papers relax the assumptions that make the basic profile brittle. Furtado Silva and E. A. P. A. Batista (2018) extend it beyond Euclidean distance and fixed subsequence length to elastic measures, and Alaee et al. (2020) give exact motif discovery under DTW. Zhu et al. (2021) handle missing data, a daily reality for tick archives with gaps and halted names. Der et al. (2022) propose a distance for comparing long series with cyclical behaviour, where Euclidean distance and DTW are less meaningful, and Gharghabi et al. (2018) argue that the common distances are less robust than believed and propose MPdist. On the symbolic side, Fuad and Marteau (2013) speed up the symbolic aggregate approximation (SAX), a dimension-reduction representation for similarity search. Approximation is a theme too: Yeh et al. (2021) give error-bounded approximate joins using compact dictionary representations.

The anomaly side is where this half of the chapter first touches risk. Yeh et al. (2024) apply the matrix profile to anomaly detection in multidimensional series, and Yeh et al. (2023b) sketch multidimensional series so that discord mining has runtime independent of the number of dimensions, with reported speedups of at least an order of magnitude on water-treatment and transportation data. Their examples are not financial, so the finance reading is ours: a multi-asset panel is just a multidimensional series.

Desk note. Streaming dislocation screen (data → regime/risk). Input: intraday returns or normalised mid-prices for a few hundred instruments, windowed to a fixed length. Output: a per-instrument discord score, built on multidimensional matrix-profile ideas such as those of Yeh et al. (2024) and the sketching of Yeh et al. (2023b), that serves as a risk filter (reduce size when the current window is far from everything seen in the last months) or as a flag for manual review. What can go wrong: the profile is distance-based, so volatility scaling and z-normalisation choices decide what counts as unusual; a discord is not a direction, so it is a filter, not a signal; and in-sample thresholds tuned on a single crisis will not generalise. Always compute the nearest-neighbour search using only data available at the decision time.

5.2 Motifs, chains, shapelets and the question of length

If the profile is the engine, motifs are the most-used output. Yeh et al. (2017b) ask what a meaningful multidimensional motif is, and Dau and Keogh (2017) propose a generic way to incorporate domain knowledge so that discovered motifs match the user's intent. The recurring practical problem is length. Linardi et al. (2018) introduce VALMOD for variable-length motifs, Linardi et al. (2020b) package it as a suite for exact detection, and Linardi et al. (2020a) extend the idea to variable-length motifs and discords together. Wesenbeeck et al. (2023) (LoCoMotif) find time-warped motifs, removing the limitations of fixed length, of only the two most similar occurrences, and of univariate input. For traders, a "pattern" that must have exactly 30 bars is arbitrary, so the length question is the real one.

Three papers bring this to markets. Rodrigues et al. (2019) study motif discovery and analysis in financial time series with VALMOD, and flag the evaluation and significance of motifs as the issue. Cremona et al. (2023) embed asset prices in a functional-data framework to find functional motifs in stock prices, motivated by recurrent shapes such as the surge of a bubble.

Cartwright et al. (2022) combine SAX, minimum description length and the matrix profile in a side-length-independent method, and link motif discovery to volatility analysis. All three are research results. None should be read as evidence that a motif earns excess return.

Bridge. Motif vs. technical pattern. What the mining silo calls a "motif" is what technical analysts call a chart pattern, and what the strategy literature (Chapter 10) tests as a rule. The mining view adds two things the chart view lacks: a distance threshold, so that pattern membership is not a matter of taste, and a significance question, which Rodrigues et al. (2019) raise explicitly. Conversely, mining people can borrow the backtesting literature's warning (Chapter 14) that a pattern found by searching many lengths and many assets needs a multiple-testing correction before it counts.

Other primitives answer different questions. Zhu et al. (2017) introduce time-series chains, ordered sets of subsequences in which each resembles its predecessor while the first and last may be arbitrarily dissimilar, which is a formal way of describing gradual drift and so connects to regime-shift ideas. Imani et al. (2018) propose snippets as a representative summary of a long series, arguing that motifs, shapelets, cluster centres and random samples are poor answers to "show me typical data". Lee et al. (2024) formalise the smallest set of representative series for a similarity threshold. Ye and Keogh (2009) introduced shapelets, discriminative subsequences for classification that can be read by a human. Mercer et al. (2022) define the contrast profile, a primitive that contrasts two classes of subsequences for real-world classification, and Yeh et al. (2017a) use weakly labelled series to predict outcomes without needing precisely extracted, correctly labelled subsequences. That last assumption is close to the trader's situation, where events are fuzzily dated.

Two papers look at patterns on a coarser footing. Wu et al. (2023) mine top-k contrast order-preserving patterns, i.e. frequent relative orders that differ between two classes of series. Aghamohammadi et al. (2014) use a permutation approach on tick-to-tick exchange rates to examine how the variety of micro patterns changed as high-frequency trading emerged. Rank-order patterns are robust to scale, which suits prices.

5.3 Clustering series: shape, dynamics and mechanism

Clustering asks which series behave alike. The survey of Warren Liao (2005) remains the entry point. Choices are the representation (raw, features, model parameters), the dissimilarity and the algorithm. Papers in this section differ mainly in the dissimilarity.

Shape-based methods compare waveforms. Fahiman et al. (2016) extend c-shapes to fuzzy clustering of finite waveforms, and Begum et al. (2016) give a density-based framework with an admissible pruning strategy so that DTW clustering stays tractable. In finance, D'Urso et al. (2021) use trimmed fuzzy clustering with DTW on financial series, with the trimming meant to cope with outlying series, and Xiao (2025) use DTW-based K-means within a causal-inference framework (additive noise model mixtures) to cluster return mechanisms. The causal claim is the author's to defend, and a desk should treat inferred directions as hypotheses.

Dynamics-based methods compare the generating process. Yue et al. (2025) cluster vector autoregressive series through system identification, Cerqueti and Mattera (2023) cluster by a time-varying Hurst exponent, so that long memory is allowed to change, and Cerqueti et al. (2024) cluster multivariate series with time-varying parameters in a multiway framework. *Frequency-domain* variants include Caiado et al. (2020), who cluster big collections using a fragmented periodogram around the driving cycles, and Ghosh (2025), who apply k-means to polyspectral means of 49 Indian stocks, bringing bispectral information that captures nonlinearity beyond the power spectrum. Caiado and Crato (2010) combine threshold-GARCH parameters with the periodogram of squared returns to cluster stock returns, a volatility-aware dissimilarity.

Feature-based methods are the bridge to the next section. Wang et al. (2007) represent each series by a fixed-length vector of statistical features and cluster the concatenation, which also

reduces dimension, and Bonifati et al. (2022) (Time2Feat) learn interpretable feature subsets for multivariate clustering. Machado et al. (2015), from accelerometer-based activity recognition, shows how sensitive unsupervised results can be to feature-extraction parametrisation, a useful warning for any feature-based grouping of assets. Oriona et al. (2023) treat ordinal-valued series, with economic applications, which resembles discretised ratings or state labels. Kim and Hwang (2024) use sparse functional PCA to find hidden dependence structure in sparse longitudinal data.

Desk note. Dynamics-aware peer groups (features → portfolio). Input: daily returns and realised volatility for the tradable universe. Output: cluster labels, from a volatility-and-spectrum dissimilarity such as that of Caiado and Crato (2010) or a trimmed fuzzy DTW scheme such as that of D'Urso et al. (2021), used as (i) hedge baskets for relative-value trades, (ii) a risk-concentration limit per cluster, or (iii) a neutralisation group for cross-sectional signals. Fuzzy memberships give soft limits instead of hard sector walls. What can go wrong: clusters estimated on the full sample leak the future into any backtest, so re-estimate on rolling windows and track label stability; fuzzy and trimmed variants still need a choice of cluster count; and clusters formed in calm periods can merge in a crisis. Treat cluster stability as a monitored quantity.

5.4 Feature libraries and the interrogation of series

A different answer to "how do I describe a series?" is to compute everything. Fulcher et al. (2013) organise annotated collections of over 35 000 real-world and model-generated series against over 9000 analysis methods, and study the empirical structure of both. Fulcher and Jones (2017) turn this into hctsa, a framework for automated phenotyping through massive feature extraction, and the same group's Fulcher et al. (2019) (CompEngine) offers a self-organising, living library of series intended to connect fields that rarely talk to each other. Fulcher (2026) supply a controlled testbed: 1000 simulated series from 133 mechanisms, including long-memory, multifractal, chaotic and noise-driven processes. In atlas terms this is alpha-mining at the feature level: generate thousands of candidate descriptors, then select. The multiple-testing concept attached to Fulcher and Jones (2017) and Fulcher et al. (2013) is therefore not a footnote. With thousands of features and a few years of data, some feature will correlate with next-week returns by luck.

Classification papers show how features and shape complement each other. Alaee et al. (2019) argue that it is a false dichotomy: some classes are best separated by features and others by shape, so a classifier should be able to use both. Baldán and Benítez (2020) study complexity measures as features for classification, in the setting where a hierarchical-vote ensemble dominates the state of the art. For anomaly detection at scale, Hyndman et al. (2015) compute a vector of features on each of thousands of monitored series and look for unusual ones in feature space. This is a design a desk could use to monitor thousands of strategy P&L streams or data feeds. Hudson et al. (2021) use subgroup discovery to find interpretable local exceptional behaviour in time series, outside finance, and the interpretability is the attraction.

Finance-flavoured descriptors appear in a few places. Hosking (1981) generalise ARIMA models by letting the degree of differencing take fractional values, which yields processes with long-term persistence. Practitioners have since used fractional differencing as a route to features that keep memory while removing a unit root; that use is our gloss, not a claim of the 1981 paper, and Chapter 14 returns to such window-safe transforms. Lacasa and Flanagan (2016) use visibility graphs to quantify time irreversibility of 35 financial indices, and show that the measure complements volatility-based ones, which they exploit for classification. The graph viewpoint is itself a bridge: a series becomes a network and network tools apply (Chapter 7).

Desk note. Feature-bank screen with a hard multiple-testing budget (features → validation). Input: a rolling window of returns, volume and realised volatility per instrument. Output: a few hundred descriptors in the style of Fulcher and Jones (2017), with fractionally differenced inputs per Hosking (1981) and graph-based irreversibility per Lacasa and Flanagan (2016) as candidate additions, scored for stability across assets and sub-periods. What can go wrong: selection on the full sample leaks the future; descriptors that are highly collinear inflate the apparent number of independent discoveries; features that need long windows have few independent observations; and a descriptor tied to a single regime (a volatility regime, say) will decay. Log the number of features tried, and deflate performance accordingly (see Chapter 14).

5.5 Labels and the modern edge

Two recent papers show where the tradition is heading, and each carries a warning. Yang et al. (2025) argue, for forecasting, that not all data are good labels. They re-label series by self-supervised reconstruction, using intermediate outputs as pseudo-labels that improve generalisation for any predictor. For finance, where the target is a noisy return, the idea of denoising the label is attractive but unproven. We suggest it as an experiment, and a leak-free implementation would fit the reconstruction network only on the training fold. Gao et al. (2026b) bridge clinical time series into a pretrained vision-language model with a compact adapter. It is a medical paper and peripheral here, but it shows the direction of travel: irregular numerical series are being fed to foundation models, a theme of Chapter 6 and the LLM chapter.

Bridge. Mining vocabulary for the fund. A "discord" of the matrix-profile school is what a risk manager would call an out-of-distribution day, and a "snippet" (Imani et al., 2018) is a representative regime exemplar. "Time-series phenotyping" (Fulcher and Jones, 2017) is feature engineering, and "weak labels" (Yeh et al., 2017a) are the fuzzy event dates and triple-barrier-style targets that the backtesting chapter discusses. In the other direction, finance contributes what the mining silo mostly lacks: costs, capacity and a hostile non-stationary adversary. A discord detector that is 50 times faster is only useful if the flagged window is actionable before the market reprices it.

Two cautions run through the chapter. First, nearly every method here is unsupervised or exploratory, so its output is a hypothesis, to be tested out of sample with realistic costs. Second, evaluation conventions come from domains where the data are stationary and labelled, which makes benchmark gains on those archives a weak guide to performance in markets. The distinction between a better distance measure and a better trading signal should be kept explicit throughout.

5.6 Who leads whom: causal structure and lead–lag

A forecasting desk rarely asks only "what will this series do?". It also asks which other series carry information about it, and whether that information is stable enough to trade. The papers in this group attack that question with tools that come from climate science, cardiology, econometrics and machine learning, and they use one shared idea: replace a symmetric correlation by a directed, conditional statement about dependence.

The econometric root is the testable definition of causality and feedback between two series, collected with the rest of Granger's work in Granger et al. (2001). The modern, high-dimensional descendant is the causal-discovery pipeline of Runge et al. (2019), which combines linear or nonlinear conditional-independence tests with a graph-discovery algorithm and is validated on climate and heart-rate series. Löwe et al. (2022) ask a different question: when many samples share the same dynamics but have different underlying graphs, can one amortised model infer the graph for each sample rather than refitting every time? The broader review in Runge et al.

(2023) places these tools within a causal-inference framework for time series, and is useful mainly for its vocabulary of assumptions (what must hold before an arrow can be read as a cause).

Finance has its own, more modest versions. Wilinski et al. (2018) build a Hilbert-transform complex correlation that works on unevenly spaced intraday quotes and picks up lead–lag relations at several scales without knowing the lag in advance; on Tokyo Stock Exchange data they report that sectors and subsectors form market components that follow each other with small delays. Zhang et al. (2023a) detect lead–lag relations in lagged multi-factor models by clustering sliding-window subsequences. Kenett et al. (2014) strip out the influence of a third stock with partial correlations, which is the static, non-directed cousin of the same conditioning idea. Koopman et al. (2015) model dependence between tick-level price changes with time-varying discrete copulas, and Patton (2012) survey copula models for economic series, whose appeal is that marginals and dependence can be specified separately. Two further papers apply clustering or classification to market structure itself: Dehao (2022) combine a graph-attention autoencoder with a self-organising map to cluster multidimensional market series, and Yeh et al. (2023a) classify foreground subsequences hidden inside background noise using nearest-neighbour graphs, a setting that resembles finding a pattern in a long price history.

Desk note. Lead–lag screen for input selection (signal stage). A natural use is a nightly job that takes returns, or intraday mid-quote changes, for a universe of related instruments, runs a conditional-independence or partial-correlation screen, and outputs a sparse directed candidate graph. The graph is a *feature-selection filter*: only instruments with a stable incoming edge enter the signal model for a target. What can go wrong: with realistic sample sizes and heavy tails, nonlinear tests have low power and many edges are false; a common driver (the market factor) can masquerade as lead–lag; and an edge found on quotes may vanish after costs and latency. Treat every edge as a hypothesis to be validated out of sample, and track edge stability across rolling windows as a diagnostic in its own right.

5.7 Small, fast and frequency-aware forecasters

A large part of the recent forecasting literature, written for traffic, energy and weather benchmarks, asks how little model is enough. For a systematic fund this is the relevant question, because the signal-to-noise ratio in returns is low and large networks mostly fit noise. Before any of these, the classical baseline deserves a sentence: Scrivano (2025) review ARMA models as a computationally cheap and statistically principled tool for stationary series. A desk should be able to beat that baseline net of costs before it spends engineering time on anything below.

The frequency-domain family starts from the observation that a series can be manipulated spectrally. Xu et al. (2023) interpolate in the complex frequency domain, discard high-frequency components, and reach performance comparable to larger models on forecasting and anomaly detection with roughly ten thousand parameters. Eefsen et al. (2024) replicate that result and then show, on a wide range of datasets, that the tiny model especially excels at capturing periodic structure while missing trends. That is an honest and useful caveat: a model that captures seasonality will look strong on intraday volume or volatility seasonality and weak on drift. Yi et al. (2023) apply MLPs in the frequency domain to overcome point-wise mappings and information bottlenecks, Fei et al. (2025) amplify neglected low-energy spectral components, and Huang et al. (2025) decompose mixed-frequency series and model each component with Kolmogorov–Arnold blocks. Zhang et al. (2025a) do the multi-resolution version with wavelets inside an MLP-Mixer, and Ferdaus et al. (2026) review twenty-six KAN-based architectures, noting practical deployment challenges.

A second sub-family mixes scales or removes decomposable parts before learning. Wang et al. (2024) treat the series as a mixture of patterns that differ by sampling scale. Choi et al. (2026b) make the dominant cycle depend on the input window rather than the dataset, Wang

et al. (2026c) separate period-aware components from trend and noise, and Liu (2026) combine decomposition with a lightweight state-space design to mitigate non-stationary distribution shift. Wang et al. (2025d) push the same multi-scale, multi-resolution idea into one backbone meant to serve forecasting, classification, anomaly detection and imputation.

Two papers argue about the architecture itself. Naghashi et al. (2025) reconsider the GRU as a cheaper alternative that still captures cross-channel dependence, and Oreshkin et al. (2020) remain the reference for the interpretable, fully connected, residual N-BEATS stack; Binte Habib et al. (2025) extend it with multi-head self-attention, but report their results mainly on a weather dataset, with brief generalisation checks in finance and healthcare; the abstract does not say what financial target was forecast, so any trading relevance is ours to test, not theirs.

Desk note. Cheap-forecaster ladder (forecast stage). Input: a panel of daily or intraday features (returns, volume, realised-volatility proxies). Output: a point forecast or, more usefully, a seasonality and trend decomposition used as a feature. A sensible workflow is a ladder: ARMA, then a frequency-domain linear model, then a multi-scale mixer, with each rung required to beat the previous one *after costs* on purged, walk-forward folds. Lightweight models suit latency-limited intraday signal generation and are easy to retrain. Risks: benchmark datasets here are periodic by construction, whereas returns are close to unpredictable in the mean, so the gains seen on electricity or traffic will shrink sharply; a model that captures seasonality may simply restate a known intraday volume curve; and normalisation or decomposition steps fitted on the whole sample leak the future.

5.8 Transformers, foundation models and exogenous information

The Transformer branch of forecasting has gone through its own swing of fashion. Liu et al. (2024) invert the usual tokenisation so that attention runs over variates rather than time stamps, and Cheng et al. (2024) pursue a related two-dimensional view with vanilla Transformers. Wang et al. (2026i) make the scale of the multivariate model adaptive, and Shen et al. (2025) ask which of encoder-only, encoder-decoder or decoder-only designs works best once time-series-specific tricks are stripped away. Lim et al. (2020) remain the practical reference for multi-horizon problems that mix static covariates, known future inputs and observed exogenous series, with interpretability built into the architecture. Liu et al. (2025b) is the one most relevant to markets because it treats non-stationarity as something to be separated into short-term and long-term effects rather than simply removed, an idea we return to in the next section.

The growing exogenous-information theme is where these ideas touch finance most directly. Dong et al. (2026) design a financial multi-scale Transformer that fuses exogenous drivers, on the premise that a target asset is coupled to other series. Outside finance, Wang et al. (2025b) fuse text, time series and satellite imagery for demand forecasting and H et al. (2026) align multilingual text with numerical streams for risk monitoring; their data differ from ours, but the fusion problem (aligning slow, sparse, textual evidence to fast numerical series without leaking future timestamps) is exactly the one a news-aware signal faces.

Foundation-model work changes the question from "which architecture?" to "can we skip training?". Hoo et al. (2026) treat forecasting as tabular regression with temporal featurisation on top of a pretrained tabular model, reporting strong zero-shot results on covariate-informed forecasting. Wang et al. (2026h) route series to experts guided by a regime predictor, which in a multi-instrument setting is a way to let one model specialise by forecastability. Fechete et al. (2025) let the forecaster re-weight its loss toward an application-specific range at inference time, a natural fit when only tail moves matter for stop placement. Marusov et al. (2026) supply a contrastive generative model for probabilistic forecasting, closer to what risk needs than a point forecast. Two other papers show how far the same machinery travels: Allam and McEwen (2021) classify astronomical transients with a Transformer that is agnostic to channel structure, and

Dong et al. (2024) forecast wind power with a dynamic hypergraph over linked producers. Both are peripheral to equity trading, but the first resembles tagging irregular event bursts across instruments, and the second resembles energy-market trading, where physical spatial coupling exists and the hypergraph has something real to encode.

Desk note. Zero-shot and exogenous forecasters as challengers (forecast and signal stages). A tabular or time-series foundation model could serve as a *challenger* that runs beside the house models on the same covariates (term-structure points, factor returns, lagged flow). Its output is a forecast distribution and a disagreement score against the incumbent. Strengths: no training pipeline to leak, easy to rerun daily. Risks: pretraining corpora may include the very market series being evaluated, so a clean test needs data dated after the model's release; exogenous covariates must be lagged to what was knowable at the decision time; and capacity is small, since a signal that looks good only in tails may be untradeable at size.

5.9 Non-stationarity, regimes and anomalies

If one fact separates finance from the benchmarks above, it is that the process changes. Several papers here attack this directly. Liu et al. (2023a) use Koopman theory with a Fourier filter to disentangle time-invariant from time-variant dynamics, and Alkharashi et al. (2025) propose a self-learning type-2 fuzzy system that prunes its rule base as forecast accuracy deteriorates after adaptation, an online, interpretable response to shifting dynamics. He et al. (2026) is explicitly financial: it argues that log-returns are non-stationary with very low signal-to-noise, and that a fixed context window is mismatched to regime-dependent dependence, so it builds a regime-aware expert network with variable context. A parallel in a different domain is Dong et al. (2025a), which clusters load and price jointly into market states before adaptive forecasting; the same joint-state idea transfers to a price–volume state for equities. In the pattern-mining tradition, Cartwright et al. (2021) apply matrix profiles to financial series and propose several ways to find similar behaviour patterns, linking this chapter to the motif material earlier in it.

Regime labels are only useful if one can tell a regime change from an outlier, and the anomaly papers address that. Li et al. (2021) propose a clustering-based anomaly detector for multivariate series in which scores account for both temporal and variable relationships. Der et al. (2024b) go one step further and ask for an explanation after an anomaly is flagged, so that a human can triage it. The same concern drives the explainable-clustering papers: Fragkathoulas et al. (2026) use counterfactual explanations to show what minimal temporal change would move an instance across a cluster boundary, Schlegel et al. (2025) survey explainable deep clustering, and Boniol et al. (2025) present a visual system for graph-based clustering. For a regime classifier, the counterfactual question (what would have to change for the label to flip?) is the right monitoring question.

Desk note. Regime tags with explanations (regime and risk stages). Input: multivariate daily or intraday features. Output: a regime label plus a nearest counterfactual ("how far is today from the boundary?") and an anomaly score, consumed by the risk layer as a leverage or signal-weight filter. Failure modes: labels are fitted in-sample and flip with a refit, so the regime history is not the history the desk would have seen live; clusters that look stable may be a volatility proxy in disguise; and a filter that turns off signals after a drawdown has already begun costs return in exactly the rebound.

5.10 Evaluation, hygiene and tooling

The last group is about whether we should believe any of the above. Its common idea is that evaluation machinery deserves scrutiny as much as the models do. Wu and Keogh (2020a)

argue that popular anomaly-detection benchmarks are flawed and create an illusion of progress. Spiliotis et al. (2020) ask whether forecasting-competition datasets are representative of reality, and Goktas et al. (2026) propose a framework for time-series foundation models because the field lacks an accepted evaluation protocol. Bagnall et al. (2017) is the classification equivalent: a bake-off of recent algorithms on a standard archive, which shows what honest comparison across many datasets looks like. Wu et al. (2021) ask when early classification, deciding from a prefix of a pattern, is meaningful at all, which is the right question for any trader who wants to act before a pattern completes. Wu and Keogh (2020b) is a small but instructive case of the same attitude: an approximate speed-up for dynamic time warping that turns out to be generally slower than the exact algorithm.

Other papers give tools for specific validation questions. Zhang et al. (2025b) propose influence-function data contribution for time series, which could help find which stretches of history drive a model; Hounyo and Lahiri (2023) show that the standard pairwise bootstrap is invalid for model averaging and propose a fixed-design residual-based bootstrap, a result relevant when combining forecasts; and Chatrath et al. (2002) test for low-dimensional chaos in commodity futures, finding strong nonlinear dependence but no long-lasting chaotic structure, a useful reminder that nonlinearity is not the same as predictability. Finally, process and tooling: Goodwin et al. (2023) document a lag between forecasting advances and organisational implementation, and Bacry et al. (2017) provide an open-source Python library for point processes and time-dependent models, the kind of infrastructure that makes research automation cheap.

Desk note. Validation checklist for time-series models (validation stage). Before any model from this chapter is promoted: (i) test on purged, walk-forward splits; (ii) include a trivial baseline and report the gap net of costs; (iii) log the number of architectures and hyper-parameters tried, so the multiple-testing burden is explicit; (iv) use a bootstrap suited to the forecast combination in question; (v) keep a held-out period untouched. A flawed benchmark gives a false sense of progress in a lab; a flawed backtest gives the same in production, only with money attached.

5.11 Bridges out of this chapter

Bridge. Lead–lag and the order book (Chapter 12). What TS mining calls "time-warped, multi-scale lead–lag" is, in the microstructure literature, the way information propagates from a liquid instrument to a slower one across venues. Complex correlation on uneven timestamps and time-varying discrete copulas on tick data speak to the same object as the order-flow papers in that chapter: dependence that changes within the day. A natural joint project is to test whether a causal-discovery screen finds the same edges that order-flow-based lead–lag models find, and whether either survives costs.

Bridge. Non-stationarity and regimes (Chapter 7). What the forecasting community calls a "distribution shift" or "non-stationary window", and the regime chapter calls a "change point" or "state", are the same phenomenon viewed from two ends. The mixture-of-experts and regime-aware designs here supply forecasters; the regime chapter supplies the labelling and detection tools that decide which expert to trust. Likewise, what finance calls a "dependence breakdown" is what anomaly detection calls a "discord".

Bridge. Validation and research practice (Chapter 14). The benchmark-critique papers translate directly into backtest language: "flawed benchmark" means "selection bias in the test set", "leaky normalisation" means "look-ahead", and "number of architectures tried" means "trials" in the multiple-testing sense. Researchers from the time-series community will find a ready-made vocabulary there; traders will find that the classification bake-off is a worked example of the discipline they are asked to apply.

Bridge. Text, features and foundation models (Chapters 15 and 6). The multimodal fusion papers pose the alignment problem that news-driven signals face, while the pre-trained and self-supervised forecasters connect to learned features. Whenever a vendor says "embedding", the TS-mining word is usually "feature vector"; whenever TS-mining says "phenotyping", the trader's word is "factor".

Chapter 6

Representation Learning and Feature Engineering

Four communities meet in this chapter and rarely read one another. Deep-learning researchers speak of *embeddings*, *self-supervised pretraining*, *contrastive views* and *foundation models*. Rough-path mathematicians speak of *signatures* and *iterated integrals*. Topologists speak of *persistent homology*, *point clouds* and *landscapes*. Time-series miners speak of *feature libraries* and *phase-space reconstruction*. Smaller groups add wavelets, complex networks and computer-vision imports.

In the shared vocabulary of the atlas, all of them answer one question: what fixed-size description of a window of data should downstream models see? A learned embedding, a signature, a persistence diagram and a hand-built indicator vector are all *features*. Which of them sits upstream of a *signal* or a *regime* label is a design choice. The vocabulary translates as follows. A "pretrained encoder" is a feature extractor fitted on data that is not the trading target. A "foundation model" is such an encoder, or a forecaster, trained on very many series. A "signature" is a systematic family of ordered cross-moments of a path. A "topological summary" is a shape statistic of a sliding window.

In the fund pipeline (data → features → signal → regime/risk → portfolio → execution → validation) this chapter sits at the features stage. It touches data when synthetic series are used for pretraining, and it touches forecasting and risk at its edges. Chapter 5 treats classification and mining of series, and Chapter 8 treats the forecasters that consume these features.

6.1 Signatures: ordered moments of a path

The signature of a path is a graded sequence of iterated integrals. Chevyrev and Kormilitzin (2026) give a primer on its definition, properties and machine-learning use, and Fermanian (2019) treats embedding and learning with signatures. Király and Oberhauser (2016) use signature features to build "sequentialized" versions of any static kernel. Their abstract describes signatures as an ordered variant of sample cross-moments. That phrase is the most useful translation for a quant: a signature records not only how far a series moved but in what order its components moved together.

Finance entered early. Levin et al. (2013) frame regression with a stream as input and use the signature's universality so that linear regression can be applied to streams. Lyons et al. (2014) propose the signature as a feature set for streams, with an application to high-frequency tick data. Gyurkó et al. (2014) apply it to market events such as order placement and cancellation. Order flow then becomes a path, and its signature is a candidate summary of it.

Newer work makes the transform trainable or compact. Bonnier et al. (2019) combine the signature with deep-learning frameworks by learning an augmentation before the transform. Li and Liu (2022) use signatures inside phase-space reconstruction, because ordinary reconstruction

needs a large embedding dimension to be useful. Yin et al. (2022) show the same machinery on landmark-based human action recognition. We cite it as evidence of portability, not as a financial result.

Desk note. Signature features for order-flow and trend models (features/signal). A natural use is to compute truncated signatures over rolling windows of a multichannel path, for example mid-price, signed volume and spread. The resulting vector feeds a linear or gradient-boosted model. Choose the truncation depth by out-of-sample validation. The feature count grows quickly with depth and channels, so regularise heavily. What can go wrong: window length and time-parametrisation choices act as hidden hyperparameters and invite leakage if tuned on the test span. Transaction costs at tick frequency can erase thin edges, and the features are hard to explain to a risk committee without a worked example.

6.2 Topology: shape statistics for crashes, bubbles and co-movement

Topological data analysis builds a point cloud from a sliding window and summarises its shape. Gholizadeh and Zadrozny (2018) survey TDA for time series and systems analysis, and Bailey (2026) reviews topology as a language for multiscale structure and early-warning signals in complex systems. In finance, Gidea and Katz (2017) track persistence landscapes of daily index returns through the 2000 technology crash and the 2007–2009 crisis. Akingbade et al. (2023) offer a heuristic argument, based on the log-periodic power law singularity model, for why TDA detects bubbles. Mounir (2025) take a neighbouring route and build a stress measure, the Signature Fragility Index, from signatures of daily returns in four equity markets, which links the first two sections.

Two papers use topology for prediction and grouping. Huang et al. (2024) study stock-index movement classification and examine point-cloud construction, topological representation and classifier choice. Jeneret and Diamantis (2025) compare clustering of currencies using TDA-derived features against traditional statistics. The shared message is that these summaries are most defensible as slow-moving stress and structure indicators, not as stand-alone alpha.

Desk note. Topological stress monitor (regime/risk). A natural use is a daily job that computes persistence-based norms on a rolling window of index or cross-asset returns and raises a flag when the norm rises above its own history. The flag could scale gross exposure down or trigger human review. What can go wrong: crash episodes are few, so thresholds are fitted on very small samples. The window and embedding parameters are free, and the series may mainly restate realised volatility. Compare it against a plain volatility or correlation-eigenvalue baseline before trusting it.

6.3 Hand-built and transform-based features

Not every representation is learned. Lubba et al. (2019) distil a large time-series feature library into 22 canonical, interpretable characteristics selected by highly comparative analysis. Silva et al. (2022) propose a complex-networks feature set, NetF, as an alternative to linear and nonlinear measures. Razdan (2004) use wavelets on two strongly correlated financial series and show that correlation can depend on scale. For a pairs desk that is a direct warning: a pair can co-move at weekly scale and diverge at daily scale.

Image encodings borrow from computer vision. Pandey et al. (2026) compare Markov transition field and Gramian angular field images for seq2seq trend prediction, and Li et al. (2025c) propose ChartNet, a unified chart representation meant to reduce the subjectivity of technical chart reading. Their value is as a way to feed convolutional models. They should be judged against the plain indicator vector they replace.

6.4 Self-supervised and contrastive encoders

Contrastive methods learn without labels by pulling together views of the same series. Eldele et al. (2021) propose TS-TCC, which uses weak and strong augmentations with temporal and contextual contrasting. Yue et al. (2021) propose TS2Vec, which contrasts hierarchically so that any sub-sequence has a representation. Meng et al. (2023) review the unsupervised family. Kusupati et al. (2022) contribute a general idea from outside time series: nested, flexible-capacity embeddings, so one vector can be truncated to suit a downstream budget.

Finance adaptations follow. Sun et al. (2022a) propose Self-FTS for intraday stock trading, motivated by overfitting of ordinary deep learning on validation sets. Demirhat et al. (2026) combine TS2Vec with 22 engineered features including technical indicators for BUY/HOLD/SELL classification on Turkish BIST stocks. Hwang et al. (2024) learn temporal embeddings for stock similarity with SimStock, motivated by the ambiguity of regional and sector classifications. Dolphin et al. (2022) learn distributed representations of assets, in the spirit of word embeddings, to model asset correlations. Feng et al. (2025) embed series in context using state-space reconstruction and a Bayesian nonparametric method. Merchant et al. (2026) argue that financial representation learning needs state representations useful for planning across tasks.

Data scarcity is the obstacle. Der et al. (2024a) report that, in their experiments, most self-supervised time-series models were beaten by simple supervised ones, and that pretraining on a larger volume of generated samples gave a noticeable improvement. Chapter 3 treats the generators.

Desk note. Asset embeddings as a universe and neighbour map (features/portfolio). A natural use is to train a stock-similarity encoder on rolling return windows and use nearest neighbours to build peer groups, replace stale sector labels, or seed pairs-trading candidates. Retrain on a schedule and freeze the encoder inside each backtest fold. What can go wrong: an encoder fitted on the full history leaks the future into every fold, and embeddings drift between retrains, which breaks identity mapping. Neighbours found during one regime may not hold in the next, and crowded pair trades have limited capacity.

6.5 Foundation models: pretrained forecasters and tokenisers

The largest group imports the language-model recipe. Ansari et al. (2024) (Chronos) tokenise scaled and quantised values and train language-model architectures with cross-entropy. Zhou et al. (2023) reuse a pretrained language model across time-series tasks. Shi et al. (2024b) scale to billions of parameters with a mixture of experts. Feng et al. (2024) cast multivariate forecasting as next curve shape prediction for zero-shot use. Liu et al. (2026b) move cross-variate modelling out of the raw variate space. Gillman et al. (2025) propose a Fourier head for modelling complex distributions over non-linguistic tokens, which matters for the heavy-tailed returns a tokeniser must handle.

Finance-specific models make the case that general models underperform on candlestick data. Shi et al. (2025b) introduce Kronos for K-line data, covering tasks such as volatility prediction and synthetic generation. Zhu et al. (2025a) build FinCast to address pattern shifts from non-stationarity and multi-domain diversity. Ding et al. (2025) present DELPHYNE as a pretrained model for general and financial series. One study, held as two records of the same title (Rahimikia et al., 2025a,b), evaluates time-series foundation models in finance. We cite no figures from any of them. A desk should treat all claims of zero-shot skill as hypotheses until tested with its own costs and its own universe.

Desk note. Zero-shot forecaster as a benchmark and feature (forecast/validation). A natural use is to run a pretrained model over daily or intraday bars to produce a forecast distribution. Treat that output as one more feature, or as a challenger in the model-comparison harness, never as the sole signal. Log each model version and its pretraining cut-off. What can go wrong: pretraining corpora may contain the test period, which is silent leakage. Inference cost scales with universe size, and gains may vanish after costs. Also check calibration of the quantiles, not just point error.

6.6 Bridges out of this chapter

Bridge. Signatures and order flow (Chapter 12). What microstructure calls an "order-flow path" or "event sequence" is what rough-path theory calls a multidimensional path whose signature gives ordered cross-moments. Microstructure asks whether imbalance at one level precedes the next trade. A depth-two signature term records exactly that kind of ordered co-movement, so it could serve as a drop-in feature set for imbalance models.

Bridge. Topology and regimes (Chapter 7). A "topological early-warning signal" is, in change-point language, a statistic whose level shift marks a candidate regime break. Regime researchers could benchmark persistence-based norms against their usual detectors, and TDA researchers could borrow detection-delay and false-alarm metrics from that literature in place of case-study plots.

Bridge. Foundation models, generated data and validation (Chapters 3 and 14). What the deep-learning silo calls "zero-shot transfer" or "pretraining on generated paths" is what a systematic fund calls a model whose training data is not its trading data. That makes leakage tests, purged cross-validation and deflated-performance checks from the backtesting literature mandatory before any pretrained feature enters production.

Chapter 7

Regimes, Change Points, Networks and Systemic Risk

Five communities talk about the same object in this chapter: a market that behaves differently from one period to the next. Econophysicists and random-matrix people speak of *market states*, eigenvalue spectra and correlation clusters. Time-series machine-learning (TS-ML) people speak of *subsequence clustering*, segmentation and state classification. Statisticians speak of *change points*, control charts, particle filters and instabilities in forecast performance. Ecologists and climate scientists speak of *tipping points*, *critical slowing down* and early-warning indicators. Financial economists and network scientists speak of *contagion*, spillovers, systemic risk and community structure. Pure mathematicians add path signatures and persistent homology as feature maps.

In the shared vocabulary of the atlas, almost all of this is *regime-shift*, *change-point*, *graph-network* and *network-contagion*, with a supporting cast of *cross-section* (a correlation matrix is a cross-sectional object), *order-flow*, *realized-volatility*, *topology* and *path-signature*. A "market state" is a cluster label on a window of data. A "change point" is the boundary between two such labels. A "tipping point" is a change point that a theory says can be anticipated. A "systemic risk indicator" is a regime variable computed on a graph rather than on a single series.

In the fund pipeline the chapter sits between signal and risk: its outputs are rarely alphas. They are filters, gates, state labels and limits that tell the forecasting models of Chapter 8 and the portfolio layer of Chapter 13 how much to trust the recent past. The same ideas also serve validation, because a backtest is only informative about regimes it contains (Chapter 14).

7.1 Market states from correlation structure

The oldest strand treats the correlation matrix of many assets as the state of the market. Drożdż et al. (2001) apply the correlation-matrix formalism to high-frequency DAX data, and report that memory effects are shorter than the standard autocorrelation function suggests and that short repeatable intraday structures exist. Tan et al. (2016) take the same idea to the minute timescale, combining random matrix theory with complex-network methods to study stock and sector interactions and a market state. Münnix et al. (2012) define a state of the market from the similarity of correlation structure and use it to locate points of drastic change in S&P 500 data, and Pharasi et al. (2020) cluster correlation matrices from sliding epochs for the S&P 500 and Nikkei 225 into named market states. Domínguez-Monterroza et al. (2025) show the econometric end of the same toolkit: the eigenvalue spectra of IBEX correlation matrices differ across periods around COVID-19 and deviate from the random-matrix prediction.

One study on NSE intraday data, held as two records in the library (Sharma et al., 2018a,b), uses kernel and functional PCA to find stocks that move together beyond what regression and correlation capture. It is a cluster tool rather than a regime tool, but the clusters are the building

block of a state definition for an Indian universe.

The common recipe is: choose a window, compute a correlation matrix, reduce it by eigenvalues or clustering, and label the window. The recipe has two well-known weaknesses. The window length trades noise against lag, and the labels are assigned after the fact. Every state is therefore visible only in hindsight unless it is assigned by a rule that uses past data only (see the online classifiers below).

Desk note. Correlation-state gate (stage: regime/risk). Input: daily or intraday returns of the tradable universe. Output: a state label, plus the largest eigenvalue and its share of total variance, computed on a trailing window. A natural use is to feed the label as a conditioning variable to the signal-combination layer and to scale gross exposure when the market-mode eigenvalue is high, since diversification then fails. What can go wrong: states are fitted on the whole sample (look-ahead leakage), the window is tuned on the same history that is later backtested, and the state count is chosen post hoc. Fix the window and the clustering rule in a pre-registered config and refit only on expanding data.

7.2 Latent states and online classification

A second strand fits an explicit state model to returns, volatility or order flow. Salvi (2025) model order-flow imbalance with an asymmetric hidden Markov model whose transition structure captures persistence and skewed liquidity, and Koukorinis et al. (2025) combine an HMM with a kernel machine (SVM/MKL) in a generative-discriminative scheme to classify high-frequency regimes and predict trade direction. The idea in both is that the generative model supplies temporally coherent features while the discriminative layer does the labelling. For a desk, this is the closest link to microstructure: the state is defined on flow, so its output can condition execution as well as signals (Chapter 12).

Nystrup et al. (2020) address the latency problem directly with a greedy online classifier, built on clustering, that assigns a hidden state to each new observation without parsing the whole history while keeping states persistent, using realized intraday volatility features. Plümer (2026) take a particle-filter route calibrated to the shape of market stress rather than to specific historical episodes, and Faris (2026) embed persistent homology in the Hamilton filter so that volatility regimes are detected from phase-space geometry and not return magnitude alone, aiming at earlier warning than a second-moment filter.

The path-signature papers offer a nonparametric alternative. Bilokon et al. (2021) classify regimes by encoding time series as path signatures, so that separation of regimes corresponds to clustering, and Issa and Horvath (2023) turn this into an online detector with a path-wise two-sample test based on a maximum-mean-discrepancy metric using signatures as the feature map. Both belong to the representation toolbox of Chapter 6.

Three more papers use ordinary supervised or data-mining machinery. Kurzman et al. (2023) combine several models to predict the "state" of the market, and Ionescu and Dinsoreanu (2013) mine markets through market-state classification with a design that gives the analyst control over which market aspects are analysed. Their value is mainly to show how the problem is framed as classification, with technical indicators as inputs. Doğan and Aytekin (2026) combine the Bai–Perron structural-break test with multifractal detrended analysis of S&P 500 prices and volume, reading the multifractality as evidence against market efficiency. A natural reading for a desk is that breaks and scaling behaviour are properties to monitor, not tradable signals in themselves.

7.3 Change points and segmentation

Where a state model asks "which regime are we in", change-point detection asks "when did the process change". Jebali (2026) apply Bayesian online changepoint detection (BOCPD) and two extensions to order flow, motivated by the need for real-time detection of structural breaks in any high-frequency system. This is the bridge between this chapter and the order-flow literature: a break in flow is a candidate trigger for a spread or sizing change.

The statistical papers supply methods that a desk could borrow. Cribben (2019) detect change points in the network structure of heavy-tailed multivariate series using precision matrices, via the graphical lasso and the tlasso, with an unknown number and placement of change points. Kim and Hwang (2026) build control charts and change-point detectors for multivariate functional data, using a nonparametric conditional distribution and functional PCA to remove variance inflation from cross-correlation. Boulton and Lenton (2019) detect abrupt shifts by searching for gradient changes over a short span, a method conceived for climate series. Banerjee and Guhathakurta (2020) apply change-point analysis directly to financial networks, linking market crashes and contagion across countries.

The TS-ML papers segment without an explicit probabilistic model. Gharghabi et al. (2017) propose a domain-agnostic online semantic segmentation built on the matrix profile, which needs few parameters. Köhne et al. (2023) use an autoencoder and iterative modelling to find change points and cluster subsequences in transient multivariate data from industrial condition monitoring, and Wang and Hsieh (2021) extend hierarchical factor segmentation to several dimensions to dissect a multivariate series into phases. The common thread is that each segment is a candidate regime label, obtained without a return forecast.

A smaller group addresses non-stationarity as a parameter-inference problem. Owens and Fulcher (2024) and its extension Owens et al. (2026) infer time-varying parameters of a non-stationary process through statistical features and slow feature analysis, without learning a generative model. Saggioro et al. (2020) reconstruct causal relationships that themselves differ by regime. The proposal for a desk is to treat a lead–lag or causal link as regime-dependent rather than global.

Desk note. Online break monitor on flow and volatility (stage: signal to risk). Input: one-minute order-flow imbalance, realized volatility and spread series per instrument. Output: a run-length posterior from a BOCPD-style detector, thresholded into a "new regime" flag that shrinks position limits for a cooling-off period and widens execution limits. What can go wrong: false alarms at the open and around announcements, hazard-rate and prior choices that quietly encode the history being tested, costs from turnover induced by the flag, and capacity when many instruments flip together. Validate by counting flags per day out of sample and by measuring the P&L impact of the action taken, not the detector's accuracy.

7.4 Adapting forecasts to shifts

If regimes cannot be detected reliably, the alternative is to let models adapt. Wong and Barahona (2023) build deep incremental learning from decision trees for financial tabular data under distribution shift, and Zhao et al. (2024a) use meta-learning (DoubleAdapt) to update stock trend forecasters incrementally as new data arrive. Both fit the retrain-and-roll loop of a production system; both inherit the usual risk that adaptation speed is tuned on the evaluation period. Rossi (2021) supplies the discipline: a guide to evaluating and improving forecasts in the presence of instabilities, including the 2007–08 crisis. The message, as we use it, is that average out-of-sample loss hides time variation in relative performance, so a desk should report forecast performance by sub-period and test for instability, not rely on a single pooled score (Chapter 14).

7.5 Early warning and critical transitions

Ecology and climate science developed indicators for approaching tipping points: rising variance and autocorrelation, termed critical slowing down. Scheffer et al. (2009) lay out generic early-warning signals across systems that, as their abstract notes, include financial markets. Dakos et al. (2012) compare detection methods on simulated ecological data, Lenton et al. (2012) compare methods for climate tipping points to improve robustness, and Dakos et al. (2015) discuss prospects and limitations of resilience indicators. Boettiger and Hastings (2012) quantify the limits to detection, a cautionary result for any user. For trading, the lesson is that the indicators are transferable but their false-alarm rate must be measured on the target series.

The finance papers do exactly that. Guttal et al. (2016) report no critical slowing down before financial meltdowns, so they are not behaving as critical transitions in the ecological sense, yet rising variability could still signal systemic risk. This is a negative result with a constructive use: variance, not autocorrelation, is the candidate indicator. Fabozzi and Phoa (2026) make a related practitioner argument, that volatility changes character before magnitude, with rising autocorrelation, volatility of volatility and unstable cross-asset correlations as diagnostics. Bouchaud (2026) supplies the methodological stance behind the whole strand, taking scaling, intermittency and extremes as stubborn features of the data to be modelled instrumentally.

Desk note. Rolling variance-and-autocorrelation risk dial (stage: risk). Input: index and sector returns, plus cross-asset correlation. Output: a composite of rolling variance, lag-one autocorrelation and correlation instability, mapped into a risk-limit multiplier. Following the finance evidence above we would weight variance above autocorrelation, and treat the dial as a limit, not an alpha. What can go wrong: indicators rise in benign trending markets, windows are tuned on the crisis they are meant to predict, and a limit tightened too late costs spread and impact when the whole fund de-risks together. Report hit rate and false-alarm rate on episodes withheld from design.

7.6 Networks, contagion and systemic risk

The last strand moves from one market to a system. Neela (2025b) model markets as multi-layer graphs in a "Systemic Risk Radar" for early crash warning. Pawanesh et al. (2024) embed the Indian stock market, treated as a heterogeneous scale-free network, in hyperbolic space and cluster it there, while David et al. (2026) detect communities in financial hypergraphs, which capture higher-order interactions that pairwise networks miss. Dahake and Chakraborty (2026) model correlations as signed networks and study the loss of structural balance during systemic stress, and Xiu et al. (2021) use anomalies in time-evolving correlation networks, together with visibility graphs and a log-periodic power-law extension, to diagnose crashes and predict rebounds. Rai et al. (2024) detect extreme events with topological data analysis across continental indices in one pass. Caraiani and Lazarec (2021) measure how monetary policy shocks change Dow Jones networks using entropy, and Konwar and Dutta (2026) study spillovers between Indian stock markets and monetary policy.

The economics papers address transmission. Rajwani and Kumar (2019) measure contagion between the US and Asian equity markets with time-varying copulas, baghi et al. (2019) use nonlinear Granger causality for Iranian financial sub-systems, and Rouz et al. (2025) study spillover dynamics and causal structure in composite stress networks. Zhu et al. (2025b) examine how bank liquidity hoarding affects the spillovers of Chinese nonfinancial firms. A last pair concerns the fund's own industry: Sabkha and Jbir (2025) investigate whether AI diffusion drives herding and systemic stress, and Meng and Chen (2026b) model AI-driven systemic risk through performative prediction, herding and cognitive dependency. These matter for a systematic fund because crowding in similar models is itself a regime variable.

Desk note. Network concentration monitor (stage: risk). Input: rolling correlation or partial-correlation graphs of positions and factor exposures. Output: a graph-based fragility score (for example community concentration or balance loss) used as a cap on gross exposure within a community. What can go wrong: graph estimates are noisy at short windows, thresholding choices change the topology, and the fund's own positions move the correlations. Treat the score as an additional limit next to a conventional covariance stress test, not as a substitute.

7.7 Bridges out of this chapter

Bridge. To time-series mining (Chapter 5). What the matrix-profile community calls a "semantic segmentation boundary" or a "regime change", and what the statistics community calls a "change point", are the same event. A TS-ML person can supply scalable online algorithms; a statistician supplies the false-alarm control they usually lack. The shared test is the same: how many flags per year, and what did acting on them earn net of costs.

Bridge. To microstructure (Chapter 12). An "order-flow imbalance regime" in microstructure language is a hidden state in an HMM and, in the segmentation vocabulary, a stretch between two change points. Chapter 12 describes the observable quantities that these detectors consume; this chapter supplies the latent-state machinery that can sit on top of them.

Bridge. To volatility and portfolios (Chapters 11 and 13). Volatility regime indicators here play the role that "state variables" play in conditional volatility models, and correlation-state labels are inputs to covariance shrinkage and risk budgeting. A portfolio engineer can treat the state label as a switch for the covariance estimator; a volatility modeller can test whether the topological and signature features add information beyond realized variance.

Part IV

Alpha

Chapter 8

Return Forecasting

Return forecasting is the stage of the fund pipeline where prepared data and features are turned into a statement about what an asset, or a cross-section of assets, will do next. It sits after data and features and before regime and risk controls and portfolio construction: its output is a forecast, a class label, a rank or a conditional distribution that later stages must size, cost and validate. The papers in this half of the chapter come from many communities that rarely cite each other. Time-series machine learning ("tsml") people think in terms of classifiers, kernels and tree ensembles benchmarked on index or single-stock data. Deep-learning authors speak of architectures, encoders and losses. Financial economists speak of risk premia, predictors, the cross-section and out-of-sample R^2. Text and language researchers speak of sentiment and event propagation. Derivatives researchers bring option-implied information. A few mathematicians contribute path geometry and chaos theory.

The atlas translates this vocabulary into a small set of shared terms. What deep learning calls a "trend label" and tsml calls a "target definition" is label design (Chapter 14 treats it as a validation matter). What economists call a "return predictor" and practitioners call a "feature" is an input to a signal. What finance calls the "information coefficient" and learning-to-rank calls a ranking metric is a forecast-evaluation measure. Where a paper reports accuracy or F1 and another reports a Sharpe ratio, the atlas asks which stage of the pipeline each number actually informs, and which costs and capacity limits stand between the number and a fund's P&L.

The sections below group the papers by the problem they attack. First come data preparation and denoising, then the design of targets, then indicator and pattern models, then conditioning information from options, liquidity, volume and text, and finally cross-sectional ranking, benchmarks and probabilistic forecasts. Throughout, a recurring caution applies: most abstracts report accuracy, backtest returns or Sharpe ratios on a single market and period, and these are claims by the authors, not evidence of tradable alpha.

8.1 Preparing the input: thinning, differencing and denoising

Many papers argue that the way raw prices are transformed matters more than the model that follows. Zhang et al. (2023c) make this point for high-frequency data: they propose three strategies for processing raw intraday data, including data thinning and feature engineering, and report F1 gains for SVM mid-price classifiers on Dow Jones 30 stocks. Stempień and Gajda (2025) compare ordinary log returns with fractional differencing and its tempered extension as preparation for LSTM models on four stock indices, evaluating both forecast errors and risk-adjusted trading metrics. The idea, standard in the field, is to remove non-stationarity while preserving as much memory as possible. A natural use of both is as a pre-processing layer that a research team can test independently of the forecaster.

A second family decomposes the series before modelling. Li et al. (2026d) split prices into trend and seasonal components with a wavelet transform and feed a BiLSTM, adding daily price

deviation as a feature. Gao (2026) decompose minute-level A-share prices with CEEMDAN into intrinsic mode functions, forecast each with an Informer and recombine. Bashir et al. (2026) likewise decompose with CEEMD before a bagged ensemble of tree models whose hyperparameters are tuned evolutionarily. Akkas et al. (2026) take a related denoising route, with attention-based autoencoders applied to indicator-augmented BIST 30 data before an XGBoost trader, and G et al. (2026) use wavelet and multiscale entropy as interpretable descriptors of informational complexity inside a cross-attention network. The shared bet is that noise lives at particular scales. A desk should note the standing risk: if decomposition is fitted on a full sample, later data leak into earlier components, so any such step must be refitted inside each training window.

A third family re-represents the series. Wu et al. (2022) map price series to graphs and use the graph embeddings and node weights as inputs to a temporal-attention predictor. Pandey and Kumar (2026) reconstruct a phase space of the Nifty 50 using embedding dimension and time delay before an LSTM, a chaos-theory idea whose abstract carries only an initial and cautious evaluation. Berrada (2026) ask whether a price window can be summarised by its terminal return and realized variance, and show with purged out-of-sample tests on 50 ETFs that path-shape features reject that sufficiency. Da Costa and Gebbie (2020) learn low-frequency patterns on JSE data with a stacked autoencoder and online updating, and they measure backtest overfitting with combinatorially symmetric cross-validation and deflated Sharpe ratios. That last point is unusually careful for this literature. Chapter 6 covers these representation ideas in more depth.

Two papers address structure rather than transformation. Fu et al. (2026) decompose stock data into market, sector and idiosyncratic layers, inspired by asset-pricing theory, to separate systematic risk from idiosyncratic alpha. Xu et al. (2025) use causal representation learning and domain generalization, with a bound linked to multi-factor models, to find invariant features that generalise out of sample. Noseda et al. (2025) address a different obstacle, data that cannot be pooled: desks train a shared LSTM trend classifier by exchanging model updates rather than raw data, with differential privacy.

Desk note. Window-safe preprocessing (data and features stage). Inputs are raw OHLCV or intraday bars. A research pipeline can wrap fractional differencing, wavelet or EMD decomposition and autoencoder compression in a single transform that is fitted only on the training window and then applied forward, so it can be swapped without touching the forecaster. The output is a feature matrix plus a stationarity report. What can go wrong: full-sample decomposition leaks future information; fractional-difference weights and EMD modes change when the window rolls, which breaks feature stability; and thinning rules tuned on one set of tickers may not transfer. Report any gain against the same model on plain returns.

8.2 Designing the target: labels and what a "hit" means

If the preceding papers change the input, the next group changes the question. Zhao et al. (2017) give a formal trend definition based on financial practice and weight training samples by temporal nearness before an LSTM. Akşehir and Kiliç (2022) propose a rule-based labelling algorithm and feature selection to cure class imbalance in a CNN that predicts next-day trade actions for Dow 30 stocks. Feng et al. (2026) use extrema labelling with oversampling and a focal loss for buy, sell and hold classes on 63 A-share stocks, and Kakad and Shinde (2023) use an adaptive labelling strategy and test how much training data an intraday LSTM needs. Zeng et al. (2021) go further: their model selects potentially profitable samples by attention and iteratively refines noisy labels. These papers all treat the label as a modelling choice rather than a given, which is the central lesson of this section.

The triple-barrier method, a standard device in which a position is labelled by whichever of a profit barrier, a loss barrier or a time limit is hit first, appears in several papers. Kang (2025) tune the window and barrier width to balance classes for an LSTM on raw OHLCV data

in Korean markets. Bieganowski and Ślepaczuk (2025) and Bieganowski and Ślepaczuk (2024) combine it with supervised autoencoders and noise augmentation, on indices and FX and on cryptocurrencies respectively; the latter adds fractionally differenced features. MarekKašpar (2026) apply it with volatility-index predictors on Czech stocks and report that, after transaction costs, tree models are usually unable to beat buy-and-hold or momentum benchmarks. That honest negative result is as useful to a desk as any of the positive ones. Gami and Jha (2026) instead target a rare event, a 50% gain in 30 days on NSE stocks, and argue that hard binary thresholds create instability near the boundary, which is the label design problem in its sharpest form.

Other targets concern timing. Guangliang Gao et al. (2015) cast buy-and-sell prediction as a survival problem with a Cox hazard model on technical covariates. El-Yaniv and Faynburd (2012) predict turning points of small swings with Fourier-enriched support vector regression, and koosha et al. (2022) predict Bitcoin tops and bottoms with forest and recurrent models, scored on precision. Merello et al. (2019) change the objective rather than the label: they regress returns, weight samples by magnitude and select the best asset each period. Adriko Debo and Ciresi (2024) add a Bayesian CNN-LSTM whose uncertainty feeds a Deep Q-learning meta-labeller that chooses side and size. Meta-labelling, in the practitioner sense, means a second model decides whether and how much to act on a primary signal.

Bridge. Time-series mining calls a "turning point" or "swing boundary" what statisticians call a change point and what regime researchers call a regime switch. Chapter 7 treats these as detection problems, scored by detection delay, while El-Yaniv and Faynburd (2012) and koosha et al. (2022) treat them as prediction problems scored by trading performance or precision. A fund could use one team's labels to bootstrap the other's benchmark.

Desk note. Label audit and meta-label layer (signal stage). Input: bar data and a primary signal. Output: a per-bar label plus a sizing or filter score. Triple-barrier parameters (width, horizon) become hyperparameters that must be chosen inside the training fold; the Korean-market tuning to balance classes and the Czech cost result together show both sides of this choice. A natural use of the Q-learning meta-labeller idea is to scale position by predictive uncertainty. What can go wrong: overlapping barrier windows create serial label dependence that inflates cross-validation scores, balanced classes do not imply balanced economic payoffs, and a reported 111% annualised return on A-shares (Feng et al., 2026) says nothing about capacity.

8.3 Indicators, patterns and model zoos

Technical-indicator classifiers are the workhorse of this literature. Rai (2026) build an XGBoost three-class (down, neutral, up) model on 36 indicators for three large-cap US stocks, stressing interpretability and hyperparameter tuning. Sharang and Rao (2015) feed deep-belief-network features of technical indicators into a weekly direction classifier for Treasury note futures. Friday et al. (2026) add structural features, fair value gaps and rolling Hurst exponents, to a multi-horizon gated-attention model. Jiang et al. (2021) are the most ambitious of the group: instead of pre-specified patterns they let a CNN learn from chart images which patterns predict returns, and report that the patterns differ from standard trend signals and carry across time scales and countries. Wang and Xie (2024) compare SVM, logistic regression and random forest with sentiment features on the CSI 300, and Leon et al. (2024) search model pipelines automatically for a USD/COP limit-order-book trend classifier. Jabbar and Jalil (2024) screen 41 models on Bitcoin with backtests and forward tests. Grice (2026) tune hyperparameters across three market regimes and find that no tabular deep architecture beats gradient-boosted trees, though a rank-aggregated ensemble does well.

Read together, these papers show a division of labour. Indicator models answer whether a cheap, auditable feature set carries any short-horizon information; learned-pattern models such

as the chart-image CNN ask whether a flexible learner finds structure that hand-built indicators miss; and the model-zoo studies ask which learner to trust when the answer is unclear. For a research-automation stage, the pipeline search and the zoo screen are the most reusable, because they can be rerun on every new market. Their weakness is selection: screening 41 models, or searching a genetic pipeline space, multiplies the number of trials, so any reported best model needs a multiple-testing correction before it enters a signal stack. Chapter 14 covers this in depth, and Chapter 10 collects the rule-based counterparts of these indicator signals.

A recurring pattern is that headline accuracy depends on the horizon, the market and the cost assumption. The tabular-versus-trees result and the Czech cost result are the most transferable lessons, because they say how much complexity the evidence supports.

8.4 Conditioning information: options, liquidity, volume and text

A different strategy adds information that prices alone do not contain. Chinco et al. (2019) apply the LASSO to the whole cross-section of lagged returns for one-minute-ahead forecasts and find predictors that are unexpected, short-lived and sparse, often tied to news. Eksi and Roy (2025) show that an index-inclusion shock raises implied-volatility skew, which then predicts price reversal. Sharma et al. (2026) combine HMM regimes, a temporal convolutional network and option-chain microstructure features for 15-minute index direction, reporting higher accuracy under regime filtering. These connect to Chapter 11.

Liquidity and volume enter next. Panigrahi (2026) relate a conditional liquidity-volatility state to short-horizon NIFTY 50 returns and find the one-day association economically modest. Yang and He (2026) use the ratio of opening to total prior volume as an uncertainty proxy with a threshold regression for the last half-hour of Chinese stock trading. Sheikh et al. (2026) build a composite sentiment index from implied volatility, volume, Bitcoin, crude oil and FEARS across seven economies.

Text-based forecasts form a distinct cluster, covered in Chapter 15. Yokoyama and Chen (2026) fuse BERTweet sentiment with price indicators in a hierarchical cross-ticker transformer. Chen and Kawashima (2025) weight sentiment from connected firms by learned correlation. Li et al. (2024c) learn lag-dependent causal graphs between stocks alongside a denoised news encoder. Wu et al. (2026c) observe that 57% of stocks lack posts on a given day on the ACL18 benchmark and that information takes several sessions to propagate, so they model event lags. Lis et al. (2026) turn Twitter emotion scores into a three-class overreaction signal for intraday AAPL trading.

Bridge. What the text silo calls an "event-lag distribution" is, in market microstructure, the decay profile of price impact after a news shock; what economists call "slow information diffusion" is the same object at daily frequency. Wu et al. (2026c) estimate it from data, while the volume-regime work of Yang and He (2026) implies a similar time-of-day dependence. See Chapter 12.

Desk note. Conditioning gates (signal and regime stages). Input: option-chain skew, open interest, opening-volume ratios or a liquidity-volatility state. Output: a filter that switches an existing intraday or reversal signal on or off. The opening-volume threshold of Yang and He (2026) and the regime gating of Sharma et al. (2026) are examples of the form. Failure modes: thresholds estimated on the full sample are look-ahead; option-derived features need clean timestamps and are costly to store; filtered subsets of trading days shrink the sample and capacity.

8.5 Cross-sections, benchmarks and probabilistic forecasts

Financial economics asks a different question: which assets will outperform, not what a single price does. Gu et al. (2018) compare machine-learning methods for measuring risk premia and trace gains to nonlinear predictor interactions, with momentum, liquidity and volatility dominant. Kelly et al. (2022) prove that models with more parameters than observations can capture more return predictability than simple ones. Practical rankers follow: Nakagawa et al. (2020) train a network with rank information coefficient as stopping criterion and transfer across regions, and Lin et al. (2026b) optimise rank IC directly inside XGBoost. Duan et al. (2025a) learn hypergraph factors that supplement human-designed ones. See Chapter 9 and Chapter 13.

The benchmark studies temper the enthusiasm. Kim (2003) showed early that SVMs are a promising alternative to neural networks for an index. Qian (2018) report that machine learning beats ARIMA and GARCH on index data, while Reikard (2026) find only small improvements over a random walk for monthly exchange rates, which are dominated by serial correlation. Pokou et al. (2024) hybridise ARIMA with learning models using Student-t errors to respect heavy tails, and Maung and Swanson (2025) survey forecasting methods around the M6 competition. Fiedor (2015) take an information-theoretic view, using entropy-rate estimates as a model-free measure of predictability. Nag (2025) apply forecasts to fundamental valuation, a different horizon from the rest.

Generative models give distributions rather than points. Zhou et al. (2018) train an adversarial LSTM-CNN on high-frequency index data and study how often to update. Vuletić et al. (2024) produce conditional return distributions with an economics-driven GAN loss, and Liu et al. (2023b) classify volatility trends with an ordinal-regression GAN. Chapter 3 treats generators more broadly.

Taken together, the forecasting-method papers in this section suggest a simple discipline for a desk. Every candidate model should be compared with a persistence or random-walk forecast and a regularised linear model on the same data, with the same costs, before any architecture is admitted; and probabilistic outputs, whether from a GAN or a Student-t hybrid, should be scored on the quantity the next stage consumes, such as calibrated quantiles for position sizing, not on point error alone.

Desk note. Rank-IC research loop (forecast stage). Input: a daily panel of features for a liquid universe. Output: a cross-sectional score handed to the optimiser. A natural use of the rank-IC objectives is to train and judge with the same metric, then compare against a random-walk or linear baseline as the benchmark studies demand. What can go wrong: IC decays quickly after costs, turnover and capacity limits bind, and gains concentrated in small illiquid names disappear at size.

8.6 Learning who moves with whom: graph models of the cross-section

The largest cluster in this half of the chapter starts from one complaint: a per-stock sequence model treats each name as an island, while a desk trades a cross-section. The remedy is a graph whose nodes are stocks. The papers differ mainly in where the edges come from, and that is the most useful way to read them.

The first family takes edges from outside knowledge. Xu et al. (2022) mine shared information through web-derived stock concepts and add "hidden concepts" beyond the manual labels. Ye et al. (2025) build a corporate knowledge graph and fuse it with alpha-factor libraries and sentiment factors in a Time2Vec transformer. Sawhney et al. (2021) motivate a scale-free relation graph from the financial literature and embed it hyperbolically, framing stock selection as learning to rank. Liu et al. (2026a) use industry, valuation and price-movement graphs, refreshed quarterly,

to correct the residuals of a LightGBM factor model. That last design is a modest and auditable way to add a graph to a tabular baseline.

The second family learns the edges. Uddin et al. (2021) use attention to learn a dynamic firm network and diffuse accounting fundamentals through it. Zhu et al. (2026a) learn a hypergraph, with a causal convolutional encoder meant to prevent future information leakage. Yao et al. (2026) smooth rolling-window correlations to stabilise a dynamic graph and fuse it with a structural view. Wu et al. (2026a) couple multi-view dynamic graphs with a multi-period temporal encoder, and Zhang et al. (2027) argue that the order of intra-stock and inter-stock modelling matters, so they run two paths. Li and Zhang (2026) worry about "crosstalk": trend, fluctuation and shock components entangled in one representation, and heterogeneous relations fused indiscriminately.

A third family asks what an edge should mean. Liu et al. (2026c) want directed, causal influence from symbolic movement patterns rather than symmetric correlation. You et al. (2026) model delayed lead–lag dependence in continuous time with a graph neural stochastic differential equation, and Zhou et al. (2025b) find time-varying lead–lag pairs by sparse cross-attention and use the leader's lag-aligned features to forecast the laggard. Zhang et al. (2026g) put game-theoretic interactions among heterogeneous investors into the graph. Two papers aim for the opposite of heavy side information. Zhuang et al. (2024) extract inter-stock structure from stock factors alone, and Zhang and Lu (2024) add a Laplacian correlation penalty to the loss of a trend network.

Desk note. Graph overlay on a tabular alpha (forecast and portfolio stages). Inputs are the existing daily factor panel plus one slowly refreshed relation set: industry codes, rolling return correlations or supplier links. The output is a residual correction to the baseline score, in the spirit of Liu et al. (2026a), or a rank for a long–short book as in Yao et al. (2026). A natural use is to run the graph model as a challenger against the plain baseline and promote it only if its out-of-sample gain survives costs. What can go wrong: correlation graphs estimated on the same window as the labels leak the target; a learned graph can quietly reproduce sector exposure, so check neutralised returns; and quarterly or concept data must be stamped with the date it became public.

8.7 Sequence architectures: transformers, state-space models and decompositions

A second cluster is about the temporal encoder itself. Transformers lead. Li et al. (2023) guide a stock transformer with market information that re-weights features and allows momentary cross-time stock correlations. Yoo et al. (2021) learn asymmetric, time-varying correlations along the data axis with multi-level market contexts, and Zu et al. (2023) use sparse static and dynamic attention combined with predefined relations. Zhang et al. (2026b) transfer a sentiment-trained transformer to numeric inputs to build factors for Chinese stocks. Surveys help with orientation: Coelho E Silva et al. (2024) catalogue attention networks for trading across asset classes, Bhuiyan et al. (2025) review recurrent, convolutional and hybrid deep models together with their overfitting and interpretability problems, and Pardeshi and Deshmukh (2025) compare deep, reinforcement and hybrid forecasters.

Cost and stability form a quieter thread. Li et al. (2025a) replace self-attention with agent attention, a mixture of experts and LSTM decoding to cut training time, and Mehrabian et al. (2025) and Hu et al. (2026) pair Mamba blocks with graphs for near-linear cost. This matters for a desk because a model that retrains cheaply can be walked forward often. Song et al. (2026a) provide a useful caution: in 1600 runs with a purged chronological split, naively inserted self-attention collapsed to single-class predictions, while residual attention did not. Architecture choice is therefore a validation question as well as an accuracy question.

Several papers decompose the input before modelling. Huang et al. (2026b) separate trend,

cyclical and high-frequency streams for futures. Lu and Zhang (2026) split returns into low-, intermediate- and high-frequency branches fused by a reliability gate, for horizons of one to twenty days. Bohra and Vijay (2026) fuse penultimate-layer features of several LSTM variants across global indices, while Gil et al. (2024) benchmark TCN, N-BEATS, TFT, N-HiTS, TiDE and an xLSTM variant, with wavelet denoising, on S&P 500 and Brazilian ETF data. Noel (2023) and Dixon et al. (2016) are earlier, simpler cases: a dynamic neural network for daily closes, and deep classifiers of five-minute futures direction backtested with a simple rule. Pillay and Ezugwu (2019) train a feedforward network with a metaheuristic against particle-swarm, genetic and ARIMA competitors. These are best read as benchmarks and history, not as templates. Their value for a modern pipeline is as baselines: a research team that cannot beat a small recurrent or feedforward network on the same features and splits should not be adding graph or attention machinery on top.

Desk note. Architecture bake-off harness (research-automation and validation stages). Inputs are one fixed feature panel, purged chronological splits and several random seeds. The output is a table in which every deep model must beat a naive forecast and a tuned linear or boosted model before it is allowed near production. A natural use is to run the multi-seed collapse check of Song et al. (2026a) as an automatic gate, rejecting models that emit one class. What can go wrong: price-level error metrics such as MAPE flatter near-random-walk forecasts; reported gains from one index, one test year or a few stocks may be regime luck; and compute budget, not accuracy, is often the binding constraint on retraining frequency.

8.8 What to predict: labels, horizons and classification

Several papers change the target rather than the model. Friday et al. (2025) predict buy, hold and sell classes at 1-, 3-, 7- and 10-day horizons jointly, so the signal can be matched to a holding period. Jiang et al. (2025) recast index regression as classification of binary-encoded digits, fusing constituent-stock embeddings. Bai and Pukthuanthong (2026) go furthest in portfolio terms: classification into return deciles beats matched regression in long–short construction, and the advantage persists after transaction costs, according to their abstract. Yang et al. (2026b) predict direction, volatility and volume jointly through a liquidity-driven mixture of experts, with a portfolio mechanism on top. Kim and Song (2026) rank stocks with vector-quantised latent factors combined with expert priors.

These ideas share a message: the label is a design variable, and a cross-sectional ranking objective is often closer to what a book needs than a squared price error. The neighbouring papers work with deciles, ranks and movement classes, not point price forecasts.

The proposals below are our synthesis, not claims of the papers. A desk that trades a monthly rebalanced book could try the decile framing of Bai and Pukthuanthong (2026) as a challenger to its regression target, keeping features, universe and cost model fixed so that only the loss changes. A desk that holds positions for several days could use the joint horizons of Friday et al. (2025) to check whether the one-day and ten-day views agree before sizing up; disagreement between horizons is itself a usable filter. The multi-task design of Yang et al. (2026b), which predicts volatility and volume alongside direction, hints at a third use: the auxiliary outputs can feed the sizing and cost stages directly, so the forecaster stops being a single number and becomes a small bundle of inputs to the portfolio and execution layers. The caveat is class imbalance and overlapping labels: multi-day labels overlap in time, so any validation scheme must purge and embargo around them.

8.9 Non-stationarity and regimes inside the forecaster

Return models degrade when market behaviour changes, and a large group builds the response into the model. Lin et al. (2021) add a temporal routing adaptor with optimal transport that sends each sample to one of several pattern-specific predictors. Sarkar (2026) gate attention by a latent market regime, group 95 features into 11 semantic categories and train with a Sharpe-aware objective; the paper also reports correcting alignment and backtesting issues that can inflate trading results. Ceci et al. (2027) limit how much stock-specific signal is written into hidden states during market-wide moves. Wang and Lera (2026) condition a prior-data fitted network on recent feature–return pairs to adapt without regime labels. Liu et al. (2025a) build meta-learning tasks from Gaussian-mixture regime clusters for zero-shot forecasting, and Xu et al. (2026c) select training samples from historical environments whose causal drivers resemble the present one, with a quantile-focused regulariser against non-implementable alphas.

Not every treatment is neural. Anshuman et al. (2022) choose among predictors of the Indian equity risk premium according to the level of policy uncertainty, and Sharma (2025) map Nifty valuation to horizon-specific return distributions over 34 years. Cherifi et al. (2017) forecast inter-stock distances by tensor decomposition as an early signal of herding-driven correlation spikes. H and P (2021) test crypto ensembles trained in turmoil and tested in a bear market. Alaminos et al. (2026) combine ARMA–GARCH with neural and fractal components for the GameStop short squeeze on intraday data.

The sharpest statement of the premise is Farmer et al. (2018): predictability is local in time, with short pockets of significant coefficients between long stretches of none. If true, a regime gate is a way of switching a signal off, not only of improving it.

Desk note. Predictability gate (regime and signal-weighting stages). Inputs are a signal's recent out-of-sample errors and a few state variables such as volatility level or policy uncertainty. The output is a multiplier on that signal's weight, possibly zero. A natural use is to estimate it as a rolling, time-varying-coefficient diagnostic in the spirit of Farmer et al. (2018) and compare it against the learned routing in Lin et al. (2021). What can go wrong: gates tuned on the same history that defines the regimes overfit badly; a gate that switches on after the pocket has started adds lag; and few regime episodes means few effective observations.

8.10 Uncertainty, noise and probabilistic output

A forecast without a confidence is hard to size. Lin and Sun (2026) combine a CNN–Transformer with a Gaussian process prior to obtain calibrated predictive intervals; their abstract states identifiability, consistency and coverage results and an uncertainty-penalised Kelly rule with a CVaR constraint. Yu et al. (2026c) address a bookkeeping problem in probabilistic candle forecasts, where quantiles cross or the forecast low exceeds the open or close, and project forecasts post hoc so that they are consistent without retraining. Together they suggest a forecast should be passed on as a distribution that is internally coherent, not a bare number. For a desk the practical consequence is a handoff contract between the forecast and portfolio stages. A forecaster that emits a mean, a calibrated interval and a flag for coherence gives the optimiser something to penalise, and gives the risk layer a quantity to cap. The same logic applies to stop and target distances, which are distances in a predictive distribution and not arbitrary round numbers. What should worry the reviewer is calibration drift: intervals fitted in a calm year are too narrow in a stressed one, so coverage needs to be monitored live and not assumed from the training sample. The frequency-decoupled forecaster of Lu and Zhang (2026) also includes quantile and likelihood terms in its objective, which places it in the same family.

8.11 Pattern and shape-based approaches at the margin

The time-series-mining silo appears here in a small way. Kim et al. (2025) extract recurrent, amplitude- and warping-invariant patterns and classify direction with interpretable shapelets on Bitcoin and large equities. Karbasian et al. (2025) add a persistent-homology loss to a generative adversarial forecaster so generated five-minute series keep their topological structure. Sen and Chaudhuri (2021) combine technical-analysis concepts with statistical and machine-learning classifiers on five-minute NSE data. Their neighbours in the graph and transformer sections rarely cite them, and the reverse is equally true; see Chapter 5 for the mining view of the same ideas. A natural use is to treat shapelet or motif libraries as interpretable feature generators for a short-horizon classifier, with each retained pattern reviewed by a human before it enters production. The topological loss is more exotic; its proposed role here is as a regulariser that keeps synthetic or forecast paths realistic, which links to the generator work in Chapter 3. The risk with all three is that a pattern found on one instrument over a short sample is easy to over-read, so out-of-sample pattern stability across assets deserves its own test before any of these features is trusted. Finally, the earlier machine-learning comparisons are mostly narrow in universe and period, so they are better treated as hypotheses to replicate than as evidence.

8.12 Does the forecast beat a hard benchmark?

The most important papers for a skeptical desk are the benchmark papers. Guo et al. (2026b) find that even ridgeless and ridge models with huge parameter counts fail to outperform a simple historical average, consistent with asymptotic results under zero true slopes. Bandi et al. (2026) argue that the zero forecast is too easy a benchmark (in their words, excessively conservative), because beating it may reflect only market-level timing, and propose the more stringent panel historical mean; relative to it, only flexible nonlinear models show consistent gains. Puoti et al. (2024) find that crypto series resemble Brownian noise and that naive models beat complex ones. ROY (2022) test randomness on intraday Nifty data with mixed verdicts, and Shah et al. (2026b) compare classical, deep, reinforcement and quantum learners under a leakage-free pipeline. The conceptual frame is offered by Alonso (2026a), whose bounded-learner notion of structure penalises description length, and the historical backdrop by Cutler et al. (1989), a classic study of what moves prices (no abstract is available, so nothing more is claimed).

Desk note. Benchmark-first model approval (validation stage). Inputs are candidate forecasts, realised returns and the panel mean computed with information available at each date. The output is a pass/fail check: a candidate must beat both the historical-mean benchmark of Bandi et al. (2026) and a naive forecast, after costs, in a majority of sub-periods, before any capital is assigned. A cheap pre-screen is a complexity or randomness test along the lines of Puoti et al. (2024) and ROY (2022) to decide whether a market deserves modelling effort. What can go wrong: a benchmark that is too weak passes level effects rather than cross-sectional skill; the screens are low-power; and passing them once says little about the next regime.

8.13 Bridges out of this chapter

Bridge. Graph learners and the risk model (Chapters 13 and 7). What the deep-learning silo calls a "learned dynamic stock graph" (Uddin et al., 2021; Yao et al., 2026) is, for the risk team, a time-varying covariance or dependence structure, and "rolling-window correlation smoothing" is a shrinkage estimator by another name. In the network literature of Chapter 7 the same object is a contagion network, and the tensor-distance forecasts of Cherifi et al. (2017) are an early-warning indicator on it. Conversely, risk people could ask the graph modellers for a cheap out-of-sample test: does the learned graph predict next quarter's covariance better than a shrunk sample estimate?

Bridge. Lead–lag, order flow and execution (Chapter 12). The "lead–lag pair" of Zhou et al. (2025b) and You et al. (2026) is what microstructure writers describe as information diffusing from a liquid asset to an illiquid one, or as price discovery. The microstructure term "latency arbitrage" covers the fastest end of the same spectrum. A shared vocabulary lets the forecaster ask for the delay at which a pair stops being tradable after costs, and lets the execution team see that the same lag estimates could schedule child orders in the laggard.

Bridge. Regime language across silos (Chapters 7 and 5). The forecasting papers speak of "routers", "gates", "meta-tasks" and "domain discovery" (Lin et al., 2021; Liu et al., 2025a; Xu et al., 2026c); the time-series-mining silo speaks of change-points and discords; and Farmer et al. (2018) speak of pockets of predictability. These are three vocabularies for one question: when has the data-generating process changed? A change-point detector from Chapter 7 could supply the labels a router lacks, and a router's assignment entropy could serve as a change-point score. The second wish is a proposal and not a finding of these papers.

Bridge. Targets, costs and the backtest (Chapters 14 and 10). The classification-versus-regression result (Bai and Pukthuanthong, 2026) and the multi-horizon labels of Friday et al. (2025) connect to the labelling schemes (for example triple-barrier) of Chapter 14, where a label is a trade outcome. The skeptics (Guo et al., 2026b; Bandi et al., 2026) supply a null hypothesis that the approval checklists in that chapter should adopt by default.

Chapter 9

Factors and Relative Value

Three communities meet in this chapter and rarely read one another. Financial economists (the `fin-econ` silo, nearly half of the papers here) speak of *anomalies*, *risk premia*, *factor models*, *alpha* measured against a benchmark set of factors, and *limits to arbitrage*. Their unit of evidence is the sorted portfolio, the Fama–MacBeth cross-sectional regression and the t-statistic. Machine-learning and decision-theory researchers speak of *alpha mining*, *formulaic alphas*, *search spaces*, *rewards* and *Markov decision processes*; their unit of evidence is the information coefficient and the backtest of a pool of expressions. A smaller group from derivatives, time-series mining and statistics speaks of *implied-volatility spreads*, *lead–lag clustering*, *cointegration*, *copulas* and *spreads*.

In the shared vocabulary of the atlas these dialects collapse to a few terms. An anomaly, a formulaic alpha and a latent factor are all a *cross-sectional feature*: a number computed per asset per date that is meant to rank assets by future return. A factor model is a *risk and attribution layer* that says which part of a signal's return is already available from known exposures. Pairs trading, statistical arbitrage and relative value are *spread signals*: a feature defined on a group of assets whose relative price is assumed to revert. Limits to arbitrage, crowding and alpha decay are *capacity and regime dependence* of the signal.

In the fund pipeline the chapter sits between features and signal (a feature is turned into a ranked forecast), and it reaches forward into regime and risk (factor exposure, crowding), portfolio construction (long–short legs, beta neutrality) and validation (is an anomaly real, or data mining?). Chapters 8 and 10 cover forecasts and signals built on time-series dynamics; this chapter covers the cross-sectional and relative-value side.

9.1 Mining formulaic alphas

A formulaic alpha is a short algebraic expression over price, volume and cross-sectional operators that returns one score per stock. Kakushadze (2016) publish 101 such expressions as code, report average holding periods of roughly 0.6 to 6.4 days and low average pairwise correlation among them, and find returns tied to volatility rather than turnover. That paper is the natural corpus and baseline for the rest of this section, which asks how to search the space of expressions automatically.

The first generation used evolutionary search. Zhang et al. (2020) build a hierarchical evolutionary algorithm that exploits a pattern in alpha structure to locate promising regions of the space. Cui et al. (2021) try to unite two classes: formulaic alphas generalise well and form weakly correlated sets, while machine-learning alphas on vector and matrix features are more predictive but harder to diversify. A related symbolic route is Zuo and Jiang (2025), who apply symbolic regression to asset pricing, searching function forms rather than fixing a linear one.

The second generation casts construction as sequential decision-making. Yu et al. (2023) generate *synergistic* collections by reinforcement learning, rewarding an expression for what it

adds to the set; Shin et al. (2024) extend the search space and seed it with a pretrained alpha set. Xu et al. (2024) argue that earlier genetic-programming methods are sensitive to their initial population and pursue logical formulaic alphas through deep reinforcement learning, while Zhao et al. (2024b) bound the variance of REINFORCE to obtain steadier factors. The central difficulty these papers name is a sparse reward, since feedback arrives only when a formula is complete. Zhao et al. (2025b) attack it with trajectory-level reward shaping learned from expert factors, Zhu and Zhu (2025) use distributional reinforcement learning against a non-stationary, reward-sparse process, and Yang et al. (2026a) constrain the search with an alpha-oriented context-free grammar so that every candidate is syntactically valid and interpretable.

A third family replaces reinforcement learning with generative models. Chen et al. (2025a) use generative flow networks for structure-aware exploration, and Jin et al. (2026) generate formulas with pool-aware hierarchical discrete diffusion, valuing each formula by the complementary information it adds to a pool. Shi et al. (2024a) stress that fixed-weight combinations perform inconsistently and mine factors together with a dynamic combination. Zhao and Sha (2026) move the problem to order-book and price–volume data with a quality-diversity multi-agent search whose stated aim is robustness after execution costs and diversity of factor families. Yu et al. (2026b) survey the field from an evolutionary-computation view, listing noisy fitness, nonstationarity, costly backtests and semantic redundancy as constraints, and Ding et al. (2026) propose an evaluation framework because backtesting is expensive and correlation measures are narrow.

Desk note. Alpha-mining loop with a quarantine. Pipeline stage: signal generation with research automation. Inputs: adjusted daily or intraday price–volume panels, an operator grammar, and the desk's existing live-alpha pool. Output: candidate expressions, each scored by its marginal contribution to the pool rather than its stand-alone information coefficient, as the synergy papers above suggest. A natural use is to run a mining method such as the grammar-guided or pool-aware searches only on an early window, then pass survivors to a frozen, cost-aware holdout evaluated by a framework like AlphaEval. What goes wrong: every candidate evaluated is a trial, so the multiple-testing burden grows with search effort (Chapter 14); mined formulas often chase high-turnover patterns that die after costs; and pools mined in one regime inherit its dependence.

9.2 Factor models, learned and priced

Classical factor models explain returns by a few exposures. The learned versions replace the fixed exposures. Duan et al. (2022) propose FactorVAE, a probabilistic dynamic factor model that uses a variational autoencoder to predict cross-sectional returns under a low signal-to-noise ratio. Zhao et al. (2026b) argue that such latent-factor models capture overall market conditions but neglect the temporal patterns of individual stocks and the richness of multiple factors, and respond with a dual vector-quantised autoencoder. Gopal (2024) go in the generative direction, enhancing classical factor models with deep generative modelling of equities. In the atlas vocabulary these are representation learners (Chapter 6) whose latent layer plays the role of a covariance and attribution model.

The finance-economics response is a warning about scale. Martin and Nagel (2022) model investors facing thousands of potentially relevant characteristics and show how conventional efficiency tests must be reassessed when prediction is high-dimensional, with Bayesian shrinkage or sparsity governing what investors can learn. For a desk this is a statement that the same dimensionality that makes machine-learned factors attractive also limits what any one of them is worth. At the other end, Holcblat et al. (2022) give simple tests to separate a risk-based explanation of high expected returns from frictions or behavioural bias, a question that decides whether a factor is a premium to hold or an anomaly to harvest and expect to fade. Buraschi et al.

(2010) show the same point for hedge funds: apparent market-neutral alpha is often associated with significant exposure to correlation risk.

9.3 Option-implied information in the cross-section

Options data give a second set of cross-sectional features, and the derivatives silo supplies them. An et al. (2012) find that stocks whose call implied volatilities rose over the prior month earn high future returns, while those with rising put implied volatilities earn low ones, with spreads of about one percent a month. Bali and Murray (2019) turn related quantities into factors, namely implied minus realised volatility, the call–put implied-volatility spread and their changes, to explain optionable-stock returns. Chordia et al. (2021) document a positive link between risk-neutral skewness and later returns and tie it to informed trading around news. Shafaati et al. (2026) move to the options themselves, using LASSO with post-selection inference over 130 characteristics of delta-hedged option returns. Zhou et al. (2025a) split implied skewness from Chinese index options into upper and lower parts to test lottery preferences.

The timeliness caveat comes from Hosseinkhani (2026), who find the implied-volatility spread robust across 1996–2024 while the sign of the option-to-stock volume ratio reversed after 2020. Options features are therefore usable signals, but their parameters belong to the regime layer (Chapter 11).

9.4 Beta, lottery, volatility and liquidity anomalies

The best-documented anomaly cluster here is low beta. Frazzini and Pedersen (2014) explain it with leverage constraints: constrained investors bid up high-beta assets, and a betting-against-beta portfolio is tested across equities, bonds and futures. Replication and critique follow. Agarwalla et al. (2014) find a BAB factor earning significant returns in India that dominates the size, value and momentum factors, whereas Rakhyani (2021) find the beta anomaly present but the premium small on raw returns. Novy-Marx and Velikov (2018) attribute much of the performance to non-transparent construction that effectively equal-weights stock returns; Liu et al. (2018) argue the anomaly stems from beta's correlation with idiosyncratic volatility and is significant only among overpriced stocks; and Buchner and Wagner (2015) suggest that linear CAPM regressions are misspecified for levered firms. Huang et al. (2015) add a timing dimension: arbitrage activity produces booms and busts in the strategy's abnormal profits.

Neighbouring effects share the lottery and risk interpretations. Bali et al. (2008) find that the maximum daily return in the past month negatively predicts returns, and Aziz and Ansari (2018) test the minimum daily return in India, finding an effect concentrated among smaller stocks. Idiosyncratic volatility gets mixed treatment: Aziz et al. (2017) report a positive but fragile relation in India, sensitive to weighting, size group and sample, while Kumari (2026) revisit it in 1,356 firms with a dual measurement framework. Liquidity is the fourth leg: Bekaert et al. (2007) show that a zero-return measure predicts emerging-market returns where turnover does not, and Sethy and Tripathy (2024) price systematic liquidity risk in India with a liquidity-adjusted CAPM.

Desk note. Anomaly screen for a long–short book. Pipeline stage: feature and signal, then validation. Inputs: monthly or weekly universe with betas, trailing maximum and minimum returns, idiosyncratic volatility, a zero-return liquidity proxy and market capitalisation. Output: ranked features and an exposure report against the desk's factor set. A natural use is to build each anomaly with transparent weights, then check whether its return is explained by BAB, size and value after costs. What goes wrong: results depend on weighting scheme and size bucket (the Indian volatility evidence is explicitly sensitive to both), short legs may be hard or costly to borrow, and construction choices can carry hidden leverage.

A practical reading of these four sections together is that a factor is a claim with three layers: a construction (weights, universe, lookback), an explanation (risk, constraint or behaviour) and a decay story. Papers in this chapter usually supply one or two. The mining literature supplies constructions in bulk but little explanation; the anomaly literature supplies explanations but rarely a deployable construction with costs; and the crowding literature supplies decay stories. A desk that wants durable signals should require all three before capital is assigned, and should record which layer each live signal is missing so that its review cadence matches its fragility. A signal with a clean behavioural explanation but no capacity estimate deserves small size; one with a strong backtest and no explanation deserves a short leash and a pre-agreed kill rule.

Emerging-market replication also deserves its own place in that reading. Several papers here test Western anomalies in India or China, where retail participation, settlement rules and short-selling constraints differ. For a desk trading those markets the useful output is not the sign of the effect but the conditions under which it appears: small stocks, particular weighting schemes or a given sample period. Those conditions are the first things to carry into the feature definition and the cost model.

9.5 Calendar effects and lead–lag structure

Calendar effects are the weakest evidence in the chapter. Nath and Dalvi (2008) study the day-of-the-week effect in Indian equities with high-frequency and daily data and find that the significant days changed after rolling settlement was introduced in 2002, a case in which a market-structure rule change moved an anomaly. Brusa et al. (2003) test whether a reverse weekend effect is spread across industries, Elangovan et al. (2022), Chawla et al. (2024) and Tripathi (2026) examine month-of-the-year effects in Indian broad and sectoral indices, with Chawla et al. (2024) adding volatility models. On their own they are descriptive checks and, for a desk, mainly examples of what a validation stage must treat with suspicion: small samples, many tested calendars and structural changes. Lead–lag evidence is less seasonal and more structural. Sahoo and Kumar (2025) use Granger causality, impulse responses and variance decomposition to find that oil and gas leads several other Indian sector indices, which could serve as a sector-rotation input if it survives out of sample.

9.6 Crowding, decay and the limits to arbitrage

A cross-sectional signal does not stay alone. Barroso and Santa-Clara (2015) show that the risk of momentum is predictable and that managing it nearly doubles the Sharpe ratio while virtually eliminating crashes; this is the template for a volatility-scaled factor sleeve. Arnott et al. (2019) list three blunders in factor investing: exaggerated expectations from data mining, crowding and unrealistic trading-cost assumptions; downside shocks far larger than naive risk management would suggest; and diversification that vanishes when factor returns become more correlated in some economic conditions. Deng and Sammon (2026) show that passive inflows reduce realised returns of long–short accounting anomalies through price impact, especially on short legs, and Unterberg (2026) connect passive growth to a decline in active-fund alpha after 2010. Meng and Chen (2026a) formalise the machine-driven case with an alpha half-life that shortens as AI adoption rises.

Desk note. Crowding and decay monitor. Pipeline stage: regime and risk. Inputs: live signal returns, rolling factor exposures, passive-flow or ownership proxies where available, and realised factor volatility. Output: a position-scaling rule (volatility targeting in the style of risk-managed momentum) plus a decay flag when a signal's rolling information coefficient falls below a threshold set at research time. What goes wrong: the monitor lags the damage, flow proxies are noisy, and decay estimates from a short window are indistinguishable from an ordinary drawdown.

9.7 Relative value and pairs trading

Pairs trading replaces the question of which stocks will rise with which relationship will revert. The pipeline has two stages, choosing pairs and trading the spread, and the papers divide accordingly.

For selection, the clustering literature begins with Mantegna (1998), who derives a hierarchical tree and an economic taxonomy from the correlation matrix of daily stock log-price differences. Millington and Sánchez Granero (2022) compare filters (minimum spanning tree, planar and triangulated maximally filtered graphs) on stock returns from five markets. Gupta and Chatterjee (2020) add a lead–lag-aware distance to pair selection, and Deng and Zhang (2026) compare clustering algorithms for lead–lag relationships, aiming to fix the cost and cluster-count sensitivity of DTW-KMedoids. Korniejczuk and Ślepaczuk (2024) combine graph clustering with the Kelly criterion and classifier ensembles in a multi-pair US strategy designed to withstand transaction costs.

For trading the spread, Mudchanatongsuk et al. (2008) model the log relationship as an Ornstein–Uhlenbeck process and solve a closed-form stochastic-control problem. Stander et al. (2013) replace linear correlation with bivariate copulas to identify mispricing. Guo et al. (2025) use the path-signature transform for an interpretable high-frequency futures pair strategy. Because cointegration can fail, Dai et al. (2025b) filter VECM-based trades using violations of asymptotic properties, and JY et al. (2021) combine structural-break detection with deep reinforcement learning. Wilkens (2025) supply the sober check by studying whether pairs trading still works in Germany from 2000 to 2023.

Desk note. Pairs book with break filter. Pipeline stage: signal, with portfolio and risk. Inputs: adjusted daily or intraday prices for a liquid universe and a sector map. Output: a pair list from cluster-then-test selection, spread z-scores as entry and exit signals, and a stop that flags suspected structural breaks. What goes wrong: selection and testing on the same window; costs and borrow on both legs; and pairs that decouple in precisely the stressed periods when the book is most levered.

9.8 Bridges out of this chapter

Bridge. Alpha mining and the backtest-overfitting literature (Chapter 14). The machine-learning phrase "search space exploration" is what statisticians call a multiple-testing problem, and a "reward" is a backtest metric. The mining papers here mostly state the overfitting danger in their own terms, such as noisy fitness; the backtesting chapter's tools could supply the missing deflation.

Bridge. Correlation filtering and regime networks (Chapter 7). What time-series mining calls a "filtered correlation graph" or "subdominant ultrametric", econophysics and systemic-risk work call a "network of contagion". The same maximally filtered graphs that choose pairs could track when clusters merge, a regime signal for the risk layer.

Bridge. Options-implied features and volatility (Chapter 11). A derivatives desk's "call–put implied-volatility spread" is an equity factor-builder's "informed-trading characteristic". The factor-sorting framework gives volatility surfaces a standard test; the volatility chapter gives the surface its dynamics.

Bridge. Crowding and market structure (Chapter 4). What fin-econ calls "inelastic demand from passive flows" is, in microstructure vocabulary, price impact from non-informational order flow. Cost and capacity estimates for the long–short legs belong with the execution chapter, Chapter 12.

Chapter 10

Strategies and Signals

Strategy research is the loudest and least coordinated corner of the trading literature. Finance economists write about "technical trading rules", "time-series momentum" and "anomalies"; evolutionary-computation people write about "rule discovery" and "fitness"; reinforcement-learning (RL) groups write about "agents", "rewards" and "policies"; time-series-mining authors write about "pattern recognition" and "motifs"; practitioners write about "liquidity sweeps", "volume profile" and "support and resistance". In the atlas's shared vocabulary these are all attempts at the same step: turning a stream of prices and volumes into a *signal*, meaning a position or a forecast that a portfolio layer can size. A "technical indicator" is a hand-built feature, a "trading rule" is a thresholded feature, and an RL "policy" is a learned mapping from state features to positions.

This half of the chapter covers the stage between features and portfolio: signal definition and signal search. It sits after the data and feature stages (Chapters 5 and 6) and before regime, risk and portfolio logic (Chapters 7 and 13). Many of the papers below report a backtest of a complete "strategy", so the line between signal, sizing and execution is blurred, and a reader should expect to decompose each strategy into its parts.

The second thing to hold in mind is validation. Almost every paper here searches over indicators, windows, architectures or reward functions, which makes it a multiple-testing problem in disguise (Chapter 14). We read each paper for the idea it contributes, and we treat reported profitability as a claim to be re-tested on our own data with our own costs.

10.1 Trend and technical rules: the economics of simple signals

The oldest question is whether simple price rules carry information. Bessembinder and Chan (1995) test simple technical rules on Asian stock indices and find them more successful in emerging markets such as Malaysia, Thailand and Taiwan than in Hong Kong and Japan. That pattern, that rules work where markets are less developed, is a reminder that signal value is tied to market structure (Chapter 4). Han et al. (2013) apply a moving-average timing rule to portfolios sorted by volatility and report abnormal returns relative to CAPM and the Fama–French three-factor model, with the effect strongest for high-volatility portfolios. For a desk this reads as a conditioning result: the same rule behaves differently depending on the risk of the instrument it is applied to.

Trend following has a more developed literature. Baltas and Kosowski (2013) show that better volatility estimation and trend detection can cut time-series momentum turnover by more than a third without a statistically significant performance penalty. This is the cleanest example in the chapter of a paper whose contribution is cost engineering rather than alpha. Basanisi and Torresetti (2023) test a four-moving-average trend strategy with a trailing stop on 205 European stocks over 2015–2022 and propose a novel variant, and Dong et al. (2025b) combine technical rules with a liquidity-premium strategy in the Chinese market and report that the combination

beats either component alone. The last two are single-market, short-sample studies, so we treat them as candidate designs, not as established premia.

The cautionary side is a breakdown. Kurth et al. (2026) document that short-term trends in about 100 liquid futures have stopped delivering reliable returns since roughly 2009, and offer a microstructural account that depends on signal speed and asset class. T and T (2022) examine market conditions and large downside risks in trend following, motivated by diminished equity profitability, and suggest remedies for trend reversals. Read together, these two papers say that a trend signal is a time-varying asset whose decay should be monitored (Chapter 7).

Desk note. Trend-signal monitor (features → signal → validation). A natural use is to run a small bank of trend signals at several speeds on a futures universe, each with a volatility-scaled position and a turnover estimate, and to track net-of-cost performance by speed on a rolling basis, in the spirit of the speed and asset-class dependence described by Kurth et al. (2026) and the turnover reduction of Baltas and Kosowski (2013). The output is a per-speed on/off flag or a risk-limit multiplier for the portfolio layer. Failure modes: the monitoring window is short relative to the decay it tries to detect, so switching rules chase noise; cost estimates taken from daily data understate the cost of fast signals; and trend and carry exposures overlap with the factor book (Chapter 9).

A different tradition treats trading as a control problem. Abbracciavento et al. (2022) use a feedback approach, in which price is an exogenous disturbance to a loop and a controller is synthesised, avoiding an explicit model of non-stationary dynamics. A natural use is as a benchmark for adaptive sizing against learned policies. Two papers on intraday and indicator design fit here. Tian et al. (2012) apply the ACD rules and pivot-point system, a practitioner method, to Chinese intraday data, and Li et al. (2019) learn stock-specific technical-indicator settings using stock embeddings, a bridge between per-instrument parameter tuning and shared representation learning.

10.2 Searching the rule space: evolutionary and swarm methods

If rules are hypotheses, evolutionary computation is a way to generate and test them at scale. Hu et al. (2015) give the systematic review of evolutionary computation for rule discovery in stock trading, screening 650 articles to 51; it is the right map of this area. Martín et al. (2019) compare genetic programming and grammatical evolution configurations, both attractive because their evolved rules remain interpretable and flexible in structure. Corazza et al. (2024) use particle swarm optimisation to set indicator parameters, rule definitions and the weights used to aggregate signals in one pass. Atiah and Helbig (2019) study how five decision models affect dynamic multi-objective optimisation algorithms for foreign-exchange trading, which raises the question of how to pick one rule from a Pareto front.

A distinctive link comes from Kacprzyk et al. (2008), who ask whether genetic-programming trading rules are sensitive to the entropy of the series, and first whether price series show temporal dependencies measurable by entropy estimates. Entropy as a predictability gauge could serve as a pre-screen for where a rule search is worth running at all.

Bridge. Rule search ↔ alpha mining ↔ multiple testing. What the evolutionary-computation silo calls "fitness" and a "population of rules" is what the backtesting silo calls a specification search, and the overfitting risk is identical (Chapter 14). Conversely, the factor and LLM chapters (Chapters 9 and 15) describe "alpha mining" with formula generators; the evolutionary papers here are the ancestors of that work, and their interpretable rule trees (Martín et al. 2019) are a design pattern modern formula miners could reuse.

A practical reading rule for the next sections: ask of every paper which stage it really improves. Some improve the feature (denoising, state design, stock-specific indicator settings), some the

decision rule (thresholds, reward, policy class), and some only the evaluation. Papers that bundle all three in one backtest are the hardest to reuse, because a desk cannot tell which component carries the result.

10.3 Patterns, charts and levels

A large body of work tries to make chart reading computable. Neela (2025a) take the deterministic route: extract structural monotonic price runs and attach public events in a window, producing auditable explanations rather than forecasts. Y et al. (2021) apply machine learning to candlestick patterns with several feature types and windows of one to ten days, and Agah et al. (2025) combine deep-learning trend prediction with candlestick recognition. Arévalo et al. (2017) propose a flag-pattern rule with dynamic stop-loss and take-profit windows and an EMA filter, and Pal (2024) study LSTM recognition of the Wyckoff accumulation pattern in currency trading. In the same family, Lee et al. (2022) combine an attention-based BiLSTM with technical indicators.

Image-based methods treat the chart as a picture. Birogul et al. (2020) apply a YOLO object detector to 2D candlestick charts to produce buy and sell decisions, and Jang et al. (2026) turn 1-minute KOSPI200 futures data into candlestick and Bollinger-band images for model-free visual pattern mining with similarity metrics. What this literature calls "pattern recognition" is the time-series-mining silo's "motif discovery" and "similarity search" (Chapter 5); the desk question is always whether the pattern label adds information beyond the raw window, and whether it survives a cost model. Kriuk et al. (2025) apply attention-based correlation analysis to identify support and resistance levels dynamically, attacking a structural-level task rather than price prediction. Jóźwicki and Trippner (2025) examine the volume profile as an alternative to turnover-based indicators for investment decisions.

Desk note. Pattern labels as audited features (features → signal). A pattern detector such as the run extraction of Neela (2025a) or a candlestick classifier as in Y et al. (2021) can be run offline to emit a categorical feature (pattern type, age, run length) plus the raw window. The output is a feature column consumed by a downstream model, never a direct order. Check three things: whether the pattern is defined using only data available at bar close; whether the detector's label frequency drifts across years; and whether a null model on shuffled windows yields comparable "hit rates". Chart-image methods multiply the number of tunable choices, so the capacity for leakage grows with them.

10.4 Liquidity-centred discretionary ideas

Practitioner concepts are entering the academic record. Gokarna Bhusal (2025) discuss liquidity-based "stop hunt" traps in Forex and US indices using ICT and Elliott-wave concepts. Hammer and Patel (2025) document a live intraday ES futures strategy built on Fair Value Gaps, liquidity sweeps and breaker blocks. These are documentation of discretionary practice, and their evidence should be graded accordingly. Mittal and Choudhary (2026) offer a more formal framing, recasting breakouts as a question of liquidity structure and arguing that quiet-looking moves travel further than visually impressive breakouts.

Bridge. "Liquidity sweep" ↔ order-book language. The ICT vocabulary of sweeps, gaps and traps describes behaviour near resting orders. Microstructure would phrase this as clustered limit and stop liquidity, depth thinning, and price impact (Chapter 12). A natural proposal is to test the practitioner labels against order-book depth and trade data, so that a "sweep" becomes a measurable event with a base rate, and the microstructure silo gains a catalogue of hypotheses it has not yet formalised.

At the other end of rigour, Amaral (2026) model the mid-price at second scale as a stationary mean-reverting error around a latent efficient price and solve for the trading rule that maximises

profit net of the bid-ask spread. Cheevirot et al. (2026) study Forex grid trading with martingale lot scaling and propose strategy-embedded labelling for machine learning, with attention to the drawdown that scaling creates. Grid and martingale schemes earn small gains with exposure to rare large losses, so their risk limits matter more than their signal.

10.5 Reinforcement learning as strategy design

The largest group, with roughly twenty papers, casts trading as sequential decision making. Sun et al. (2023b) give a survey of RL for quantitative trading, which is the best entry point. The applications are mostly Q-network variants on single instruments. S and J (2024) use multi-level deep Q-networks for Bitcoin, Huang et al. (2023) propose a dual-action, dual-environment DQN for stocks, Saepudin and Rauf (2025) apply DRL to the Indonesia Stock Exchange, and Yu et al. (2026a) augment DQN with an LSTM to handle partial observability. Indian-index studies include Mishra et al. (2025b), who compare DQN, double DQN and dueling double DQN on NIFTY 50, Borkar and Jadhav (2026), who compare Q-learning, DQN and DDQN intraday, and Verma et al. (2027), who use volatility guidance for intraday index trading. Vishwakarma et al. (2026) focus on state representation for day trading with technical indicators and chart patterns, and Park et al. (2024) condition DRL "robots" on market conditions and US interest rates.

Several papers add information to the state or change the learning target. Casares (2025) apply wavelet denoising to DIX, GEX and VIX indicators before DRL on S&P 500 futures. Novotny and Hajek (2026) combine support-vector-regression forecasts with PPO and sentiment and volatility indices for precious-metal ETFs. Pillai et al. (2026) assemble a hybrid system of EMA and MACD trend capture, machine learning and sentiment for regime-adaptive equity strategies. Shen and Huang (2025) bring meta-RL and cognitive game theory to non-stationarity, and Tan et al. (2024) learn options-trading positions end to end, directly from market data, without an option pricing model, with transaction costs in the picture.

The sharpest recurring problem is the reward. Orra et al. (2025) shape the reward with expert feedback, Goluža et al. (2024) use an imitation reward for noisy environments, Orra et al. (2026a) learn from several heterogeneous expert strategies by adversarial inverse RL, and Orra et al. (2026b) discover auxiliary tasks automatically to stabilise training. These papers share a diagnosis: with a low signal-to-noise ratio, the raw profit reward is a poor teacher. Transfer is the other fix for scarce data. Poh et al. (2023) fuse a source-market encoder with a target-market module to rank crypto assets for cross-sectional momentum, and Koshiyama et al. (2022) transfer learning across trading strategies to build global rather than per-market models.

Desk note. RL as a research harness, not yet a production signal (signal → validation). A cautious use is to run an RL agent on top of a frozen feature set, with the reward set to net-of-cost, volatility-scaled return, and to compare it against the fixed rules of the previous sections on the same walk-forward splits. The output is a comparison table and, if the agent wins, a candidate position signal that enters the portfolio layer with a hard risk cap. Hazards: many reported agents are tuned on one instrument and period, seed variance is rarely reported in abstracts, simulators ignore impact and capacity, and a policy trained in one regime can fail in the next (Chapter 7).

10.6 Do classical technical rules survive scrutiny?

The first half of this chapter met rule-based systems as engineering objects. A second family of papers asks the older, blunter question: do such rules earn anything once data snooping and costs are respected? The classical bootstrap evidence for simple moving-average and trading-range-break rules on Dow Jones data from 1897 to 1986 (Brock et al., 1992) set the baseline, and the later literature is largely a sequence of attempts to repeat that exercise with better

multiple-testing control. Rink (2023) test 6406 rules across 23 developed and 18 emerging equity indices with a superior-predictive-ability test. They report in-sample outperformance, much of it in emerging markets, but also decaying predictability, sensitivity to moderate costs and no persistence out of sample. Chuang et al. (2024) reach a similar verdict on two Chinese indices with 38,456 rules and stepwise error-rate control: after the correction only a few complex rules survive, and the lucrative ones are high-turnover rules whose out-of-sample returns turn negative after costs. Yang et al. (2020b) add a bound on cumulative return and a bootstrap across international markets, and conclude that rules are hard to beat against a random trading benchmark.

Not every author is so negative, and the dispersion is informative. Comparative indicator studies on Indian equities (Pedrycz et al., 2024), on the volatility–return trade-off in NSE 500 portfolios (Bhama, 2024) and on multi-indicator combinations (Sukma and Namahoot, 2025) report profitability for Bollinger bands, MACD, RSI and their blends. Such studies typically test a handful of hand-picked parameterisations, which is exactly the setting where the multiple-testing studies above warn that apparent edge is cheap to find. We treat these three papers as a catalogue of rule definitions and a source of market-specific hypotheses, not as evidence of tradable alpha.

Why then does technical analysis persist? Two papers offer behavioural answers. Ebert and Hilpert (2019) model how the market timing of technical rules induces lottery-like, positively skewed profits, which a skewness-loving investor would find attractive even without a premium. Hoffmann and Shefrin (2014), using Dutch brokerage records matched to surveys, find that individual investors who report using technical analysis hold more concentrated portfolios, turn them over faster, trade more options and earn lower returns. For a desk, these results define the counterparty side of the trade and warn that the popularity of a pattern is not evidence for its profitability.

Desk note. Rule-zoo audit (validation stage). Input: a library of candidate rules (moving-average crossovers, channel breakouts, RSI thresholds) with their parameter grids, plus daily or intraday bars and a realistic cost model. Output: a pass/fail report per rule family, using a data-snooping-aware test over the *whole* grid and not over the winner, followed by a walk-forward check with costs. A natural use of Rink (2023) and Chuang et al. (2024) is to make the grid size an explicit input to the test, so that adding more rules raises the bar. What goes wrong: the audit is run after the grid was shaped by looking at the data (leakage by researcher), turnover is underestimated, and a rule that worked in the early sample is promoted although the evidence points to decay.

10.7 Analytical models of technical rules

A smaller group of papers replaces simulation by mathematics, which is where the maths and control communities meet this chapter. Lorig et al. (2019) study trading with exponential moving averages and obtain closed-form solutions to log-utility and growth-rate problems when the drift is an Ornstein–Uhlenbeck process or a two-state Markov chain. Blanchet-Scalliet et al. (2007) compare a moving-average style rule against model-based strategies that must detect a change in drift at an unknown time, and find the technical rule beats an optimal allocation under mis-specified parameters but not the Model and Detect strategies. The lesson for a desk is conditional: a crude rule is a robust fallback when the model is wrong, not a substitute for a model that is right.

Support and resistance receive the same treatment. De Angelis and Peskir (2016) pose an optimal-prediction problem for a stopping time under geometric Brownian motion with a hidden aspiration level, and Henderson et al. (2026) analyse a three-state path-dependent price model through optimal stopping, proving smoothness of the value function. Neither says that levels exist in real data; both show under what dynamics such a rule would be optimal. The empirical

counterpart is Chung and Bellotti (2021), whose heuristic discovery algorithm finds intraday levels that reverse price statistically significantly, with more prior bounces predicting further bounces and a decay of bounce probability over time.

Control theory contributes a different language. Barmish and Primbs (2016) propose model-free feedback controllers whose investment level is a function of the gain–loss process, aiming at robustness certificates in place of backtests. Azhmyakov et al. (2023) apply a PID controller to trading and tune it in a data-driven way, and Hsieh (2025) formulate robust gain selection for a double linear policy so that expected profit is positive across a range of uncertain market conditions. These are the nearest relatives of the stop-and-sizing logic in a risk layer, and we return to them in the bridges below.

10.8 Levels, patterns and market structure as features

Many practitioner papers turn chart folklore into machine-readable events. Elliott-wave labels feed a classifier for option entries in Bhattacharya et al. (2026); short-term fast–slow patterns in Nasdaq-100 futures are identified in Montes-Gómez et al. (2024); León-Camacho et al. (2026) correct Bollinger bands for skewness and kurtosis with an Edgeworth expansion and use them as contrarian thresholds; and Lin et al. (2026a) augment MACD with volume, volatility and intraday structure plus a sensitivity calibration. Salman et al. (2025) use a genetic algorithm to combine several strategies in the directional-changes paradigm, an event-driven alternative to fixed-interval bars, and Sharma et al. (2024) track open interest, spot price and delivery data in Indian derivatives to infer the movement of large operators.

The most recent cluster concerns Smart Money Concepts and Inner Circle Trader vocabulary. Fumanal-Andres et al. (2026) automate detection of change-of-character, range boundaries and liquidity voids, and Fumanal-Andres (2026) use generative AI to build an MQL4 expert advisor that detects fair value gaps. Oukhouya et al. (2026) detect such events on OHLC data, learn calibrated probabilities that they resolve favourably and size trades with linear programming under a risk constraint, while Boutgajouft (2026) compare a formal liquidity theory against these heuristic concepts. Levine (2026) report a paper-traded comparison of opening-range breakout and ICT-style setups in E-mini futures; the sample is a single trader over a short period, so we read it as a hypothesis generator.

Andriushchenko et al. (2026) wrap polynomial moving regression bands in an economic-security policy for proprietary cryptocurrency traders, a reminder that a band rule also needs firm-level limits around it; we read the paper for that governance framing and for the idea of curve-fitted bands as an adaptive channel, and not for any performance claim. Such bands are, in effect, a local polynomial smoother with a residual envelope, so the usual caveats about end-point revision apply: the fit at the latest bar is the least reliable one and is exactly the one a live rule uses.

Two papers are good models of discipline. Brown (2026) pre-register four predictions of Gann angle theory, test them on 2016–2022 in sample and 2023–2025 held out, and apply false-discovery-rate and block-bootstrap checks; they report significant first-revisit and near-touch effects but negative net returns for fan crossings. Mesfin (2026a) build a day classifier from overnight gap, first-30-minute return and relative first-bar volume with expanding-window thresholds; it finds a recurring intraday profile, yet none of the derived strategies pass walk-forward, after-cost validation. A negative result reported this carefully is worth more to a desk than a positive one reported loosely.

Desk note. Structure events as features, not as trades (feature and filter stage). Input: intraday OHLCV and open-interest bars. Output: a table of event flags (level touch, gap fill, range break, day-type label) with causal timestamps, consumed by a downstream classifier or used as a regime filter. The proposal, building on Mesfin (2026a), Oukhouya et al. (2026) and Chung and Bellotti (2021), is to treat each folk concept as a candidate feature with a pre-registered hypothesis and an expanding-window threshold. Failure modes: swing points and levels that are only confirmed after the fact (lookahead), thresholds fitted on the full sample, event definitions drifting between the research and live code paths, and thin capacity on micro contracts once spread and slippage are charged.

10.9 Fusing rules, filtering signals and meta-labels

If single rules are weak, an obvious response is to combine or filter them. Evidential-reasoning methods are the older tradition: Dymova et al. (2016) build a Forex expert system on rule-base evidential reasoning, and Lepskiy and Suevalov (2019) compute belief and plausibility for buy, sell and hold recommendations via fuzzy inference on Russian equities. Their contribution is a formal way to represent disagreement between indicators instead of averaging it away.

The modern descendants are secondary models that decide whether to act on a primary signal. Hu and Zhou (2024) treat the signal as the start of an observation window and a profit target as the event of interest, so that deep survival models filter false signals. Zhang et al. (2026a) evaluate meta-labeling of moving-average crossovers with XGBoost on 15-minute Chinese futures under purged walk-forward and triple-barrier labels, and report ROC-AUC near 0.5, mixed contract-dependent outcomes and no significant paired confidence interval. Mikhailov (2026) similarly apply a meta-labeling approach to daily Bitcoin. Read together, these papers say that a filter helps only if the secondary features carry information the primary rule did not. Vancsura et al. (2026) make the related point for forecasts: RNN, LSTM and GRU predictions feed a rule-based trader with volatility-adjusted stops, a rolling-error confidence filter and error-based sizing, tested across twelve instruments and three regimes. Forecast accuracy is an input to, not a proxy for, strategy performance.

10.10 Reinforcement-learning traders

A different silo reaches the same decision problem from the other end: skip the explicit rule and learn the policy. Kabbani and Duman (2022) argue that deep reinforcement learning can merge the prediction and allocation steps, and Z et al. (2021) extend the action space to variable position sizes with an LSTM-DDPG agent. Tsui et al. (2026) train an agent to sell or buy at perceived peaks and valleys, and Kar et al. (2025) bypass reward design altogether by training with profit-guided loss functions. Go et al. (2026) borrow model predictive control to add inference-time planning with price forecasts to a static policy.

At shorter horizons, Cartea et al. (2021) derive statistical-arbitrage policies in an FX triplet with double deep Q-networks and a new reinforced deep Markov model. For cryptocurrency, Qin et al. (2024) use hierarchical RL for second-level trading, Zong et al. (2024) add memory and context awareness to the same setting, and Sun et al. (2022b) build risk-awareness into an intraday framework. The shared premise is that the agent should respect risk rather than only maximise return. Control-engineering papers above and RL here are cousins: both choose an exposure as a function of a state, one with guarantees and few parameters, the other with flexibility and weak guarantees.

Desk note. RL agent as a candidate, never a deployer (signal and validation stages). Input: bars or limit-order-book features, a cost and latency model, and a held-out regime set. Output: a policy that is promoted only if it beats a transparent baseline such as a volatility-scaled trend rule after costs. Our proposal is to run the agents of Sun et al. (2022b) or Zong et al. (2024) as research challengers inside a shared simulator, and to demand regime-by-regime results as in Vancsura et al. (2026). Risks: simulator fills that are more generous than real queues, non-stationary reward, seed sensitivity, and policies that quietly rely on a lookahead feature.

10.11 Humans, platforms and automation

The last group concerns who builds and runs strategies. Chivu et al. (2024) compare human and algorithmic trading with a Pine Script bot using learning vector quantisation; Guarino et al. (2022) evaluate autonomous, adaptive automated traders during cryptocurrency bubbles, asking whether learning helps at all. Bohrinovtseva et al. (2026) propose a human–AI co-creative framework in which the weight of AI signals and human input varies with volatility, task complexity and trust, which is a good description of the semi-automated desk this atlas targets. AlDhaen et al. (2026) describe a cost-effective smart algorithmic trading system for small professionals, and Sun et al. (2023a) offer TradeMaster, a platform for reproducible RL-based trading research. Together with the generative-AI-built expert advisor of Fumanal-Andres (2026), they point at research automation: the same pipeline that implements a strategy can implement its test harness.

10.12 Bridges out of this chapter

Bridge. Event-based time and "discords". The directional-changes paradigm in Salman et al. (2025) samples price when it has moved by a threshold, not when a clock ticks. Time-series mining would call that a form of adaptive segmentation, and a recurring Elliott-wave or flag shape a motif; the day-type outliers of Mesfin (2026a) resemble discords. Microstructure readers will recognise "liquidity voids" and "stop-hunt zones" as regions of thin resting depth. A shared vocabulary lets the pattern-similarity tools of Chapter 5 be tested on these labelled events, and lets order-book data from Chapter 12 supply the depth evidence that chart traders only infer from OHLC.

Bridge. Data snooping is a backtesting problem. The multiple-testing designs of Rink (2023) and Chuang et al. (2024), and the pre-registration of Brown (2026), are the strategy-side face of what Chapter 14 calls selection bias and multiple-hypothesis control. Translated: "the number of rules tried" is the same quantity as the number of trials in a deflated-Sharpe style correction, and a "rule zoo" is a hypothesis family. The same chapter's purged walk-forward machinery is what Zhang et al. (2026a) apply to meta-labels.

Bridge. Feedback controllers are position-sizing rules. What control engineers call gain selection and robustness certificates (Barmish and Primbs, 2016; Hsieh, 2025) is, in portfolio language, a sizing function of running profit and loss with a worst-case guarantee; Chapter 13 treats the stochastic-control and Kelly-style counterparts. Conversely, optimal stopping results on levels (De Angelis and Peskir, 2016; Henderson et al., 2026) give the stopping-time view of stop-loss and take-profit, which the regime machinery of Chapter 7 could condition on drift states, the same two-state structure used by Lorig et al. (2019).

Chapter 11

Volatility and Derivatives

Four communities share this chapter and rarely read each other. The first is the derivatives-pricing school (quant-deriv), whose native words are *smile*, *skew*, *risk-neutral density*, *Greeks*, *risk premium* and *rough volatility*. The second is financial econometrics (fin-econ and stats), which speaks of GARCH, realized variance, bipower variation, jumps, spillovers and superior predictive ability. The third is a small market-microstructure group that treats volatility as a by-product of discrete, noisy, event-driven trading. The fourth is the machine-learning group, who see volatility as a regression target and ask about architectures, underspecification and deployment.

The atlas vocabulary lets us translate between them. What derivatives people call an implied-volatility surface or a risk-neutral density is, in the shared terms, a market-implied *forecast* of a distribution, and it is also a set of *features* for a signal. What econometricians call a volatility forecast evaluated by a loss function is a *forecast* judged by *forecast evaluation*, and the superior-predictive-ability test is a *multiple-testing* control. A *variance risk premium* is the gap between a market-implied and a realized quantity, so it is a *factor* with a sign and a price. A *spillover* is an edge in a *network*. An expiry-day effect is a calendar *anomaly* in *liquidity* and *realized volatility*.

In the fund pipeline this chapter sits in the middle. Volatility and option data enter as raw inputs (data), become features (implied volatility, skew, realized measures, jump indicators), feed forecasts of risk and of returns, and then drive regime flags, position sizing, hedging and risk limits. Its papers mostly serve the forecast and risk stages, but several matter for validation, and a minority are direct signals. About a third of the sources (20 of 64 by a keyword scan of titles and abstracts) study India (Nifty, Bank Nifty, single-stock options), which makes the chapter a practical resource for a desk trading NSE derivatives.

11.1 Implied volatility as information and as fear gauge

The starting point is that option prices carry information that prices of the underlying do not. Christoffersen et al. (2013) survey how implied volatilities, skewness, kurtosis and densities can be extracted for forecasting, and note that any forecast target that is a smooth function of the future price can use option-implied information in a principled way. They also stress adjusting for risk premia first. The rest of this section asks who trades on that information and what it says in an emerging market.

On the Nifty, Pathak (2015) study whether option volume and open-interest changes respond to expected volatility, with a common implied-volatility measure and a vector autoregression, and report a feedback relationship between the two. Chakrabarti and Kumar (2020) use model-free implied volatility on five-minute data to test leverage, volatility-feedback and behavioural explanations of the short-run return–volatility link. Shaikh (2018) find an asymmetric, negative relation between the India VIX and returns, and read the index as a gauge of fear and portfolio-insurance cost rather than optimism. Blau and Brough (2015) look from the equity side and find

that put-call ratios are not a proxy for short-sale constraints, and that their return predictability is driven by ratios following positive-return periods. Gupta (2018) test put-call parity on Nifty options against moneyness, maturity and volume, treating violations as a measure of illiquidity as much as of arbitrage.

Structural studies of the Indian surface follow. Shaikh and Padhi (2014) document smile, skew and term-structure patterns of Nifty implied volatility. Jain et al. (2019) find that a smile-adjusted Black model fits well, that implied volatility adds predictive power for future volatility, and that the premium in single-stock options is higher than in other markets. Agarwalla et al. (2021) use implied measures from single-stock options to argue that the COVID-19 repricing was gradual and rational. Jayapalan (2013) propose a simple Nifty call pricing model built on the correlation with the market index, and Singhal and Bhattacharyya (2024) backtest short-theta strategies on Indian index options, using delta moves as a parameter and noting that gamma and delta can overpower theta.

A second thread asks how far implied volatility can be predicted from the past. Wen et al. (2024) show a clear pattern of past dependence up to twenty days, in both linear and nonlinear models. Guyon and Lekeufack (2023) find that a simple, power-law-weighted function of past returns explains most of the variance of equity-index implied volatility and a large share of realized volatility. Together they say that a good part of the implied-volatility level is endogenous to price history, so a feature built from option data may repackage what returns already contain.

Taken together, these studies show a recurring pattern. Each starts from a clean theoretical object, such as parity, a smile or a risk-neutral density, and then measures how far an emerging, retail-heavy market departs from it. The departures are where a desk might look for edge, but also where costs, thin strikes and settlement rules are most likely to erase it. A reader from a machine-learning background should note that none of the Indian studies above is a trading backtest in the modern sense. They are mostly descriptive tests of efficiency, so any signal derived from them needs its own walk-forward evaluation.

Desk note. Implied-volatility feature block (features, then signal and risk filter). Inputs are daily or intraday Nifty and single-stock option chains, the India VIX, and put-call ratios. A natural pipeline computes model-free implied volatility, a smile-adjusted level and slope, and the put-call ratio, and regresses each on a path-dependent baseline of past returns in the spirit of Guyon and Lekeufack (2023) and Wen et al. (2024), keeping only the residual as a candidate signal. The output is a volatility-regime filter and a position-size scaler. Failure modes: stale or illiquid strikes that make parity and smile fits noisy (Gupta, 2018), confusing a risk-premium effect with a forecast effect (Christoffersen et al., 2013), and trading costs on thin single-stock options, which the cited studies do not net out.

11.2 Risk premia, surfaces and pricing models

If implied volatility is information, it is also a price, and prices include compensation. Duarte and Jones (2008) estimate a volatility risk premium from a cross-section of equity-option returns and find it rises with overall market volatility. Andersen et al. (2015) show from index option surfaces that priced left-tail risk is not spanned by volatility and introduce a tail factor that predicts equity and variance risk premia rather than future volatility. Christoffersen et al. (2009) explain the shape and term structure of the index smirk with multifactor stochastic-volatility models. For a desk, these are the factor-model view of option returns: selling volatility earns a premium that is time-varying and tied to crises.

Whether such premia exist locally is a testable question. Pillai (2026) apply a finite-sample simulation test, originally for US index puts, to Nifty puts from 2015 to 2025 under Black–Scholes, Heston and a jump model, and the abstract itself warns that the jump-model estimation did not fully converge. That is a model for how a validation check on an option-selling idea should look: a null model, many simulated paths, and a confidence interval, not a single backtest.

The modelling literature supplies the engine room. Ahlip et al. (2018) give a semi-analytical call price under a double-Heston jump-diffusion with stochastic rates. Lu (2026) compare option models from Black–Scholes to fintech-enhanced approaches, and Black (1975) is an early, plain-language account of what options do and do not offer. Chen and Kay (2026) review the history of probability measures in asset pricing, from state prices to stochastic discount factors, and Abedi (2026) derive jump-diffusion option pricing from information constraints. These are reference works: they define the measures under which implied quantities are computed, and the distinction between risk-neutral and real-world measures underlies every premium above.

Newer papers bring machine learning to the surface. Shikhman et al. (2026) show that accurate option prices do not imply an accurately recovered risk-neutral density, report that many pricing directions are numerically null, and test on a chronological Nifty benchmark. Han et al. (2026) generate dynamic implied-volatility surfaces with diffusion models and judge them through data-driven hedging. Green (2026) describes deep hedging, in which a network chooses hedging positions directly, and Bloch (2026b) propose a path-signature swap for path-dependent risk. Daniluk et al. (2024) reduce Monte Carlo error in pricing and Greeks in local-stochastic-volatility models. Abi Jaber and Li (2025) report that rough-volatility models are inconsistent with the global shape of SPX smiles and underperform simple Markovian alternatives of the same size. Finally, Sahu (2026) show that aggregate dealer gamma exposure, quoted by vendors, is only bounded by exchange open-interest disclosure, and that its sign is undetermined for the participants examined.

Desk note. Option-selling risk control (risk stage and validation). Suppose a desk runs short-volatility structures on Nifty weeklies. Inputs are the surface, open interest, and a premium estimate in the manner of Duarte and Jones (2008) and Andersen et al. (2015). The output is a risk limit on short-gamma exposure plus a validation check that compares the realized P&L distribution with a simulated null as in Pillai (2026). Gamma-flip levels from vendors should not be used as a trigger without the caveat of Sahu (2026). What can go wrong: premia that are positive on average but crash-concentrated, margin and cost drag, capacity limits in far strikes, and a generated surface (Han et al., 2026) that has never seen a true stress regime.

11.3 Forecasting realized volatility

On the econometric side the benchmark is Bollerslev (1986), the GARCH model, and the question of whether anything beats it. Reinhard Hansen and Lunde (2003) compare 330 GARCH-type models with a superior-predictive-ability test and ask exactly that; on IBM realized volatility, GARCH(1,1) is beaten, mainly by models with a leverage effect. Measurement improved with high-frequency data. Aït-Sahalia et al. (2005) show that with microstructure noise there is a finite optimal sampling frequency, and Hoffmann et al. (2010) estimate the volatility function nonparametrically under such noise with wavelets. Vipul and Jacob (2007) compare extreme-value volatility estimators for the Nifty against a two-scale realized-volatility benchmark. Andersen et al. (2007) add jump components using bipower variation, and Qu et al. (2024) revisit the puzzle of null jump effects by using jump intensity inside a heterogeneous autoregression.

Latent-state methods give another view. Johannes et al. (2009) derive optimal filtering for jump-diffusions to extract latent volatility and jump states, Virbickaitė and Lopes (2019) add Markov-switching regimes and a flexible error distribution to stochastic volatility, and Woitschig and West (2026) combine a dynamic gamma process for realized volatility with Bayesian dynamic linear models for prices. Dai et al. (2025a) build a GARCH-type model for fuzzy-set-valued returns. Johansson et al. (2023) propose a simple method for predicting covariance matrices, which is the multivariate counterpart.

The recent cross-currents link ML and rough volatility. Rosenbaum and Zhang (2024) train a pooled LSTM on hundreds of stocks and find that rough-volatility parametric forecasters match

it, which they read as universality in the volatility formation process. Mouti (2026) measure roughness across equities, futures, rates, FX and commodities and report realized volatility rough everywhere, with implied estimates informative only where the skew term structure is clean. Wang et al. (2026f) give a Hawkes-process microstructural foundation for a rough Hawkes–Heston model, which is the nearest this chapter comes to order-flow modelling. Cortesi et al. (2026) warn that networks with identical test loss on volatility learn different functions and produce portfolios with sharply different turnover. Ye and Wanjiku (2026) study post-training quantization of forecasters, whose calibration data can matter for deployment. S et al. (2024) apply seven machine-learning methods to VIX constant-maturity futures from term-structure information, with a walk-forward design and a constrained mean-variance portfolio.

Desk note. Volatility forecast as a shared service (forecast stage, feeding sizing and risk). A fund can maintain one volatility forecast per instrument: a HAR or GARCH baseline from noise-robust realized measures (Aït-Sahalia et al., 2005; Andersen et al., 2007), challenged by a pooled neural model (Rosenbaum and Zhang, 2024). The output is a variance forecast for position sizing, stop distances and covariance inputs. Gate every challenger with a superior-predictive-ability test (Reinhard Hansen and Lunde, 2003), and compare models on decision metrics such as turnover, not loss alone (Cortesi et al., 2026). Risks: leakage through a pooled training window, realized-volatility estimates that change with sampling frequency, and VIX-futures roll costs, which the strategy study may not fully capture.

11.4 Expiry, calendar and session effects

Derivatives expiry is a scheduled event, so it is a calendar feature. Illueca and LaFuente (2006) use realized volatility to study expiry effects in Spanish index futures. Xu (2014) exploit a change in expiration day in Sweden. In India, Agarwalla and Pandey (2013) study cash settlement on the final half-hour volume-weighted average price and how short trading breaks affect intraday volatility, Mahalwala (2016) study volume, return, volatility and reversal around expiry, and Suryanarayana et al. (2024) fit an EGARCH on Nifty futures volume and price. Chhimwal et al. (2025) study the shift of Bank Nifty expiry day and its effect on volatility, volume and connectedness. Ahluwalia et al. (2023) do the same for another scheduled event, index rebalancing, with asymmetric GARCH models of volatility persistence. Day-of-week patterns in volatility and volume appear in Kiymaz and Berument (2003). Zielkowski (2026) show that first-passage times scale with volatility with an exponent far from the diffusion value, largely because of first-bar overnight gaps, which matters for holding-period and stop design.

These papers share an idea: treat the exchange calendar as an exogenous regressor. A desk can encode time-to-expiry, expiry-day dummies, rebalance dates and session-open flags as features, so that signals do not mistake a settlement mechanism for information.

The three themes above also interact. Expiry days change realized volatility, which changes the volatility forecast that feeds the risk limit, which in turn changes position size on the day when liquidity is least reliable. A pipeline that treats the calendar, the forecast and the limit as separate modules can therefore be wrong in a correlated way. For that reason the calendar flags should enter the forecast model as regressors, rather than being patched on afterwards by a risk overlay.

11.5 Spillovers and causality in variance

Volatility moves across assets and markets. Diebold and Yilmaz (2012) define total and directional spillovers from a generalized vector autoregression with order-invariant variance decompositions. Cerqueti et al. (2023) cluster stocks by directional spillovers with a weighted fuzzy method on the

Italian market, Chai et al. (2020) model G20 comovements with an ICA-based ARMA-APARCH-M approach, and Blazsek et al. (2026) use score-driven models for return and volatility spillovers between India, the US and USD/INR. Cheung and Ng (1996) supply the underlying statistical test for causality in variance from residual cross-correlations. The shared lesson is that a spillover table is a directed graph, and that its edges rest on forecast-error decompositions that should be tested.

11.6 Drawdown and tail risk measures

Three papers concern risk summaries. Harding et al. (2003) discuss the strengths and weaknesses of drawdown as a risk statistic in managed futures, Mendes and Lavrado (2017) implement and test Maximum Drawdown at Risk, and Banerjee and Paul (2024) test a multiplicative-component GARCH with extreme value theory for intraday VaR and expected shortfall in emerging and developed markets. They connect the volatility forecasts above to the limits a risk committee actually sets.

Desk note. Event-calendar and drawdown guardrails (risk stage). Combine the expiry and rebalance flags above with a volatility forecast and a drawdown-at-risk limit (Mendes and Lavrado, 2017; Banerjee and Paul, 2024), so that leverage falls into scheduled events. Test the limits with a rolling out-of-sample coverage check, not only an in-sample fit. Cost and regime dependence matter: rule changes such as moved expiry days (Chhimwal et al., 2025) invalidate old calendar estimates.

11.7 Bridges out of this chapter

Bridge. To time-series methods (Chapter 5) and representation learning (Chapter 6). Econometricians say that volatility is long-memory, rough, or path-dependent (Guyon and Lekeufack, 2023; Abi Jaber and Li, 2025). A time-series-learning group would say that the series has a kernel-weighted look-back with power-law weights, which is a learned or fixed convolution. The power-law-weighted sums are interpretable features that a representation-learning pipeline can use as a prior, and the pooled LSTM result (Rosenbaum and Zhang, 2024) suggests that the cross-asset pooling familiar from foundation models applies here.

Bridge. To regimes and systemic risk (Chapter 7). What this chapter calls a Markov-switching stochastic volatility state (Virbickaitė and Lopes, 2019) or a latent jump state (Johannes et al., 2009) is what change-point and regime papers call a regime or segment. A directional spillover matrix (Diebold and Yilmaz, 2012; Cerqueti et al., 2023) is a network whose weights can be tracked for structural breaks, and the dealer-gamma bounds (Sahu, 2026) are a reminder that any regime label built on an unobserved quantity needs an uncertainty interval.

Bridge. To microstructure and execution (Chapter 12) and market structure (Chapter 4). Microstructure speaks of order-flow imbalance, liquidity and settlement mechanics; the derivatives literature speaks of expiry effects and hedging pressure (Agarwalla and Pandey, 2013; Zielkowski, 2026). Hawkes-based foundations for roughness (Wang et al., 2026f) connect self-exciting order flow with the Hurst exponent measured in Mouti (2026). Participants' books and their disclosure (Sahu, 2026) are where the market-structure chapter and this one meet.

Bridge. To backtesting (Chapter 14) and synthetic data (Chapter 3). A simulation-based null, as in Pillai (2026), and a superior-predictive-ability test (Reinhard Hansen and Lunde, 2003) are what the backtesting chapter calls multiple-testing control. Generated surfaces (Han et al., 2026) are what the synthetic-data chapter calls a market generator, and Shikhman et al. (2026) supply the warning that a generator should be tested on quantities not pinned down by the prices it was fitted to.

Part V

Implementation

Chapter 12

Market Microstructure, Order Books and Execution

Four research communities meet in this chapter and rarely read one another. Market-microstructure economists, who supply most of the papers here, study how the rules of a venue turn orders into prices: who demands liquidity, who supplies it, and what each trade reveals. Financial econometricians and statisticians contribute estimators for noisy high-frequency data, lead–lag tests and price-discovery decompositions. Deep-learning groups treat the limit order book as an image or a sequence and try to forecast the next mid-price move. A decision-and-control group, which mixes stochastic control, game theory and reinforcement learning, asks how to slice a parent order. Smaller contributions come from physics, rough-path mathematics, time-series mining and behavioural finance.

The vocabularies translate directly into the atlas's shared terms. "Order-flow" is the signed stream of trades and order-book events, and "trade classification" is the labelling step that signs it. A "limit-order-book" state is a set of price levels with queued volume. "Liquidity" covers spreads, depth and resilience, and "market-impact" is the price response to one's own or others' flow. "Execution" is the schedule or policy that works a parent order, and "transaction-costs" are what remain after fees, spread and impact. Microstructure people call a persistent signed imbalance "informed flow". Machine-learning people call the same object a feature, a label or a latent state.

In the fund pipeline (data, features, signal, regime and risk, portfolio, execution, validation) this chapter sits at both ends. At the front it governs data and feature construction: signing trades, measuring spreads and encoding the book. At the back it governs execution and cost, which decide whether a signal from any other chapter survives contact with the market. Whatever the holding period, the cost estimates here set the break-even for the strategies of Chapter 10, and the return forecasts of Chapter 8 are only as good as the execution assumptions behind them.

The evidence has a regional flavour worth noting. A substantial share of the papers use Indian exchange data, which suits a desk trading NSE instruments, but many findings come from US or Australian equities and from cryptocurrency venues, and transfer to a new market should be treated as a hypothesis.

12.1 Signing the tape and measuring liquidity

Almost every order-flow feature starts with a label that the raw data usually lack: was a trade buyer- or seller-initiated? Lee and Ready (1991) document two problems with quote-based classification. Quotes may be recorded ahead of the trades that triggered them, and trades inside the spread are hard to classify. They propose simple procedures to improve the labels. Aitken and Frino (1996) test the claim that the tick rule is accurate on the Australian exchange, a

robustness check outside the US. Chakrabarty et al. (2012) compare bulk volume classification with the tick rule and the Lee–Ready algorithm and find the conventional rules more accurate, a caution for anyone computing VPIN-style indicators. Jurkatis (2020) argues that established algorithms fail as quote changes accelerate and proposes an algorithm that searches for the matching quote instead of assuming a fixed assignment. Bessembinder (2003) shows that the time adjustment between trades and quotes and the signing procedure both affect estimated execution costs. For a desk, the message is that the sign of a trade is an estimate with its own error, and that error is a source of leakage-like bias when timestamps are misaligned.

Once trades are signed, the next question is how liquid the instrument is. Several reviews organise the answer. Le and Gregoriou (2020) sort low-frequency proxies into transaction-cost, volume, price-impact and multidimensional families. Cobandag Guloglu and Ekinci (2022) concentrate on high-frequency measures and conclude that these cluster around the spread and the order book, the latter being richer. Shaik and Bairaria (2026) survey bid–ask spread estimators and describe a shift from covariance-based metrics towards data-driven ones. These three give a menu from which a data team can choose by data availability and computational cost.

The empirical papers on India show what such menus look like in practice. Thomas (2003) characterise intraday NSE data and find behaviour similar to that of more mature markets. Krishnan and Mishra (2013) find U-shaped intraday volume and spread patterns on the NSE, together with the unusual feature of high volume and wide spreads at the same time, and they check for commonality across stocks. Pan and Misra (2021) test whether the classic low-frequency determinants of the spread still hold in high-frequency data. University of Hyderabad et al. (2016) relate spreads, volume and volatility at five-minute resolution, within the mixture-of-distributions and sequential-information-arrival frameworks. Taken together, they suggest that intraday seasonality must be removed or conditioned on before a spread or volume feature is used.

Two further papers refine the measurement itself. Bibinger et al. (2026) generalise a one-sided noise model for limit-order-book quotes so that the tail of the noise is asset-specific, and estimate the noise tail parameter. This matters for realised-volatility estimation at tick frequency, which Chapter 11 treats. Ibikunle et al. (2026) train models on a dataset with observed high-frequency-trading activity and apply them to public data to separate liquidity-supplying from liquidity-demanding strategies. The reported result is that their measures outperform conventional proxies. A natural use is as a participant-mix feature when public data are all a desk has.

Desk note. Trade-signing and spread feature stage (data and features). Inputs are timestamped trades and quotes. The output is a signed-flow series with a per-day misclassification check. A possible procedure: sign trades with a quote-matching rule, compare it with the tick rule on a subsample, and log the disagreement rate as a data-quality metric. Intraday-seasonally adjusted spread and depth then become inputs to the cost model and to the execution scheduler. What can go wrong: exchange timestamps and quote feeds that are not synchronised can leak future information into the sign, spreads estimated from daily data will not match what the desk actually pays, and a classifier validated on one market may fail on another.

12.2 Order-book features and short-horizon forecasts

The most visible strand of the literature turns the book into a predictor. The older and simpler idea is order flow imbalance. Su et al. (2021) generalise it to allow for changes that are not a minimum quotation unit and for a stationarised log version, tested on CSI 500 constituents. Gupta and Kumar (2025) ask whether depth beyond the best prices carries information on the NSE, and report that deeper levels contribute to short-term return predictability after controlling for autocorrelation, volume and trade imbalance. Mishra (2026) combine a Hawkes model of trade arrivals with imbalance and Kyle's lambda across the NIFTY50 universe and backtest

a strategy. Its headline Sharpe ratio is an in-paper backtest figure and should be discounted accordingly until costs, latency and selection are examined (see Chapter 14). Guda et al. (2025) extend the bivariate Hawkes process, which captures self- and cross-excitation of order arrivals, with a liquidity imbalance factor and a rolling volatility term.

A second family is statistical learning on the raw book. Sidogi et al. (2023) compute rough-path signature features from book data and compare them with autoregressions and with neural-network and random-forest learners; this links to the path-signature material of Chapter 6. The deep-learning group is represented by Sirignano and Cont (2018), who report a stationary, universal relation between the book and subsequent direction across many stocks, and by Lucchese et al. (2022), who introduce a volume representation of the book and study how far ahead predictability lasts. Y et al. (2025) apply a patch-based transformer to the book. A et al. (2025) release an open-source pipeline and show that forecasting power depends on the microstructural characteristics of each stock and does not automatically translate into trading profit. Naviglio and Lillo (2026) use a neural network on returns and signed volume as a tool for discovering structure, and contrast it with a linear VAR benchmark. Their aim is to recover effective parametric models, not only to forecast.

These papers make one point that desks should absorb. Predictability is real at very short horizons, but the horizon is often shorter than the time needed to cross the spread. The gap between forecasting accuracy and tradable profit is therefore the main lesson of A et al. (2025), and it is why the same data need the execution models discussed below. A complementary use of the book is anomaly detection: Kong et al. (2021) study micro-trading behaviour before price jumps with multivariate time-series methods and select the attributes that are most informative. H et al. (2023) apply thermodynamic ideas to the book of a Bitcoin spot market and derive temperature and entropy measures that correlate with spread-based liquidity measures. Both could serve as regime features, but neither is validated as a trading signal in the sources.

Bridge. What microstructure calls an "order-flow imbalance regime" is what time-series mining calls a "discord" or an abnormal subsequence, and what the pre-jump study of Kong et al. (2021) measures with attribute-wise abnormality scores. Conversely, what the deep-learning papers call a "spatial–temporal patch" of the book is, to a microstructure economist, simply a depth profile over time. A desk that borrows the mining toolkit (Chapter 5) gets similarity search over book states, and one that borrows the economist's view gets a reason to expect a pattern to persist.

12.3 Price discovery, lead–lag and information

A different question is where information enters first. In India the classic test is between index futures and the cash index. Sinha and Sharma (2006) use cointegration and regression and find that Nifty futures lead the cash index, and Debasish and Mishra (2008) examine the same relation from an efficiency and arbitrage angle. More recent work quantifies the role of derivatives directly: Aggarwal and Thomas (2019) report that single-stock futures hold a 55% information share during news arrivals, higher for negative news, which they connect to short-sale restrictions and a leverage–liquidity trade-off. Outside India, Frijns et al. (2015) study cross-listed shares and find that lower relative spreads and higher relative trading activity raise an exchange's share of price discovery. Chordia et al. (2005) model liquidity, returns and volatility between small and large firms in a vector autoregression and find liquidity spillovers across the two groups.

Informed trading can also hide in options and in who is trading. Anand and Chakravarty (2007) show that informed traders split orders into small or medium trades depending on contract volume, with at-the-money calls carrying the most information. Muravyev (2016) decomposes option price impact into inventory-risk and asymmetric-information parts and finds that the inventory component is larger, which matters for option-based order-flow signals (Chapter 11). Jang et al. (2023) find that foreign institutions earn large profits in index derivatives while

domestic individuals lose, and argue that the evidence is consistent with superior intraday market-wide information. Garg and Mitra (2015) look at the lead–lag relation between foreign institutional herding and Indian returns using daily data. Chaboud et al. (2014) show in foreign exchange that algorithmic trading improves two measures of price efficiency, triangular arbitrage opportunities and high-frequency autocorrelation, mainly when computers take liquidity.

For a desk, lead–lag relations are the textbook "signal" in this chapter, and also the most perishable. The estimates describe averages over multi-year samples; the half-life of an edge is shorter, and the lag at which it is observable must be longer than the desk's latency or it cannot be traded. Where a lead–lag pair has a stable cointegrating relation, Chapter 9 treats the relative-value use.

12.4 Who is trading: high-frequency and algorithmic participants

A long series of papers asks what different kinds of participants do to the book. Brogaard et al. (2015) find that price discovery happens predominantly through limit orders, most of which come from high-frequency traders, and that this falls in volatile periods. Hirschey (2021) reports that aggressive high-frequency trades lead those of other investors, and that a subset of high-frequency traders predicts the order flow of others the best. The model of Aït-Sahalia and Saglam (2013) has a strategic fast trader with an imperfect signal about future flow and predicts reduced liquidity provision when volatility rises. Easley et al. (2019) argue that the defining feature of high-frequency trading is decision-making on a volume clock, not speed, and offer advice for slower traders; time sampling by volume instead of by clock is also a feature-engineering idea in its own right.

On the NSE, where algorithmic participation is flagged, a set of papers distinguishes proprietary from buy-side algorithms. Arumugam and Prasanna (2021) find that the two groups affect quoted spreads and price impact differently, and Arumugam and Krishna Prasanna (2021) report within-group commonality and between-group contrarian behaviour. Arumugam et al. (2023) use jump-robust volatility estimates and find that heavy directional and non-directional trading by buy-side and high-frequency algorithms increases volatility. Syamala and Wadhwa (2020) show that algorithmic traders earn intraday market-timing profits, particularly when providing liquidity. The identity of the trader can matter even on a decentralised exchange: Zhai (2026) reconstruct a full-depth book where counterparties are visible by wallet address and report that informativeness is a persistent trader attribute. Anshuman et al. (2025) find that intraday proprietary traders are contrarian and correct temporary mispricing through informed trading rather than liquidity provision. Kim and Hansen (2026) document periodic bursts at one-, five- and fifteen-minute marks in cryptocurrency futures, tie them to algorithmic participation through trade-size roundness, and report out-of-sample predictability of opening returns.

Taken as a group, these studies suggest participant labels as features: the share of volume from aggressive fast traders, or the periodic clock phase, can condition other signals. They also warn about crowding. If many algorithms commonly trade the same direction, the resulting flow is itself a risk factor.

Desk note. Order-flow and participant signal (signal and filter stages). Inputs are signed trades, book snapshots and, where the venue provides it, a participant flag. A possible output is an imbalance-based short-horizon forecast used as a filter on a slower strategy: trade in agreement with the imbalance, or delay entry when it opposes. The clock-phase effects of Kim and Hansen (2026) could be added as a time-of-hour interaction. What can go wrong: published imbalance edges are measured before costs; the forecast horizon may be shorter than the desk's latency; flags for participant types exist only on some venues; and an edge found on one venue or period may reflect that venue's design, so it needs a walk-forward test by regime.

12.5 Auctions, price pressure and behaviour

Some of the most reliable microstructure effects arise from forced or predictable demand. Wu and Jegadeesh (2020) study closing auctions and report that the temporary price impact takes three to five days to dissipate, and that strategies exploiting the impact and reversals are profitable in their sample. Brown (2025) document that public retail order flow predicts reversals at the open, again with a long–short strategy in the paper. Coval and Stafford (2005) analyse fire sales caused by mutual fund flows and find that those who trade against constrained funds earn returns for providing liquidity. Peress and Schmidt (2020) use sensational news that distracts noise traders as a natural experiment, finding that trading, liquidity and volatility fall and prices reverse less. Devadutta Indoria (2024) propose a cognitive-load account of overreaction and underreaction using tick data across several exchanges.

Two papers address regimes and endogeneity. Filimonov and Sornette (2012) fit a Hawkes model to E-mini futures to measure how much of price change is endogenous, and report a significant rise over 1998 to 2010, with implications for flash-crash prediction; this ties the chapter to the regime material in Chapter 7. Ochędzan and Antulov-Fantulin (2026) build an agent-based market in which market makers with different information face self-exciting order flow, to study adverse selection against price discovery; Chapter 3 treats such simulators. Two peripheral items belong with the practitioner's folklore. Jan (2021) give microeconomic foundations for the volume profile tool, and Sharma et al. (2025) apply integral transforms to market efficiency and information asymmetry. Neither should be read as evidence of an edge without testing.

Bridge. What microstructure calls "price pressure" and "temporary impact" (the five-day dissipation in closing auctions) is what the return-forecasting literature calls short-term reversal, and what the factors literature files as a liquidity-provision premium. A desk that sees the reversal as a premium paid for absorbing someone's forced trade will size it by capacity and inventory risk. One that sees it as a statistical factor will size it by turnover cost. Chapter 9 gives the second vocabulary.

12.6 Working the order: execution, placement and liquidation

If signals tell a desk what to trade, this strand tells it how. The classical frame is Almgren–Chriss. Labadie and Lehalle (2012) derive recursive formulas for target-close and implementation-shortfall strategies in it, add minimum participation constraints and propose alternative risk measures based on self-similar processes. Schedule-based algorithms (TWAP, VWAP, percentage of volume) are the baseline, and Maciejewski (2026) propose a passive percentage-of-volume variant that sets its placement rate from observed order flow, without an explicit model. Barzykin et al. (2026) model passive execution using two observables, the roughly exponential decay of fill probability with distance from the mid-price and the linear response of price to order-flow imbalance, and combine them into a passive impact rate. Cao et al. (2008) show on the Australian exchange that the book affects submissions, cancellations and amendments, and Baruch (2005) analyse who benefits when the book is visible. Mishra et al. (2025a) study expiry-day behaviour where agency algorithms displace proprietary ones from queue priority by quoting aggressively.

A second group casts liquidation as a game or a learning problem. Fu et al. (2025) and Chen et al. (2024) solve mean-field liquidation games, the second with a major trader and periodic targeting, motivated by observed periodic volume. Graf and Mastrolia (2026) compare deep reinforcement-learning policies for liquidation that combines a limit-order book with a closing auction, tested on synthetic and historical paths in a simulator. Li et al. (2025b) use imitation learning with a flow-matching policy for high-frequency trading under varied market conditions. Maeda (2026b) solve a liquidation problem where price follows a skew Brownian motion with support and resistance levels, which is a rare formal treatment of a technical-analysis idea.

The common caution is that these models are only as realistic as their impact assumptions. The mean-field and control papers give structure and sensitivity to parameters; the learning papers give flexible policies whose validity depends on the simulator, a point Chapter 3 returns to.

Desk note. Execution scheduler and cost check (execution and validation). Inputs are the parent order, an intraday volume curve, spread and depth estimates from the first section, and an impact model. The output is a schedule or passive-placement policy with a pre-trade cost estimate and a post-trade comparison with realised shortfall. A possible design: use a participation schedule as the default, and let a passive policy based on fill-probability decay replace it only where the measured spread is wide. What can go wrong: impact parameters calibrated on one market regime, simulator-trained policies that exploit simulator artefacts, auction and expiry-day behaviour that differs from continuous trading, and capacity limits that make the paper-trading cost optimistic.

12.7 Do algorithms and high-frequency traders improve markets? Causal evidence

A desk that trades against, alongside or through high-frequency traders (HFTs) needs an answer to a plain question: does their presence make execution cheaper or dearer? The papers in this group share an identification problem rather than a data set. Algorithmic activity is endogenous: it rises where liquidity is already good. Almost every study therefore leans on an exogenous shock such as exchange co-location, a tick-size reform, a latency upgrade or a short-sale ban, and then asks what changed in spreads, depth, volatility and price efficiency. A time-series-classification reader will recognise the logic as an interventional label: the shock defines a "before" and "after" class, and the question is which features separate them.

The broad international and single-market studies mostly point the same way. Boehmer et al. (2012) use co-location as an instrument across 42 equity markets and report better liquidity and informational efficiency, somewhat higher short-term volatility, and lower execution shortfalls for institutional buy-side investors. Aggarwal and Thomas (2014) use a co-location event with matching and difference-in-differences and find lower liquidity costs, order imbalance and intraday liquidity risk. Mestel et al. (2018) reach a similar conclusion in Austria with an instrumental-variables design, consistent with algorithmic traders acting on average as market makers, and Broussard et al. (2025) classify Nordic traders into algorithmic, institutional, professional and retail groups and find narrower spreads for algorithmic and institutional traders, with retail traders surprisingly acting as liquidity providers. Smaller markets and asset classes fit the pattern with caveats: Hu et al. (2020) find that algorithmic quoting in agricultural futures improves pricing efficiency and damps short-term volatility but narrows effective spreads only in the more liquid contracts, and Riyanto et al. (2024) report a positive HFT effect on market quality in Indonesian LQ-45 stocks together with a link from tick size to HFT activity. Survey-based work such as Haralayya and Shetkar (2024), which draws on the experience of 70 trading experts, offers a taxonomy of HFT strategies but not causal estimates, and should be read as vocabulary rather than evidence.

The counter-narrative is about heterogeneity. Hu (2026) argue that market-making and strategic HFT differ in information processing, that one trader can switch roles, and that the simple dichotomy should be replaced by multi-dimensional identification. Breckenfelder (2024) find that when HFTs compete, their speculative trading increases and liquidity deteriorates and short-term volatility rises, while Brogaard and Garriott (2019) find that spreads on a new venue converge to the incumbent's as HFTs enter, which fits HFTs behaving as quantity rather than price competitors. Brogaard et al. (2017) use the 2008 short-sale ban to separate HFT and non-HFT short selling, and find that HFT short selling adversely selects limit orders and

hurts liquidity. Weller (2018) adds a different cost: more algorithmic activity is associated with less information in prices ahead of scheduled disclosures, which is a long-horizon effect that spread-based metrics cannot see. Finally Arumugam (2023) looks at intraday profitability among buy-side algorithms, HFTs and non-algorithmic traders and finds that the three groups win and lose in different ways depending on whether they supply or demand liquidity.

Two ideas make these results portable to a trading desk. The first is measurement without labels. Hasbrouck and Saar (2013) propose a low-latency activity measure computed from widely available message data and correlated with exchange-constructed HFT estimates, and Murray et al. (2016) show that a latency cut on one exchange improved liquidity but did not lastingly cut costs for institutions that already had low-latency access. The second is the structure of rents. Yao and Ye (2018) show that a uniform tick size and time priority allocate rents to the fastest liquidity providers, and that the message-to-trade ratio is a poor proxy for HFT liquidity provision. Barardehi et al. (2025) refine the Tick Size Pilot reading by finding thresholds in the number of ticks inside the spread at which a larger tick switches from helping to harming quality, a warning against treating the pilot as a clean exogenous liquidity shock for all stocks.

Desk note. Venue and algorithm-selection filter (execution stage). Inputs are message-level quote and trade data per symbol and venue, plus event dates for tick-size, latency or fee changes. The output is a per-stock regime label, for example a high-speed-competition indicator built from a low-latency activity measure, and a spread-to-tick ratio that conditions the choice between passive and aggressive child orders. A natural use is to estimate effective and realised spreads for the desk's own fills on each side of a venue event, as a validation check on routing assumptions. What can go wrong: published effects are average treatment effects on large, liquid names and may reverse for wide-spread or small stocks; the shocks are old and the competitive landscape has moved; and a measure calibrated on one exchange's message format does not transfer unchanged.

12.8 Auctions, ticks, expiries and market design

Market design determines where a systematic desk can trade without paying for immediacy. Closing and opening auctions are the clearest case. Bogousslavsky and Muravyev (2023) document that closing auctions carry a growing share of daily volume, that closing prices typically match pre-close quotes, that price impact is lower than in continuous trading, and that auction volume jumps on index additions, which they connect to indexing and ETF growth. In stress the picture changes: Foley et al. (2022) report severely dislocated closing prices during COVID-19, exacerbated by retail activity and reduced HFT liquidity provision. Older exchange-level evidence, Camilleri and Green (2009) on the suspension of call auctions at the National Stock Exchange of India, found improved volatility, efficiency and liquidity after suspension and suggests that auctions carry liquidity externalities that help the liquid more than the illiquid. A comparative study of the two main Indian exchanges, Goel et al. (2021), covers trading mechanisms, efficiency tests and liquidity statistics, and is mostly useful as a reference for the institutional facts of local venues.

Settlement and expiry mechanics create predictable flow. Hsieh and Ma (2009) compare two index futures on the same underlying with different settlement rules and find that average-price settlement minimises expiration-day effects relative to opening or closing price settlement, with foreign institutional activity increasing them. Hancock (1993), whose abstract is not in our brief, is titled as a study of the triple witching hour, and we place it here as an expiry-effects reference only. Together these papers say that scheduled liquidity events are not noise; they are known-timing features that any intraday model and any execution schedule must either exploit or avoid.

12.9 Market impact, optimal execution and transaction-cost analysis

12.9.1 Why impact has the shape it has

Several theoretical papers explain why trading large size is so costly, and they fit together. Lillo et al. (2005) show that long memory in order signs can arise when large orders, whose sizes follow a power law, are split and executed incrementally. Toth et al. (2011) propose a dynamical theory in which the supply and demand profile is V-shaped and vanishes near the current price, so that metaorders must be fragmented and small orders have a disproportionate, square-root impact. Sato et al. (2026) embed the square-root law in the order-splitting model to reconcile predictable order flow with approximately diffusive prices, and Zhou et al. (2026a) test three competing predictions for the square-root exponent in a simulated order-book model and find that none of them reproduces the measured exponents alone, concluding that order splitting and liquidity replenishment are jointly necessary. The book-length treatment Bouchaud et al. (2018) sits behind much of this literature (its abstract is not in our brief), as does Doyne Farmer 5 et al. (2004), whose title asks what really causes large price changes and which we cite for the question only. At the strategic end, Evangelista et al. (2022) derive price formation from game-theoretic and mean-field models with liquidity costs, Yang et al. (2020c) model back-runners who infer fundamental information from past order flow so that fundamental investors randomise their orders, and Yang et al. (2024) show that an informed institution can underperform naive price-taking individuals when noise trading is high. Serrano-Padial (2012) supplies the related behavioural result that prediction-market prices are mispriced when naive traders are numerous, which we include as a reminder that noise-trader share matters for any impact model.

12.9.2 From impact models to schedules and monitoring

On the practical side, Rastegar and Eghbalreihani (2019) fit an I-star market-impact function for Tehran Stock Exchange shares and use it to split a large order over the investor's horizon, an example of a simple, auditable schedule built from an estimated cost curve. Lehalle and Neuman (2019) add a Markovian signal to the transient-impact optimal trading framework and give an explicit strategy for a mean-reverting signal with exponentially decaying impact; the point for a desk is that alpha and impact decay pull the trade in opposite directions. Zovko (2026) argue that real-time slippage monitoring needs hundreds of fills to separate impact from background volatility and does not establish causality, which makes in-flight impact detection a statistical design problem. After the trade, Markov (2019) give a Bayesian framework for comparing broker algorithms on implementation shortfall, volume-weighted and participation-weighted benchmarks and short-term reversion, with fat tails and heteroscedasticity, and with hierarchical pooling across algorithms. Equally relevant to cost modelling is Sai et al. (2026), which frames the economics of an algorithmic platform through slippage and capital-budgeting measures such as net present value; its claims about order-flow shares are broad, so we use it as a framing of the business case rather than a source of parameters.

Desk note. Impact calibration and TCA loop (execution and validation stages). Inputs are the fund's own parent-order and fill records, arrival-price and interval benchmarks, and daily volume and volatility. Outputs are a per-instrument impact curve (for example a square-root form with fitted coefficient), a schedule that trades off signal decay against impact, and a ranked list of broker algorithms with credible intervals. A natural workflow is to fit the curve offline, use it as a cost term in the portfolio stage, and update the broker ranking hierarchically so that thin-data algorithms borrow strength from the group. What can go wrong: impact estimates are conditioned on the fund's own strategy mix and are contaminated by signal-driven selection of when to trade; a square-root exponent fitted on liquid names will be wrong for illiquid ones; and capacity claims inherit every misspecification of the curve.

12.10 Learning to execute and to make markets

Reinforcement learning (RL) is the natural machine-learning framing of execution: a state of inventory, time and book features, an action of order size or price, and a reward equal to shortfall. The papers here are mainly methodological and mostly simulation based. Zhang et al. (2023b) model optimised trade execution as offline RL with a dynamic context and study why existing methods overfit, which is the key concern for any deployment. Zaznov et al. (2026) propose a deterministic actor-critic variant with pessimistic backups, policy smoothing and conservative regularisation for liquidating large stock sell programs. Cheridito and Weiss (2026) build a multi-level market-making agent that submits market and limit orders of varying size across price levels while controlling inventory, with a deep-set encoder for variable-length order sets and potential-based reward shaping, tested in three simulated environments. Koulouris and Campajola (2026) place two deep RL agents in a shared liquidation game and examine whether memory lets them reach lower implementation shortfall than the game-theoretic competitive benchmark, a reminder that learning agents interact with one another. The deployment side is addressed by Hong et al. (2026), who push order-book construction and learned inference into programmable network switches to cut latency.

For a desk, the shared lesson is that the hard part is not the optimiser but the simulator and the evaluation. Each of these methods learns against an assumed impact model or a replayed book, and results are only as good as that environment's response to the agent's own orders, which replay cannot provide.

12.11 Stress, flash crashes and the fragility of liquidity

12.11.1 Who supplies liquidity when it matters

Whether electronic liquidity providers withdraw in stress is a recurring question. Brogaard et al. (2018) find that HFTs on average absorb imbalances during extreme price movements in single stocks, but that when several stocks move together their liquidity demand dominates their supply, with little evidence that they cause the movements. Nawn and Banerjee (2019) find that proprietary algorithmic traders on the National Stock Exchange of India increase limit order supply after high volatility and extreme moves, with sub-three-millisecond revisers behaving like their parent class. Arumugam (2025) study the two-way link between algorithmic activity and intraday tail risk, finding that order placement and cancellation by proprietary and agency algorithms are associated with lower tail risk, while rising tail risk reduces order activity. Panayi et al. (2015) take the design view: designated market makers have quoting obligations that do not track uneven liquidity demand, and the authors recommend adding requirements on replenishment speed measured by a threshold exceedance duration. Kumar and Misra (2018) and Cheriyan and Lazar (2019) document commonality in liquidity on Indian markets, with individual-stock liquidity co-moving strongly with market and industry liquidity; for a portfolio this means that liquidity shocks are correlated precisely when many positions need to be exited.

12.11.2 Toxicity indicators and cascades

Indicators built on order-flow imbalance, notably volume-synchronised probability of informed trading (VPIN), were proposed as early warnings. The evidence here is sceptical: Andersen and Bondarenko (2014) find VPIN a poor predictor of short-run volatility that peaked after, not before, the flash crash, and Abad et al. (2018) find that high VPIN readings rarely precede abnormal illiquidity or trading halts once realised volatility is controlled. The same measure is nevertheless useful descriptively; Bhattacharya and Chakrabarti (2014) apply it to Indian IPO aftermarkets and find adverse selection risk highest on listing day and decaying thereafter, and Teodorovic (2011) apply a standard price-impact decomposition to stocks of different liquidity

on the London Stock Exchange. A systematic desk should thus treat VPIN as a state descriptor to be validated out of sample, not as a risk trigger.

Cascade mechanisms get a more structural treatment. Osler (2005) present evidence that stop-loss orders can create self-reinforcing price cascades in currency markets, and Virgilio (2022) identifies prior volatility, scarce liquidity, large quantity and stop-loss orders as drivers of very short-time price changes. Turiel and Aste (2021) show that volume distributions during anomalous events have tail exponents below two, hence a diverging second moment, and tie flash crashes to self-organised criticality; Turiel and Aste (2022) report a phase transition between small and large co-crashes. Johnson et al. (2013) argue that a sub-second all-machine regime of extreme events emerged beyond human response time. Seuma (2026) measure branching ratios in crypto-perpetual liquidation cascades and find the largest one to be strongly subcritical, Woo (2026) show that scalar per-product monitoring of leveraged-ETF rebalancing feedback misses coupled amplification, and Gao et al. (2026a) show larger price pressure from corporate bond ETF fire sales during COVID-19. Xu et al. (2026a) train a neural detector for mechanical quote erosion in an agent-based simulator, where ground truth exists, and O'Donovan et al. (2023) link retail option flow to hedging-driven effects on market quality. A looser group covers flow by investor type: Dhingra et al. (2016) study foreign institutional flows and volatility in India, finding them to be positive-feedback traders who increase volatility when selling.

Desk note. Liquidity-stress filter (regime and risk stage). Inputs are order-book depth and spread series, signed volume and, where available, participant-type flow. Outputs are a liquidity-regime flag, a cross-asset commonality score, and a participation cap that tightens when the flag trips. A natural use is to define the cap from heavy-tailed volume models rather than Gaussian ones, in line with tail exponents below two, and to couple it with a position-level cascade check for ETF or leveraged-product exposure. What can go wrong: indicators such as VPIN may fail out of sample, so backtest the trigger against realised-volatility controls; simulated ground truth (as in the crumbling-quote detector) does not guarantee real-data precision; and regime flags that depend on a single venue can miss cross-venue stress.

12.12 Point-process and structural models of order flow

Event-time models treat orders as a self-exciting sequence. Bacry et al. (2015) review Hawkes processes in finance, covering volatility estimation from transaction data and applications to many high-frequency problems, and Jaisson and Rosenbaum (2015) show that nearly unstable Hawkes processes rescale to integrated Cox–Ingersoll–Ross dynamics, a result that explains why fitted kernels sit close to the stability boundary. The cascade papers above (branching ratios, criticality) are the same mathematics under another name. For ecology and structure, Handa et al. (1998) is titled as a study of the ecology of an order-driven market, and we place it as a background reference in the agent-based tradition without further claims.

A natural use is to estimate a branching ratio per instrument in rolling windows as a cheap feature of endogenous activity, while treating a near-unit estimate as a modelling warning rather than a signal.

12.13 Bridges out of this chapter

Bridge. Toxic-flow regimes and time-series mining (Chapter 5, Chapter 7). Microstructure speaks of "flow toxicity", "crumbling quotes" and liquidity "regimes". Time-series mining speaks of anomalies, discords and change points in a multivariate stream. The crumbling-quote detector and the VPIN critiques are, in that language, supervised and unsupervised anomaly detection with an unreliable label; the methods of Chapter 5 for similarity search and Chapter 7 for change points could supply label-free alternatives, while microstructure supplies the economic meaning of a break (a market maker withdrawing) that a generic change-point method lacks.

Bridge. Execution as decision-making under uncertainty (Chapter 13, Chapter 14). Optimal execution and RL execution are what the decision-theory silo calls sequential stochastic control, and the overfitting results map onto what Chapter 14 calls backtest overfitting. The impact curve estimated here is the transaction-cost term that portfolio construction in Chapter 13 needs; conversely, portfolio people can read "capacity" as the point where the square-root curve consumes the signal's edge.

Bridge. Event-time clocks and generative simulation (Chapter 3, Chapter 4). Hawkes branching, order-splitting long memory and agent-based books are generators of order flow. The synthetic-data chapter treats such simulators as training grounds; the lesson here is that a simulator must reproduce impact, long-memory signs and stress behaviour before an RL agent trained on it can be trusted. Chapter 4 supplies the participant taxonomy (retail, institutional, proprietary) that these simulators need as agent types.

Chapter 13

Portfolio Construction and Decisions under Uncertainty

Five communities meet in this chapter, and they rarely cite one another. Operations researchers and "decision" machine-learning people speak of *predict-then-optimize*, regret, contextual optimization and differentiable optimization layers. Statisticians speak of random-matrix noise bands, eigenmodes, efficient-frontier inference and conformal coverage. Applied mathematicians speak of stochastic control, optimal stopping, rough paths and signatures. Reinforcement-learning researchers speak of rewards, policies and distributional critics. A smaller group from language-model research speaks of calibrated confidence and abstention. Each of these words has a counterpart in the shared vocabulary of the atlas. A "decision loss" or "regret" is a *decision-focused learning* objective. An "eigenmode outside the Marchenko–Pastur band" is a *factor-model* direction in the *cross-section*. A "coverage guarantee" is a form of *uncertainty quantification*. A "signature-linear control" is a *path-signature portfolio-optimization* rule.

In the fund pipeline of the atlas (data, features, signal, regime and risk, portfolio, execution, validation, research automation) this chapter sits in the middle. Its input is whatever the signal and forecast stages emit: point forecasts, ranks, probabilities, or a covariance estimate. Its output is a vector of target positions that the execution stage must turn into orders. Its central question is how forecast quality, forecast uncertainty and cost should shape the size of a bet. Return forecasting is treated in Chapter 8, cost-aware execution in Chapter 12, and the validation of the whole stack in Chapter 14.

The chapter is organised by the problem each group of papers attacks. We first clean the covariance matrix, then size bets and choose allocation rules, then plan trades with costs, then train forecasts through the allocator, then learn policies by reinforcement learning, then attach uncertainty and abstention to the signal, and finally cover the theory papers that supply tools without a trading application of their own. Where a paper is peripheral we say so and place it in a group.

13.1 Cleaning the covariance matrix

Every mean–variance allocator inverts or otherwise consumes an estimated correlation matrix, so the first problem is that the matrix is mostly noise. Laloux et al. (1999) compare the eigenvalue density of empirical stock correlation matrices with random-matrix theory and conclude that blind use of empirical correlations for risk management is doubtful. Plerou et al. (2002) extend the test to several large panels and find that most eigenvalues fall inside the random-matrix bounds, with deviations carrying genuine market and sector structure. These two classic papers define the idea: a null model of uncorrelated series gives a noise band, and only modes outside it are treated as information.

Recent work from the same tradition reuses the idea for networks. Pawanesh et al. (2025) filter

the correlation matrix of NIFTY 200 and 500 stocks and report that core–periphery structure lives in the market mode while community structure lives in the sector mode. Ansari et al. (2026) keep only 10 to 16 structured eigenmodes of NIFTY and S&P 500 correlation matrices and report more stable core–periphery networks than the full matrix gives. Ansari et al. (2025) then use the core–periphery profile directly: portfolios of periphery stocks from minimum spanning trees and planar filtered graphs are compared with centrality-based strategies on Sharpe, Sortino and information ratios. Network people call a stock "peripheral" what a risk manager would call weakly exposed to the dominant market factor; a natural use is as a diversification screen, not as a stand-alone alpha.

Other papers replace the correlation matrix or tame its use. Smyth (2025) cluster equities with entropy-based rather than correlation-based distances to select stable minimum-variance constituents, and note that clustering can flag stationarity changes in excess returns. Hiraki and Sun (2022) propose OWL shrinkage, which gives similar weights to correlated stocks and, as the abstract reports, lands near the $1/N$ portfolio with lower turnover. Gregnanin et al. (2024) swap the Pearson matrix for a signature-based similarity matrix on S&P 500 networks and report better cumulative log-returns and Sharpe ratio than the correlation baseline. Halperin (2026) go further and drop the covariance and the expected-return vector, building the state from a correlation distance matrix and two rank-transition matrices; its out-of-sample claims should be read against the rest of this book's warnings on backtests. Banner et al. (2005) offer a stylised counterpart, rank-based Atlas models, in which growth rates depend on rank; these are a benchmark world rather than a tradable recipe.

Desk note. Covariance hygiene gate (portfolio and risk stages). Input: a trailing window of daily returns for the tradable universe. Output: a cleaned correlation matrix and a count of eigenmodes above the noise band, used as an input to the optimiser and as a risk limit. A natural use of the random-matrix papers is to compare the leading eigenvalue and the number of retained modes each week and to freeze rebalancing when the count jumps. What can go wrong: the noise band assumes independent observations, so fat tails and volatility clustering make the band optimistic; window length versus universe size sets the aspect ratio and hence the band; and a cleaned matrix that is stable in sample may still be unstable in a crisis, when correlations rise together. Cleaning does not fix biased expected returns.

13.2 Sizing bets: Kelly, universal portfolios and Markowitz at scale

Once the matrix is usable, the next problem is how much to bet. Stiffelman (2026) revisit the Kelly criterion through Cover's universal portfolio, and argue that Kelly's objective splits into a money term, an entropy term and a divergence term; this gives sizing a regret benchmark against the best constant-rebalanced portfolio. Barmish and Proskurnikov (2025) ask a practical variant: update the account value from realised open, high, low and close bars, without assuming geometric Brownian motion, and tailored to the order type. A related sizing idea appears in Hubáček and Šír (2020), who argue that a predictor trained to be decorrelated from the market price can profit even if it is less accurate than the market, by exploiting a market taker's advantage. We read this as a proposal about objectives rather than a rule to deploy.

The mainstream practical answer is convex optimisation. Boyd et al. (2024) describe a Markowitz extension that includes transaction costs and leverage limits and handles uncertainty in return statistics, still as a convex problem. Boyd et al. (2017) give the multi-period version, trading off return, risk, transaction and holding cost and executing only the first planned trade. Johansson and Boyd (2025) apply the same convex receding-horizon idea to tax-lot and account-location costs in retirement funding; it is not a trading paper, but shows how cost terms enter a convex planner. Bodnar et al. (2009) supply the inference side: under a stationary

Gaussian assumption, estimated efficient-frontier quantities are asymptotically normal, so claimed improvements can come with confidence bands. Finally, Bengio (1997) is an early argument that the model should be trained on net trading gains, including transaction costs, instead of on squared prediction error. That idea is the seed of the next section.

Desk note. Cost-aware rebalancer (portfolio and execution stages). Input: signal scores or forecast returns, a cleaned covariance matrix, current holdings, a cost model and borrow rates. Output: target trades for the next bar, with only the first step of the plan sent to execution. A natural use of the convex multi-period framework is as the production layer under any signal, with the risk-aversion and cost-aversion parameters treated as tuned hyperparameters. What can go wrong: forecast errors are amplified by the optimiser, so the uncertainty-aware variants matter; cost models fitted on small trades understate impact at capacity; and tuning parameters on the backtest is itself a source of overfitting (see Chapter 14).

13.3 Path-dependent allocation with signatures and kernels

A separate mathematical line replaces the snapshot of means and covariances with the whole price path. Bayer et al. (2026a) extend the notion of a trading strategy to signatures, solve a mean–variance criterion explicitly, and sketch an application to optimal execution. Bank et al. (2025) show that controls written as functionals of the path signature are dense in the progressively measurable controls, which licenses Monte Carlo computation of approximately optimal controls. Alonso (2026b) cast portfolio theory in the signature coordinate and link the gap between two lifts to the excess growth rate; the abstract reports an empirical dimensional trade-off, which means gains depend on how many path terms must be estimated. Futter et al. (2025) offer a kernel alternative with closed-form solutions and report that both signature and kernel approaches beat classical Markovian methods when dynamics or signals are time dependent. Bloch (2026a) is a more promotional paper that proposes path-geometry risk monitors; we treat its claims as untested until validated out of sample.

What mathematics calls a "signature feature" is, for the machine-learning side, a fixed nonlinear embedding of a window of returns that is invariant to reparametrisation. Chapter 6 treats signatures as features; here they appear as the input to an allocator.

13.4 Training the forecast through the allocator

The idea of Bengio (1997) has since become a field of its own, called decision-focused learning or smart predict-then-optimize. Mandi et al. (2024) survey it and benchmark eleven methods on seven problems; Zhang and Li (2026) classify the designs into closed-loop losses, differentiable optimisation layers and parameterised policies; Sadana et al. (2023) map the same field into operations-research language, where it appears as contextual or prescriptive optimization. For a trading desk the three surveys supply a translation table: what a quant calls "fitting the alpha model, then running the optimiser" is two-stage predict-then-optimize, and what ML calls "decision loss" is what the desk calls realised net P&L after the allocator.

Several papers refine the method. Balghiti et al. (2023) give generalization bounds for the smart predict-then-optimize loss, a guide to how much data a decision-aware return model needs. Schutte et al. (2024) replace empirical regret with expected regret to account for aleatoric and epistemic uncertainty. Wang et al. (2025c) allow feedback from the optimisation back to prediction. Kotary et al. (2023) learn the optimal decision directly from features, which could amortise a heavy optimiser across daily rebalances. Caunhye et al. (2026) train against worst-case feature perturbations, a natural fit for noisy decision-time features. Wang et al. (2025a) sample tail regions of a learned generative model of the optimisation parameters. Zhang et al. (2026h) is an inventory paper, but its CVaR-aware allocation layer under shared capacity is structurally

close to a constrained portfolio problem, and we cite it as an analogy. Boyd et al. (2024) again matters here, since a convex allocator is exactly the kind of layer through which a forecast can be trained.

Two papers bring the same logic to the feedback that our own trading creates. Cristian et al. (2025) learn under endogenous uncertainty, where decisions bias the data they later generate; in trading this is market impact. Bajpai (2025b) prove, in a stylised closed-loop model, that high predictive power need not imply profit and can accelerate P&L erosion under convex execution costs. We take from it a proposal: evaluate forecasts by their post-cost, post-feedback contribution, not by their accuracy.

Desk note. End-to-end forecast training (signal to portfolio stages). Input: features, a feature-to-return model, and the production allocator with its constraints and cost model. Output: a forecast model whose loss is the allocator's regret, or the net return it achieves. A natural pilot is to fine-tune an existing return model with a decision loss on a small universe and compare it with the same model trained on squared error, under identical costs. What can go wrong: gradients through the optimiser are fragile when constraints bind; decision losses overfit the in-sample optimum even more readily than squared error does; and the benefit may vanish once feedback from our own trading is included, as the negative result above suggests.

13.5 Reinforcement-learning allocators and regimes

Reinforcement learning (RL) treats allocation as a sequential policy rather than a one-shot optimisation. Jeon et al. (2024) feed frequency-band decompositions of stock signals to a deep RL allocator, arguing that time-domain models miss abrupt shifts. Aminrostamkolaee et al. (2025) compare seven reward functions across CNN, CNN+LSTM and Transformer agents and report that minimising standard deviation worked best across architectures and horizons; the lesson for a desk is that the reward is a design choice that deserves a validation harness. Upreti (2026) replace hard agent selection with a hidden-Markov-model soft mixture of agents and add a CVaR penalty to the reward, and Pan et al. (2026) use a regime detector to trigger knowledge transfer between allocators rather than full rolling retraining. In regime language (Chapter 7) the two papers use a posterior over states as a gate. Zerman (2022) address a neighbouring choice, the trading frequency, with a hierarchical RL agent in an FX setting. Bouaicha et al. (2022) give the control-theory counterpart, equilibrium allocation under Markov regime switching, which could supply a principled action map for a regime module.

A second group concerns where rewards and policies come from. Haußmann and Kandemir (2026) model the joint distribution of several returns, such as return and drawdown, with Stein particles. Ho and Ermon (2016) and Fu et al. (2018) learn policies or rewards from expert demonstrations, and Roa-Vicens et al. (2019) test inverse RL on a simple limit-order-book environment. Peng et al. (2019) learn from logged off-policy data with supervised regression steps. Sokota et al. (2026) solve games of hidden information, which is a loose analogy for counterparties with unobserved intent. All four are general RL tools with no trading result of their own; they are peripheral here, and any desk use is our proposal.

Stopping problems form the continuous-time cousin of RL exits. Colaneri and De Angelis (2022) solve a recursive stopping problem for trading across two venues, Maeda and Jacka (2020) model support and resistance as a stochastic process to decide when to exit, and Maeda (2026a) solve the matching entry problem under a skew Brownian motion. These are stylised models whose assumptions must be tested on real prices before any rule is adopted.

13.6 Uncertainty, calibration and abstention

The last family asks when not to trade, and how to scale a bet to the forecaster's own doubt. Hutchison et al. (2004) prove that calibration does not reduce accuracy and gives decision thresholds, which is the textbook justification for calibrating a directional classifier before converting its probabilities into sizes. Chalkidis and Savani (2021) let a classifier abstain and add a flat third class for small moves; Nabar and Shroff (2023) chain abstaining models so that predictions are made only where support is high. Lahlou et al. (2021) estimate epistemic uncertainty as excess risk and Sanderink (2026) adapt it to cross-sectional rankers, reporting that inverse-uncertainty sizing de-levers the strongest signals and hurts, and proposing a two-level policy that first decides whether to trade at all. That finding is a warning: uncertainty and signal strength can be correlated, so a naive size rule can cancel the edge.

Conformal methods give finite-sample coverage. Taheri Hosseinkhani (2026) report that machine-learning return intervals achieve near-marginal coverage while high idiosyncratic-volatility and market-regime cells are undercovered, and build a signal from that undercoverage. Singh et al. (2026) show on imbalanced tabular datasets (credit scoring, fraud, healthcare, industrial safety) that marginal conformal prediction under-covers rare classes and that class-conditional (Mondrian) prediction restores coverage. Bach (2025) give a convex loss trading set size against conditional coverage, and Zhang (2026) learn tail-aware ellipsoids for multivariate prediction, which could define joint return uncertainty sets for a robust optimiser. Noarov and Roth (2026) give online forecasts with guarantees against adaptive sequences.

Language models enter the same stack. Torres et al. (2026) combine retrieval, conformal calibration and abstention for financial forecasts, Damani et al. (2026) train models to report calibrated confidence with a Brier-score reward, and Dixon (2026b) map word probabilities to posteriors over declared states, using professional market text. Dixon (2026a) formulate how much decision authority an AI agent should receive as evidence quality changes. Chapter 15 treats these systems in depth. Finally, Sadeghi et al. (2024) learn causal links among nonstationary financial series, Jung et al. (2022) attribute outcomes to causes by do-Shapley values, and Felin and Holweg (2024) argue for theory-based causal reasoning over data-based prediction. The three are peripheral to allocation, but together suggest a research-governance gate: require a causal story and an attribution plan for each signal.

Desk note. Trade-or-abstain gate and exposure scaler (risk stage). Input: a model's score together with a calibrated probability, a conformal interval or an epistemic-uncertainty estimate, evaluated by regime cell (volatility bucket, market state). Output: a multiplier on target size, from zero to one, plus a kill-switch flag. A natural design follows the two-level idea: a slow gate decides whether the strategy trades at all, and a fast rule sizes within active trades, with the uncertainty signal first tested for correlation with signal strength. What can go wrong: calibration data are scarce in exactly the regimes that matter; calibration windows leak if built with future information; and a gate tuned on one stress episode may be nothing more than a fit to that episode.

13.7 Bridges out of this chapter

Bridge. Decision loss and the backtest. What operations research calls "regret" and the decision-focused papers call "decision loss" is what Chapter 14 calls net post-cost performance. Conversely, the "backtest overfitting" and walk-forward language of that chapter tells decision-focused learners that a regret estimate on one historical path is a single noisy sample. Both silos would gain by reporting regret with the confidence bands that Bodnar et al. (2009) give for frontier quantities, and by testing reward choices as Aminrostamkolaee et al. (2025) do.

Bridge. Noise bands, regimes and discords. What random-matrix theory calls an "eigenvalue outside the noise band" is a factor in Chapter 9, and a jump in the number of such modes is, in the vocabulary of Chapter 7, a candidate change point; in time-series mining, the same event is a "discord" in a rolling correlation profile. The regime-gated allocators of Upreti (2026) and Pan et al. (2026) could take this count as a state input.

Bridge. Own-impact and microstructure. What Cristian et al. (2025) call "endogenous uncertainty" and Bajpai (2025b) call "feedback" is what Chapter 12 calls market impact and crowding. Microstructure models could supply the counterfactual that decision-focused learners lack, while portfolio people could in turn supply the objective, net P&L, that microstructure papers often leave implicit.

Bridge. Calibrated confidence as a shared interface. The conformal, epistemic and LLM-calibration papers above all produce a number meaning "how much to trust this forecast". Chapter 8 produces the forecasts and Chapter 15 the text-derived views; a common calibrated-confidence interface would let this chapter size both in the same units.

Part VI

Research Operations

Chapter 14

Backtesting and Research Practice

Four communities share this chapter and rarely read one another. The time-series machine-learning people (the `tsml` silo) speak of *labels*, *targets*, *barriers*, *imbalance* and *classification accuracy*. The statisticians speak of *multiple testing*, *selection bias*, *size and power*, *false discovery* and *superior predictive ability*. A third group, which we call the practice silo, consists of quant practitioners, platform builders and sceptical auditors. It speaks of *backtests*, *walk-forward*, *profit factor*, *live versus paper* and *null results*. A fourth, smaller group of financial economists and control or decision scientists contributes the vocabulary of *specification search*, *evidence hierarchies* and *feedback*.

In the shared vocabulary of the atlas these become two families of concepts. Everything about what a model is asked to predict is *label design* (with the *triple barrier* and *meta-labelling* as named special cases). Everything about whether a reported result should be believed is *backtest overfitting*, *multiple testing*, *data leakage*, *forecast evaluation* and *walk-forward* validation. A third strand, *transaction costs*, is the reality check that cuts across both. A classification person who says "the F1 score on the buy class improved" and a statistician who says "the Sharpe ratio is not significant after correcting for the number of trials" are making statements about different links in one chain, and the chapter tries to connect them.

In the fund pipeline (data, features, signal, regime and risk, portfolio, execution, validation, research automation) this chapter sits at both ends of the research loop. Labels belong to the data and forecast stages: they define the supervised problem before any model is chosen. Validation sits after the signal and portfolio stages and gates what reaches capital. The final section of the chapter covers the research-automation stage, that is, the platforms that make experiments reproducible and keep backtest and live code identical. The chapter is therefore less about any single alpha than about the machinery that decides which alphas are real.

14.1 What to predict: the label as a design choice

Most machine-learning work on markets treats the target as given: the next-bar or next-month return, or its sign. A large group of papers in this collection argues that this is the weakest assumption in the pipeline. The cleanest evidence is cross-sectional. Guida and Coqueret (2026) hold the universe, the features and the model fixed and decompose the label into location, scale and shape operators, finding that reshaping the target alone changes long-short decile performance and that the label explains more out-of-sample variation than the model does. Cakici and Zaremba (2026) reach a compatible conclusion across 35 equity markets, where ordinal targets beat raw-return targets, with the caveat that rank targets discard magnitude and lose ground when dispersion or skewness is high. At the binary end of the spectrum, Balogh et al. (2027) show that the return threshold defining a "positive" class changes class separability sector by sector. For a desk the lesson is simple: a Gaussianised rank or a thresholded sign is a modelling decision with its own hyperparameters, and it deserves the same ablation discipline as

the architecture.

A second family replaces the next-bar target with a label that describes a *trend* or an *event*. Wu et al. (2020) criticise short-horizon comparison labels for missing continuous trends and propose continuous trend labelling. Han et al. (2022) label only at local extrema over N periods, so that small price changes do not generate noisy labels, and feed an XGBoost trading system. Saberi et al. (2024) label by the percentage move over a window with adaptive thresholds and use LightGBM. A et al. (2022) add a third "no-action" class so that a model may abstain from small moves. Pasterak (2026) go the other way and replace hard classes by smoothly varying buy-to-sell values fitted by evolutionary search. Wang and Gu (2025) build the interim path into the label, scoring an event by a probability that accounts for drawdown along the way and not only the end point. These are variations on one idea: the label should be an *actionable trade outcome*, not a price comparison. Their natural users are teams that already run a gradient-boosted signal and want fewer, cleaner training examples; the cost is that labelling only at extrema or breakouts shrinks the sample and makes the label depend on hindsight about the window, which must be handled in the cross-validation design.

How should the labelling rule be chosen? One answer is to simulate. Kovačević et al. (2023) define the robustness of a trend-labelling algorithm through a noise model of classifier error, so that algorithms can be compared without training a classifier each time. Another is to search: AlRashedy and Mathkour (2025) evaluate all combinations of model type, look-ahead window and feature-subset size per asset, and Song et al. (2026b) argue that the best supervision signal can sit at an intermediate horizon, different from the prediction target, and select it by bi-level optimisation. Kili et al. (2025) push adaptivity further with multi-scale, causally filtered, meta-learned labels that output calibrated probabilities. Search over labels is, however, a form of specification search, and every extra label option enlarges the multiple-testing burden discussed later in this chapter.

Two groups attack label *quality* rather than definition. Akşehir and Kılıç (2023) denoise prices with a signal-processing decomposition (EEMDAN) before labelling, and Yang et al. (2023) use a meta-learned label-correction network under a multi-task framework, borrowing from the weak-supervision literature. Rare-event settings raise the imbalance problem directly. Zhao et al. (2025c) treat label imbalance between profitable and unprofitable events once transaction costs are netted out in Chinese futures, and Zaman and Sahed (2026) extend the target in a Bitcoin limit-order-book setting to include high-volatility regimes, which enlarges the informative class. In a derivatives setting, Vaghela (2026) label intraday option bars by Black-Scholes repriced profit and loss, with theta, vega and costs inside the target, rather than by the direction of the underlying. A natural reading is that the more costs and risk are pushed into the label, the less the label and the final P&L can diverge.

Desk note. **Stage:** data and forecast. **Input:** daily or intraday bars for a cross-section of instruments, plus a cost estimate per instrument. **Output:** a label-ablation table before any model comparison. A reasonable protocol is to hold features and model fixed and vary the target among raw return, cross-sectional rank (as studied in Guida and Coqueret, 2026), thresholded sign and a net-of-cost event label, and to report each variant across folds. **What can go wrong:** labels built from a window that extends past the training cut leak the future into the validation set; rank and extremum labels can look good on classification metrics while the traded portfolio is poor; and every label variant tried counts as a trial for the deflation steps below.

14.2 Barriers, meta-labels and the gap between a label and a P&L

The triple-barrier method labels an event by whichever is hit first: a take-profit barrier, a stop-loss barrier or a time limit. It is the point where the label vocabulary meets the risk vocabulary, since

the barriers are stop and target levels. The empirical papers here mostly ask whether it helps. Li et al. (2022) compare raw-return, fixed-horizon and triple-barrier labels on one deep model for the Nasdaq-100 and find that each scheme wins on a different metric, with the triple barrier best on the buy-signal F1 score. Kovačević et al. (2022) ask the more important question, namely how classification performance maps to strategy performance, and report a statistically significant but low correlation for fixed-horizon and triple-barrier labels. The same warning arrives in a sharper form from Lima (2026): in a factor-investing setting, a positive top-decile lift on the take-profit label coexisted with a negative event-return lift, and the converted mark-to-market portfolio lost money at a 50 basis-point round-trip cost. A better classification score is not evidence of a better portfolio, and Bach (2026) make the related point that the metric a classifier is tuned on (precision on trades rather than raw accuracy) is tied to the expected Sharpe ratio.

Applications of the barrier idea extend it in several directions. Fu et al. (2024) tune barrier widths for pair trading in crypto with a genetic algorithm, targeting profit or maximum drawdown. Youssefi et al. (2026) test triple-barrier labels with tree classifiers on four timeframes for Bitcoin and find that the choice of timeframe matters. Grądzki et al. (2025) combine information-driven bars (CUSUM filter, range, volume and dollar bars) with triple-barrier labels for deep models and report that the CUSUM and triple-barrier combination performed best among their alternatives. Event-based sampling and event-based labelling are two halves of one design, and the sampling half is the part that microstructure researchers will recognise from Chapter 12.

Meta-labelling separates the *side* of a bet from its *size* or its acceptance. A primary model proposes trades, and a secondary classifier decides whether to take each one, usually by predicting whether the primary signal will succeed. Meyer et al. (2022) organise the design space into feature-driven and strategy-driven architectures and introduce inverse meta-labelling to improve the side forecasts. Thumm et al. (2022) study ensembles for the meta-model and find them most helpful when the data contain multiple regimes and are nonlinear; they point to position sizing as a further application. The caution comes from Atluri (2026), who build a heavy pipeline of regime features and a three-model ensemble on futures and find, on a held-out half-year, a meta-model not separable from random, with a structural inversion of the direction-to-profit relation during a drawdown. Their one favourable model could only be identified as best by looking at the held-out set, which disqualifies it. A meta-label is a model of the primary strategy's failure modes, and it inherits the regime dependence of the primary signal.

Regime dependence appears from a different angle in Valls and Steurer (2026), who study a volatility-regime switching rule on three equity indices and report both substantial gains over buy-and-hold and, on closer analysis, sensitivity to parameters and instability across sub-periods. Hassan et al. (2010) propose monitoring a trading system as a "machine with variable outputs" by statistical process control, which gives a production-side counterpart to these pre-deployment tests: control charts can flag when live results leave their historical range. We read the two together as a proposal that every regime filter should carry a monitor that tells the desk when the filter's own premise has failed. Regime detection itself is the subject of Chapter 7.

Desk note. Stage: signal and risk. **Input:** the event stream of a primary strategy (entry time, side, realised path, cost) and a feature set that was available at entry. **Output:** a probability that the trade will clear costs, used as a filter or a position-size multiplier in the manner of Meyer et al. (2022). **Checks:** compare the filtered book with the unfiltered book net of costs, not on AUC alone; require the AUC to be significant on data not used to choose the model, as Atluri (2026) do; and chart the live acceptance rate against its historical range, in the spirit of Hassan et al. (2010). **What can go wrong:** overlapping events inflate the effective sample, a regime change can invert the relation the meta-model learned, and a filter that rejects most trades may leave a book too small to cover its fixed costs.

14.3 Overfitting and multiple testing: the statistics of selection

The second half of the chapter belongs to the statistics silo and to practitioners who borrowed from it. The common starting point is that a backtest is the winner of a search, and the winner of a search is biased upward. Lopez De Prado (2013) states the proposition most plainly: high simulated performance is easy to reach after trying a modest number of configurations, the more configurations tried the higher the probability of overfitting, and, since researchers seldom report the count of trials, investors cannot judge the degree. Bailey et al. (2014) and Bailey et al. (2015) turn this into demonstrations, with online simulators that optimise a simple strategy over a random walk so that the "optimal" variant is by construction overfit. These are teaching tools that a risk committee can use as much as a research team can.

Three formal answers follow. Bailey et al. (2013) estimate the probability of backtest overfitting through combinatorially symmetric cross-validation, which splits the performance matrix of tried strategies in all balanced ways and asks how often the in-sample winner ranks poorly out of sample. Bailey and López De Prado (2019) deflate the Sharpe ratio for the number of trials and for non-normal returns, and the earlier Bailey and López De Prado (2011) give the probabilistic Sharpe ratio and the track-record length needed to reject a given threshold, with non-normal moments. Witzany (2021) offer a Bayesian alternative using an MCMC posterior over Sharpe ratios, and Koshiyama and Firoozye (2019) propose a covariance-penalty correction that lowers a risk metric according to the number of parameters and the amount of data. Lopez de Prado (2015) summarises the programme in the form of a practical demand: report the number of trials and treat unadjusted backtests as exploratory.

A parallel strand comes from the multiple-testing and selective-inference tradition. Harvey and Liu (2019) give simple adjustments to Sharpe ratios and related statistics, suitable for real-time evaluation of candidate strategies. Hansen (2005) proposes the test for superior predictive ability, an improvement on the reality check for data snooping that compares a large set of forecasts to a benchmark. Pav (2019) and Pav (2026) treat the same problem for the best-looking asset: the first applies conditional inference to the asset with maximum sample Sharpe ratio among correlated candidates, and the second compares shrinkage and debiasing estimators for the true Sharpe ratio of the selected asset, finding James-Stein best in the simulations reported. Chen (2022) gives a counterweight: using false-discovery-rate bounds, the author concludes that most published cross-sectional predictability claims are likely true. We read this as a statement about the prior odds of academic anomalies, not as comfort about a single team's in-house backtest, which was selected from a far larger search.

A final group moves from correcting single numbers to auditing the whole protocol. Coqueret (2023) advocate reporting results over many small variations of design choices, rather than a single specification. Nikolopoulos (2026) test whole predictive workflows against synthetic null environments and falsify those that generate significant walk-forward evidence from nothing. Inglese (2026) treat the validation stack as an estimator whose size and power can be measured with planted signals, and show that stacking two corrections against the same failure mode leaves the false-positive rate unchanged and destroys power. Wiecki et al. (2016) supply real-world evidence: across 888 strategies from one platform, reported backtest Sharpe ratios predicted out-of-sample results poorly, while higher moments and portfolio-construction features carried more information, and more backtesting by the author meant a wider gap. López De Prado (2023) proposes a hierarchy of empirical evidence ranked by causal strength.

Desk note. **Stage:** validation, immediately before capital allocation. **Input:** the full matrix of returns of every variant the team has tried for a research idea, together with a log of how many labels, features, windows and parameters were explored (the counting step Lopez De Prado (2013) asks for). **Output:** a gate with three numbers: a deflated or probabilistic Sharpe ratio, an estimated probability of backtest overfitting, and the result of a placebo run on synthetic data, in the manner of Nikolopoulos (2026). **What can go wrong:** trial counts are under-reported when several people work on one idea; the placebo may not match the real data's autocorrelation or fat tails; and, as Inglese (2026) warn, adding redundant corrections costs power without removing a new failure mode, so a stack should be justified failure mode by failure mode.

14.4 Evaluating the forecast itself

Before a forecast becomes a strategy it has to be judged as a forecast, and several papers show that the judging is easier to get wrong than the forecasting. Diao (2026) give the sharpest example. Five-minute gold-return models predicting a 288-bar forward return are ranked by tiny error differences on overlapping windows; the overlap leaves about twenty independent labels, so nominal test size overstates the independent size by orders of magnitude, and the audited networks do no better than a zero predictor. The generic lesson is an effective-sample-size correction: when labels overlap, the number of independent observations is nearer the sample length divided by the label horizon than the number of rows. Bloch (2025) make a related argument about estimation windows, showing that short look-back windows give mean, variance and Sharpe estimates with too few effective observations, so that speed of adaptation is often a statistical mirage.

For directional forecasts there are tests designed for the decision at hand. Bürgi (2025) proposes tests that penalise costly extreme misses and handle serial dependence in small samples, and applies them to EUR/USD survey forecasts, where a simple time-series model is not beaten. Sitthiyot and Holasut (2023) propose a joint test of directional skill across several variables, which suits a multi-asset signal set judged as one decision. Ozimek (2026) compare strategies not by one full-sample number but by modelling fold-level information ratios as a distribution that depends on volatility and momentum regimes, with a GAMLSS model. Together these suggest that a desk dashboard should show, for each signal, an effective sample size, a directional-skill test and a regime-conditional distribution of fold results, and not only a Sharpe ratio.

14.5 Baselines, nulls and honest negative results

The most useful recent papers in this chapter report that something did not work, and report it with enough protocol to be believed. Pandhe (2026) compare a gradient-boosted ranker with a plain 12-1 momentum rule on the Nifty 500 under the same walk-forward and cost model; momentum is numerically ahead and the Sharpe difference is statistically indistinguishable from zero, with a rank correlation of predictions and outcomes near zero. Cliff and Rollins (2020) show that some reported best-performing AI trading agents can be routinely out-performed by extremely simple strategies involving no learning, and attribute this to minimal experimental evaluation. Mesfin (2026b) test fourteen intraday signal families on Micro E-Mini Nasdaq futures and find none that survives a fixed round-trip cost with a minimum t-statistic, trade count and year-by-year consistency. Darmanin (2026) gate five retail signal families through multiplicity-corrected, net-of-cost and bankroll-survival tests, with no family supported. Castellanos Macias (2026) pre-register and audit a search over a very large space of configurations on crypto perpetuals, and find that the past-performance correlation that held in-sample is near zero on independent data and that a seemingly strong in-sample profit factor collapses below breakeven when parameters are fixed a priori.

The same discipline is applied to newer model classes. Ye et al. (2026a) argue that headline Sharpe ratios of LLM trading agents should not be treated as deployment evidence until they pass tests for temporal integrity, frictions and counterfactual robustness, a theme taken up in Chapter 15. Gort et al. (2022) formulate backtest-overfitting detection as a hypothesis test for deep reinforcement-learning agents in crypto and reject the overfit ones before deployment. One work, held as two records (Deep et al., 2025b,a), enforces information-set discipline and rolling walk-forward testing for microstructure signals and honestly reports modest and statistically insignificant aggregate results with strong regime dependence. On the constructive side, Mroziewicz and Ślepaczuk (2026) parameterise training and test window lengths in walk-forward optimisation, use double out-of-sample data and run the final test once, and Sheppert (2026) propose a composite objective, the GT-Score, that rewards consistency and penalises downside and overfitting during optimisation. Letteri et al. (2022) go in another direction, proposing to test candidate rules on future paths forecast by a neural network in addition to historical data; we would treat this as a scenario generator to be validated with the tools above, and not as a substitute for out-of-sample evidence.

Desk note. Stage: validation and forecast, as a standing gate for every new model. **Input:** the candidate's out-of-sample predictions and the predictions of three baselines: a zero or random-walk forecast, a plain momentum rule as in Pandhe (2026), and a simple non-learning trader as in Cliff and Rollins (2020). **Output:** a pass or fail on net-of-cost performance, on a significance test of the Sharpe difference, and on the effective sample size of overlapping labels. **What can go wrong:** baselines tuned less carefully than the candidate give a false pass; transaction-cost assumptions that are too light turn a gross edge of a fraction of a point into a net loss, as Mesfin (2026b) document for intraday futures; and a model that matches the baseline carries its own operational cost, so equal performance should usually mean rejection.

14.6 Platforms, surveys and the research loop

The last group is the infrastructure and the map. Yang et al. (2020a) describe Qlib, a platform that organises data, models and backtests for AI-driven quantitative research, and Yang et al. (2026c) describe FinRL-X, a modular architecture in which backtest and broker execution share a weight-centric interface so that research and live semantics are the same. Both address the research-automation stage and indirectly the leakage problem, because one code path from data to weights leaves fewer places for the backtest to differ from live trading. Shah et al. (2026a) and Chen and Singh (2025) look at the market side: the first is a narrative review of how AI-driven trading alters liquidity, volatility and price discovery, and the second combines a survey of high-frequency professionals with case studies, reporting that risk management and data quality drive algorithm choice. These two are peripheral to validation but carry a warning that matters for it: if many agents learn from the same data, historical impact assumptions may not hold. Barmish et al. (2024) offer a control-theoretic tutorial on order types, feedback, Kelly betting and the limit order book, a vocabulary through which control engineers can enter the field.

The surveys help locate this chapter in the literature. Horobet et al. (2024) map 4552 papers by science-mapping methods and note limited collaboration across scholars, which is the silo problem of this atlas in bibliometric form. Cavalcante et al. (2016) review computational-intelligence methods across financial applications, Dakalbab et al. (2024) systematically review 143 AI-in-trading papers and observe inconsistent estimation and performance metrics, Jukl and Lansky (2025) follow PRISMA to review 208 algorithmic-trading papers, and Rahaman (2026) give a graduate-level review of statistical methods. Ţiţan (2015) review the efficient market hypothesis literature and conclude that efficiency is hard to test when conditions change, a reminder that predictability may be transient. Finally, Li (2026) state five axioms for a quantitative investment system, including that whatever predictability exists is slight, and argue

that declared quantities such as a signal ceiling and an invariance defect bound what past states can yield. Such a declaration is a formal version of the pre-registration that several empirical papers above practise.

14.7 Bridges out of this chapter

Bridge. To time-series mining (Chapter 5) and return forecasting (Chapter 8). What the classification community calls *label noise* and *label correction* (for example in Yang et al., 2023) is the same problem that the forecasting chapters meet as *target choice*: a rank, a sign or a horizon. Conversely, what return-forecasting calls an "out-of-sample R^2" difference between models is what this chapter's statisticians treat as a hypothesis test requiring an effective sample size (Diao, 2026; Hansen, 2005). We propose a shared checklist: every model paper states its label, its label horizon and overlap, its baseline, and its number of trials.

Bridge. To volatility and derivatives (Chapter 11) and microstructure (Chapter 12). A "Greeks-aware profitability label" (Vaghela, 2026) is, in derivatives language, a label built from the position's own scenario P&L; the same idea would translate to microstructure as a label built from the realised cost of an execution schedule. What the label community calls an *event* sampled by a CUSUM filter (Grądzki et al., 2025) is what microstructure researchers would recognise as an information-driven clock, and the extreme-move targets of Zaman and Sahed (2026) are what a volatility researcher would call a clustering regime. A natural use is to share event definitions between the label pipeline and the execution model.

Bridge. To portfolio construction (Chapter 13) and strategies (Chapter 10). Meta-labelling's "side versus size" split (Meyer et al., 2022) is the portfolio chapter's separation of an alpha forecast from a sizing rule, and the finding that label gains may not survive conversion to a portfolio (Lima, 2026) is the decision-focused learning argument in a different jargon: train on the decision's payoff, not on a proxy. In the other direction, the strategy chapter's many reported Sharpe ratios can be read as an unfiltered sample of winners; the deflation and probability-of-overfitting tools here (Bailey and López De Prado, 2019; Bailey et al., 2013) are how a reader or a fund converts them into evidence. Chapter 15 makes the matching request for agent papers.

Chapter 15

LLMs, Text and Agentic Research

Three communities share this chapter and rarely read one another. The natural-language-processing and LLM community speaks of prompts, agents, tool use, memory, chain-of-thought, fine-tuning, reinforcement fine-tuning and benchmarks. The quantitative-investment community, which has produced most of the applied work here, speaks of alpha factors, information coefficients, alpha decay and formulaic alphas. A third, smaller group comes from security and software engineering and speaks of attack surfaces, poisoning, verification and reproducibility. Scattered among them are finance-economics authors who test whether a language model can forecast returns, and time-series people who ask how a numeric series should be fed to a model that reads tokens.

The translation into the shared vocabulary of the atlas is short. An "LLM agent" is a program that proposes hypotheses, code or orders, calls tools such as a backtester, and reads the result. In the atlas's terms it is a search procedure over candidate features and signals, wrapped around a validation oracle. "Alpha mining" is feature and signal generation under multiple testing. "Alpha decay" is regime dependence and crowding. A "benchmark for trading agents" is a backtest protocol, so its failure modes are those of Chapter 14: look-ahead leakage, survivorship, data snooping and costs ignored. "Sentiment" is a text-derived feature that must be aligned in time with the price series it is meant to predict.

In the pipeline (data → features → signal → regime and risk → portfolio → execution → validation → research automation) this chapter touches every stage, but its centre of mass is at the two ends. At the front it turns unstructured text into features. At the back it automates the research loop itself, and then asks whether the automated loop can be trusted. The middle stages (portfolio, execution, risk) are thinly covered and mostly appear as roles inside agent frameworks.

15.1 Text as data: from news to features

The oldest idea in the chapter is that text carries information that prices absorb with delay. Lopez-Lira and Tang (2026) test this directly with post-knowledge-cutoff headlines. They report that GPT-4 scores predict subsequent drift, especially for small stocks and negative news, and that forecasting ability grows with model size. They also report that strategy returns decline as LLM adoption rises, and they build a model with information-processing constraints, underreaction and limits to arbitrage to rationalise this. For a desk the last finding matters most: a language-model news signal is a crowding-sensitive signal with a visible half-life.

Earlier and smaller-scale work establishes the same pattern with older tools. Kennis (2018) relate Bitcoin news and online discourse to price and volume movements, and Burchi and Regni (2026) compare BERTweet, RoBERTa and FinBERT on tweets from 25 large-cap stocks, aggregate daily sentiment into trading signals, and compare strategies by performance rather than classification accuracy. That shift from classifier metrics to trading metrics is the right one, and it is where this literature meets the cost accounting of Chapter 10. Zhang et al. (2025d)

follow a similar route with a business-oriented sentiment labelling scheme for trend prediction, and Ntale (2026) compare five general and finance-specialised LLMs on technical market analysis. Taken together these papers differ in the model used for scoring, not in the idea: a sentiment or event score is computed per document, aggregated per asset and day, and lagged before use.

A second group builds richer structure from text. Mitra et al. (2026) introduce NEIC, a lightweight transformer that classifies Indian financial news into events, aimed at stock price prediction with small models. This suits desks that cannot send every headline to a frontier API. Li and Sanna Passino (2024) go further and use a fine-tuned LLM to generate a dynamic knowledge graph, FinDKG, from financial news, with an attention-based graph network on top for thematic trend detection. A natural use is to treat the graph's evolving edges as a slow-moving feature for sector rotation, and its entity links as a path for thinking about contagion, which links to Chapter 7.

Multimodal predictors combine text with numeric series. Du et al. (2026) combine an LLM with transformer networks for multimodal stock movement prediction, and Wu et al. (2026b) present TextGrad-Trader, which arbitrates among multimodal signals with attention to risk and uncertainty. These sit at the border with the deep-learning forecasting literature of Chapter 8, and the usual question applies: does the text channel add anything after the price channel has been modelled? A paper that reports a gain over a text-free baseline answers that question, and one that does not leaves it open.

The reverse problem, getting numbers into a text model, receives less attention but has a direct cost. Bhagat and Mudbari (2026) propose TSLN, a serialisation format using a schema-first layout, relative timestamps and differential encoding. They report a 68–73% token reduction against JSON on cryptocurrency, stock and IoT data while remaining lossless, and estimate annual API savings above $18,000 for a million records a day. Whatever one thinks of that dollar estimate, the idea is the familiar one from time-series compression, and it matters once an agent reads price tables many thousand times a day.

Desk note. News-event feature with decay control. Stage: data and features. Input: timestamped headlines and filings, with a vendor timestamp and an ingestion timestamp. Output: a per-asset, per-day score (event class times model-scored polarity), lagged at least one bar and neutralised against sector and size. A natural design, following the drift and crowding findings above, is to monitor the score's information coefficient by month and to cut gross exposure when it decays. What can go wrong: the LLM may have seen the outcome during pre-training, so the score must be tested on text after the model's cutoff; small-cap, negative-news drift is expensive to trade, so costs and borrow must be charged before judging the signal; and capacity shrinks as more desks run the same model.

15.2 Zero-shot forecasting and LLM reasoning for trading

A different line uses the model itself as the forecaster. Chen and Pu (2026) deploy an LLM daily on Russell 1000 stocks from April 2025, letting it search the web and synthesise a view without curated text. Because predictions are collected at the live edge, they are free of look-ahead by construction. They report that the model identifies top winners but not the rest of the ranking. A desk should read that as a long-only tilt candidate, not a full cross-sectional ranker. Cui et al. (2026) attack a related problem from the training side. Their Hindsight Preference Optimization trains a language model for time-series advisory, where the quality of an answer depends on an outcome unknown at prediction time. Xiao et al. (2025a) train Trading-R1 with reinforcement learning to produce structured, interpretable investment reasoning. Luo et al. (2026b) address a narrower and more verifiable task: translating a natural-language trading intent into an executable option strategy over option-chain data, which links to Chapter 11.

These papers share a theme with the formulaic work below: replace an opaque numeric forecaster with a model whose output can be read by a human. They differ in what is trusted.

In the live-nowcasting design, nothing is trusted but the timestamp. In the reinforcement-trained designs the reward definition carries the whole burden, and reward hacking against a backtest is the obvious failure.

15.3 Alpha mining with LLM agents

By volume this is the chapter's core. The problem is old: search a combinatorial space of formulas over price, volume and fundamentals for predictive factors, then keep those that survive. Genetic programming and reinforcement learning were the incumbents. The LLM idea is to use language-model priors as the proposal distribution, so the search is guided by economic meaning rather than random mutation.

15.3.1 Single-loop generation and refinement

The simplest designs wrap a model in a generate–backtest–refine loop. Wang et al. (2025e) present Alpha-GPT, in which a human steers an LLM through natural-language ideas, putting a human inside the loop. Cao (2025) propose Chain-of-Alpha, using chained model calls for generation and refinement. Li et al. (2024b) combine symbolic and neural elements in a neural-symbolic factor mining agent aimed at interpretability, and Yu et al. (2026d) describe a hybrid approach to formulaic alphas with LLM assistance. Vu et al. (2026) add a knowledge prior: a WriterAgent, a JudgeAgent and a backtest engine in nested loops, seeded with the WorldQuant 101 formulaic alphas and tested on Vietnamese stock data, with the stated aim of narrowing the search space and reducing invalid code. Duan et al. (2025b) instead use multi-agent debate for interpretable stock factors, and Weng et al. (2026) organise generation around an explicit market logic, the mechanism meant to explain why a factor should persist.

15.3.2 Search structure: trees, evolution, schemas

A second family makes the search explicit rather than leaving it to the agent. Shi et al. (2025a) combine an LLM with Monte Carlo tree search for formulaic factors. Luo et al. (2026a) use the language model as a semantically informed mutation operator in an evolutionary search, and Tang et al. (2026b) train an agent with reinforcement learning to improve how it evolves alphas. Yi et al. (2026) argue that search needs an explicit space of trading semantics rather than delegating navigation to the agent. Shi et al. (2026) combine LLMs with constraints for safe, diverse and reproducible discovery, noting that unconstrained program generation is operationally unsafe. Ye et al. (2026b) treat the evaluation budget as something to allocate, and keep a record of how each candidate was produced. Neighbouring ideas appear in strategy-level evolution. Sharma and Shroff (2026) evolve whole trading programs with an LLM as mutation operator, and Yun et al. (2025) frame strategy discovery as multi-agent evolution aimed at preserving diversity across market conditions.

15.3.3 Learning from the search history

A third family asks the agent to learn. Zhang et al. (2026c) update the generator's policy by reinforcement fine-tuning on evaluation feedback, and Huang et al. (2026a) align a model with the literature for symbolic factor construction through policy learning. Wang et al. (2026g) give an agent skills and experience memory against redundancy as the factor library grows, and Yu et al. (2026e) store structured search-process memory while warning that naively reusing past successes risks overfitting. Wang et al. (2026e) post-train local models, motivated by cost, availability and confidentiality, and to avoid research path collapse onto a few mechanisms. Zhao et al. (2026a) add multi-objective reinforcement learning for alphas that work well together, a portfolio-aware objective.

15.3.4 Discipline: decay, overfitting and testing

The most useful papers here treat the failure modes as design targets. Tang et al. (2025) observe that LLM-driven mining tends to produce homogeneous factors that worsen crowding and decay, and counter with originality enforcement via abstract-syntax-tree similarity against existing alphas, plus hypothesis–factor alignment. Kim et al. (2026) point out that automated approaches optimise directly for returns and rarely check whether a factor has an economic rationale, and add empirical validation. Huang and Fan (2026) report a closed loop with out-of-sample validation and economic-rationale requirements to mitigate data snooping, and quote a long-short Sharpe ratio of 3.11 on U.S. equities. A skeptical reader will weigh that number against the number of trials the loop ran, a point we return to in the validation section. Xia and Wang (2026) supply theory: a hybrid of structured local search and LLM-generated non-local proposals helps only under three geometric conditions, including residual signal alignment with the target, and novelty alone does not raise yield. That is a useful guard against the assumption that more diverse proposals mean more alpha.

Other papers push mining toward fundamentals and whole workflows. Koa et al. (2026) use self-evolving multi-agent symbolic discovery for financial valuation, Kou et al. (2025) describe a risk-aware three-stage multi-agent system for strategy finding, and Li et al. (2025e) jointly optimise data-centric factors and models in R&D-Agent-Quant. Guo et al. (2026a) study recursive self-improvement across experiments with AQuA, and Yuan et al. (2026) carry the workflow through cross-sectional signals toward portfolio construction in AlphaCrafter.

Desk note. Agentic factor factory with a human gate. Stage: signal generation and research automation. Input: a panel of daily bars and fundamentals, a library of accepted factors, and a point-in-time data layer. Output: candidate formulas with a stated hypothesis, a similarity score to the library and out-of-sample statistics. A workable design combines ideas above: AST-similarity and redundancy filters, a prior seeded by known formulaic alphas, and a hard count of every candidate ever tested so a deflated Sharpe or multiple-testing correction can be applied downstream (Chapter 14). A human approves anything that enters production. What can go wrong: the trial count is lost when memory modules discard failures; the agent's pre-training exposes it to well-known factors, so "novel" alphas may be recycled; and in-sample IC from a loop that re-reads its own evaluation is optimistic by construction.

15.4 Trading agents: roles, consensus and self-improvement

Alongside factor mining sits a literature that builds the whole desk out of agents. Xiao et al. (2025b) present TradingAgents, a multi-agent framework with specialised roles including analysts, researchers, a trader and risk management. Zhao et al. (2025a) apply a role-based multi-agent design to stock selection and portfolio construction, evaluated against benchmarks under several levels of risk tolerance. Miyazaki et al. (2026) argue that role prompts mimicking analyst and manager are too abstract and propose fine-grained task decomposition, with a data-leakage concern in view. Li et al. (2026a) diagnose "uniform trust", the habit of treating all retrieved sources as equally informative, and propose selective consensus to reduce decision uncertainty. Chen et al. (2026) constrain self-improvement to a human-auditable rubric rather than free-form prompt rewriting, and Mao et al. (2026a) use a verifier to guard strategy revisions against hallucinated changes. Guo et al. (2026c) treat the evolution of market modes as a modelling problem sitting between asset-centric and market-centric views, which relates to the regime material in Chapter 7. Chavan (2025) come from the microstructure side and propose a high-frequency architecture with three parallel pipelines (news LLM, order-book graph encoder with reinforcement learning, technical strategy agent), which is worth reading for its time-based scheduling and confidence-scaled capital allocation across pipelines rather than as evidence of performance (its reported daily alpha comes from a backtest on NIFTY 200 minute data).

The system-level picture is supplied by two reviews. Hua et al. (2026) survey agentic quantitative trading across factor mining, signal discovery, portfolio construction, execution and risk management. Gong (2026) lay out an architecture for AI agents in financial markets and examine its systemic implications. An engineering counterpoint is Ferrante (2026), whose AgenTrade uses a language model only where the task is genuinely open-ended reasoning over ambiguous input. That principle, selective use, also appears in the cost argument for compact formats and small models earlier in the chapter, and is the most transferable design rule in this section.

Desk note. LLM as analyst, not trader. Stage: signal and risk filter. Input: news, filings and a structured summary of current positions. Output: a veto or a size-down flag on proposed trades, plus a one-paragraph rationale logged for the risk team. Following the selective-use and rubric ideas above, the model never sets size and never sends orders; deterministic code does. What can go wrong: the model's rationale reads as convincing but may not reflect what drove the flag; a manipulated feed could trigger vetoes at the worst moment (see the security section); and the filter's value must be measured against a rule-based filter, not against no filter.

15.5 Attacks, contagion and reliability

Once agents read public feeds and talk to each other, they inherit an attack surface that numeric models do not have. Na et al. (2026) decompose a trading pipeline into Analyst, Researcher, Trader and Risk Manager roles and study how an adversarial signal entering through source data and prompts propagates toward a decision and a realised loss. Sua et al. (2026) study black-box, input-only attacks via admissible social-media feeds and how adversarial signals spread among agents. Yan et al. (2025) propose TradeTrap to test whether trading agents remain reliable and faithful under adversarial or faulty conditions. Wang and Saxena (2026) survey robustness and security failures in published LLM trading schemes, and Mao et al. (2026b) widen the lens to autonomous agents in commerce that hold and move digital assets. In the vocabulary of the atlas these are input-validation, stress-testing and model-risk questions, and the language of "contagion" here is the network contagion of Chapter 7 applied to message passing instead of balance sheets.

15.6 Evaluating LLM agents without fooling ourselves

The evaluation papers are where this chapter converges with classical backtesting. The core threat is look-ahead via memory: a model trained on data covering the backtest window can recall prices and narratives rather than reason. Zhu et al. (2026c) address this with a data-side masking protocol that anonymises tickers and dates, and a Barra-style attribution that separates market, style and selection returns. Cai et al. (2026) provide an auditable point-in-time framework in response to look-ahead leakage, optimistic execution and risk mandates that are described but not enforced. Roy and Roy (2026) attack contamination at the signal level, combining membership-inference features with cross-model disagreement that exploits different training cutoffs, as a filter that costs nothing at retraining. That filter could serve as a standard validation check on any LLM-derived signal.

Benchmarks differ in what they stress. Chen et al. (2025b) and Yu et al. (2025) evaluate agents on real-world or live markets, Qu and Chen (2026) propose a closed-loop, cost-aware benchmark that diagnoses process as well as returns, Zhang et al. (2026j) stress the behavioural instability of agents across repeated runs, and Luo et al. (2026c) extend to options with nonlinear payoffs, linking to Chapter 11. On the code side, Khoroshilov et al. (2026) benchmark whether LLMs write executable strategies, and Wang et al. (2026d) test whether the generated code behaves as the described strategy, since a run can plot a clean backtest while the risk logic is

silently wrong. Hall et al. (2026) verify numeric claims drawn from filings, which is a data-quality gate for any text-to-number pipeline.

The sceptical evidence is the most informative. Li et al. (2026b) backtest LLM timing strategies over two decades and 100+ symbols and report that previously reported advantages deteriorate under the broader cross-section and longer period, with strategies too conservative in bull markets and too aggressive in bear markets. Barton et al. (2026) report a six-month production record from two fleets, finding among other things that the operating layer, such as a risk slider, shapes behaviour more than strategy text and that sizing is blind to volatility. Both are measurements of what agents do, which is a different object from what papers claim they can do.

Desk note. Leakage audit for LLM signals. Stage: validation. Input: the signal history from any LLM-based feature, the model's documented training cutoff, and a second model with a different cutoff. Output: a pass/fail flag per signal period, based on whether the signal's predictive power is concentrated before the cutoff or disagrees across models, in the spirit of membership-inference and cross-model checks. Pair it with masked-identifier replays and factor attribution. What can go wrong: cutoffs are only approximately known; masking removes information the real desk would have; and passing the audit does not remove ordinary overfitting or cost problems.

15.7 Research automation as infrastructure

The last stage of the pipeline is the research process itself. Agent Team et al. (2026) present Mint-Agent, finance-native agentic foundation models built to be reliable on grounded evidence and sustain auditable long-horizon research. Li et al. (2026c) replace free-form messages between mining and evaluation agents with structured, replayable transport, a contract that makes a hand-off verifiable. Zhang et al. (2026i) constrain agent self-evolution with finite state machines, from the general deep-research literature, and Zimmer et al. (2026) offer a practical guide to AI-assisted research in mathematics and machine learning. The last two are peripheral to trading but share a message with everything above: the useful agent is the one whose steps are logged, bounded and replayable.

15.8 Bridges out of this chapter

Bridge. Alpha mining and the multiple-testing literature (Chapter 14). The LLM community says an agent "explores" a "search space" and "accumulates experience". The backtesting and statistics community says the researcher runs many trials and the best one is biased upward. These are the same fact. Every memory module, tree search and evolutionary loop above should report its number of trials so that selection bias can be corrected. Conversely, statisticians can supply the stopping rule and deflation that agent builders are missing.

Bridge. Factor libraries and the factors chapter (Chapter 9). What the agents call "redundancy" or "research path collapse" is what the factor-zoo literature calls spanning and crowding, and what the similarity-search community calls near-duplicate retrieval. AST similarity and orthogonality tests for a candidate formula are the same operation as asking whether a new anomaly adds anything beyond existing factors. Economists can supply the economic-rationale criterion that several papers here try to enforce by prompt.

Bridge. Contagion and regimes (Chapter 7). Security researchers speak of an adversarial signal "propagating" through agent roles. Network and systemic-risk researchers speak of shock transmission through links. A knowledge graph of entities from news and a graph of agent messages are both networks on which one can ask who is exposed. In the other direction, change-point and regime detectors from that chapter could serve as an automatic circuit breaker for agents whose behaviour was tuned in one regime, which is the failure the long-horizon evaluations describe.

Bridge. Numeric series as tokens (Chapter 6). The serialisation and "token efficiency" work treats a price series as text to be compressed. The representation-learning community treats it as a sequence to be embedded. The two meet where an LLM reads a series: whether a lossless differential encoding or a learned embedding gives the model what it needs is an open question that neither side currently tests against the other.

Chapter 16

General Methods and Cross-Domain Tools

Four communities speak in this chapter, and none of them is a finance community. Statisticians talk about outliers, change points, bandwidth-free inference and misspecified Bayesian updating. Deep-learning researchers talk about noisy labels, sample weighting, domain-invariant representations, flow-matching and tabular foundation models. Time-series-mining people talk about pattern mining, symbolic regression and instance spaces. Language-model and agent researchers talk about harnesses, skills, evolutionary coding agents and verifiable data. A few papers come from further away: remote sensing, survey design, microservice operations, quantum benchmarking and the mathematics of rough paths.

The atlas needs a common vocabulary to make these usable. A statistician's *outlier* is a bad tick or a stress print, that is, an *anomaly* in the shared concept index. A *change point* or *distribution shift* is a *regime shift*. A *noisy label* is a flaw in *label design*: the triple-barrier or forward-return target never equals the quantity one wished to predict. *Hyperparameter-free inference* and *Bayesian updating under misspecification* are two ways of asking how much a performance number can be trusted, which is *uncertainty quantification* and *forecast evaluation*. *Symbolic regression* and *evolutionary coding agents* are two routes to *alpha mining*. An *agent harness* is the research-automation layer.

Within the pipeline of data, features, signal, regime and risk, portfolio, execution, validation and research automation, this chapter sits at the two ends and in the middle. It supplies data hygiene and feature selection at the front, regime detectors and generic forecasters in the middle, and validation and automation at the back. Most papers were written for other applications, and the chapter treats them as tools to be tested. A tool is only as good as the assumptions it makes about the data, so for each group we state what the assumption is and whether markets are likely to honour it.

16.1 Cleaning the inputs: anomalies, noisy labels and dependence

Every desk has bad data. Li et al. (2020) propose COPOD, an outlier detector that builds an empirical copula and reads off tail probabilities, producing something like an anomalous p-value for each record. It has no parameters to tune and is interpretable, which suits an unattended data-quality gate on trades and quotes. Its limit is that a genuine stress event is also a tail event, so such a gate should flag and route to review, not delete. Kim and Hwang (2025) address a neighbouring problem, directional dependence in sparse multivariate longitudinal data contaminated by measurement error, using functional principal components and copula imputation. Their setting is not market data, but irregular, noisy panels are common

in alternative data, and the idea of imputing through a copula, not by carrying forward, is worth testing.

A second kind of noise lives in the targets. Several deep-learning and statistics papers treat noisy labels as a first-class problem. Kim et al. (2021) detect noisy samples with a noise-cleansing method, and Shu et al. (2022) learn a class-aware weighting of training samples, replacing hand-set schemes. Hu et al. (2025) argue the opposite of the usual intuition for imbalanced data: adding suitable label noise can adjust biased decision boundaries. The theory side is van Rooyen et al. (2015), who revisit learning under symmetric label noise and show why an unhinged loss remains useful where convex potentials can fail. A natural use is the following proposal. Labels such as "next-day direction" or "barrier hit first" are noisy by construction, and event labels are imbalanced, so sample weights and noise-robust losses are candidate tools for label design. None of these papers tests market labels, and a desk must check that the noise model (often class-conditional and independent of features) resembles the way trading labels go wrong, namely through overlapping horizons and regime-dependent noise.

Representation and selection come next. McInnes et al. (2018) give UMAP, a manifold learner that serves both visualisation and general dimension reduction; it could be used to map the state space of a feature set and spot clusters of days that behave alike. Lu and Osterrieder (2022) select features with an intervened interpolative decomposition that gives each column its own priority, and apply it to diversifying quantitative strategies; it is one of the few papers here with a trading application in its title. Nakagawa et al. (2016) prune the search for predictive patterns safely, so a superset of useful patterns is found without enumerating them all, which is relevant wherever candidate rules are combinatorial.

Several peripheral papers show how far outside finance the data stage can reach. Biswas et al. (2026) show that off-the-shelf biomedical encoders give high cosine similarity to unrelated concept pairs, a warning for any pipeline that links assets or news through embedding similarity. Buehler (2024) turn a corpus of papers into an ontological knowledge graph and analyse its degrees and communities, which is a template for a graph over the research library itself. A et al. (2026) estimate material outflows from satellite imagery and city master plans, an example of the alternative-data pattern in which an imagery signal is converted into a physical quantity that might later be mapped to a sector. Schuster et al. (2024) survey best-worst scaling as a prioritisation tool; it could be used to rank candidate research ideas by pairwise judgement. Du et al. (2025) synthesise verifiable data for language-model training, a reminder that synthetic data is only valuable when something can verify it (see Chapter 3).

Desk note. Data-quality gate (data stage). Input: tick, quote and bar streams plus vendor corporate-action tables. Output: a flag and a tail p-value per record, produced by a COPOD-style detector (Li et al., 2020) run separately per instrument and per time-of-day bucket. Records above a flag threshold are quarantined and shown to a human, not dropped. What can go wrong: stress periods are legitimately extreme, so a fixed threshold removes the very observations a risk model needs; auction prints and halts need their own calendars; and a detector refitted on contaminated history learns the contamination.

16.2 Detecting change: change points and regime shifts

The largest cluster in the chapter is change-point detection, and it is the clearest case of a statistics idea that a trading desk already half-owns. The common model is that a series is piecewise stationary and the task is to say when the generating parameters jump. Habibi (2022) and Huang (2025) apply Bayesian online change-point detection (BOCPD) to financial series. The latter builds four simulated scenarios (mean drift, variance drift, joint drift and a t-distributed mean drift) and compares BOCPD against generalised-likelihood and Kolmogorov–Smirnov tests. Musaev et al. (2024) survey adaptive algorithms for non-stationary, chaotic financial series,

arguing that stationary-model methods are ill-suited. These three share a theme: the online setting, where a run-length posterior is updated bar by bar, is the one that matters for trading, because an offline segmentation sees the future.

The remaining papers widen the data type. Sundararajan and Pourahmadi (2018) test multivariate piecewise stationary series nonparametrically through differences in spectral density matrices, which makes the method sensitive to changes in dependence structure and not only in level. Chen and Weichi (2025) model collective anomalies and change points jointly in fine-grained series with a recursive online algorithm, a distinction that matters because a burst of ten abnormal bars is not a regime change. Luo et al. (2026d) treat online change detection for multivariate inhomogeneous Poisson point processes with low-rank intensities; trades and quote updates are point processes, so this is a candidate detector for shifts in order-arrival intensity (Chapter 12). Pham et al. (2024) use multivariate BOCPD for root-cause analysis of microservice failures. Our proposal is to point the same method at the research infrastructure itself: feed latency, fill rate and data-vendor lag are multivariate series in which a change point is an incident.

Detection is only half of a regime system, because something must act on it. Li et al. (2025d) cluster markets into ten volatility states with k-means and evaluate equal-weight, minimum-variance, maximum-diversification and equal-risk-contribution allocations under those states. The theory papers say why such switching is hard. Zhao et al. (2019) analyse learning invariant representations for domain adaptation, and an invariant representation across regimes is what a strategy would want, but their analysis concerns when invariance alone suffices. Kalan et al. (2025) study Neyman–Pearson classification when both the null and alternative distributions shift, which maps naturally onto a detector whose false-alarm rate must be capped while the world changes.

Desk note. Regime and risk filter (regime stage). Input: daily realised volatility, cross-asset correlation and trade-arrival counts. Output: an online change-point posterior from a BOCPD variant (Habibi, 2022; Huang, 2025) that scales down gross exposure when the probability of a recent break exceeds a limit set from a false-alarm budget in the Neyman–Pearson spirit of Kalan et al. (2025). What can go wrong: detection delay means the filter fires after the loss; hazard-rate and prior choices change alarm timing and must be tuned on a window the strategy research never touched; and turnover from flickering states can cost more than the drawdown avoided.

16.3 Generic forecasters: tabular foundation models, trees and interpretable rules

The forecast stage in this chapter is populated by models built for general tabular data. Jäger et al. (2026) introduce TabPFN-3.5, a tabular foundation model that reports state-of-the-art results on TabArena and extends to non-i.i.d. data with temporal or grouped splits. Cho et al. (2026) present Causilo, which reports a performance-efficiency Pareto frontier with fast inference, and Zhang et al. (2026f) present LimiX-2, a structured-data model built around contextual mechanism networks and masked pretraining. The shared idea is in-context learning: the model is not fitted per task but conditions on a labelled table at inference. For a desk, a natural use is as a fast baseline for cross-sectional return or fill-probability prediction. The traps are specific to finance. Benchmark tables are close to i.i.d., markets are not; pretraining corpora may contain the evaluation period; and in-context rows are a leakage route if they include future information. Chapter 8 shows how return forecasters are judged.

Tree ensembles remain the usual comparison. Loecher et al. (2026) study hierarchical shrinkage, a post-hoc regularisation that shrinks node predictions toward ancestral means, and contrast it with the implicit regularisation of random forests. Low-signal financial data punishes variance, so shrinkage is a cheap regulariser to test. For interpretability, Lumadjeng et al. (2026)

start from a classifier written as a single equation and simplify it step by step into readable rules for financial decision making, motivated by the idea that a model that cannot be explained cannot be deployed in a regulated domain. Rezaei and Krishnan (2026) use an LLM to infer mechanisms from model residuals on tabular data, which could serve as a hypothesis generator for why a forecaster fails in particular regimes.

Other forecasting papers are placed here for their idea, not for market evidence. Hwang and Zohren (2025) fuse feature extraction and decision making in a signature-informed transformer for asset allocation, arguing that separating forecasting from optimisation creates a mismatch (Chapter 13). Zheng et al. (2023) add classifier-free guidance to flow-matching models for generation and decision making; conditional generation is a possible engine for scenario simulation. Li et al. (2024a) study when diffusion models generalise rather than memorise, which is exactly the question for a synthetic market generator. Ma et al. (2027) reason over temporal knowledge graphs to predict future events and Xu et al. (2026b) distil hypergraph networks into cheaper students; both are candidates for event and relation graphs between firms, though neither is evaluated on them.

16.4 Discovering formulas: symbolic regression and alpha mining

Symbolic regression searches for a short closed-form expression instead of a black-box fit. That is attractive in trading because a formula can be audited, costed and monitored. Cranmer (2023) describe PySR, a library built on a distributed evolutionary search; Cranmer et al. (2020) distil symbolic representations from graph neural networks by encouraging sparse latent representations and then applying symbolic regression. d'Ascoli et al. (2022) train transformers to infer the recurrence behind integer or float sequences, and Saegert and Köthe (2026) identify a simplification bottleneck in amortised neural symbolic regression, which is the fast-but-unscalable alternative to genetic programming. Basiri and Nicholson (2026) generate basis functions automatically so that sparse identification of dynamics does not depend on the researcher guessing the library in advance.

In the shared vocabulary this is alpha mining, and the warning from Chapter 14 applies in full: a search over expression space is a search over a huge number of hypotheses. Each paper here assumes noise-free or lightly noisy data from physics or mathematics, where a correct formula exists. Prices have no such formula, so a symbolic regressor will find an expression either way. The correct use is to treat any discovered formula as one candidate in a family whose size is recorded and corrected for. Nezhad et al. (2025) and Liu and Li (2026) come from LLM reasoning: the first generates verifiable code in place of prose to make mathematical reasoning checkable, the second proposes an algebraic semantics for traceable reasoning. Both support a design rule for automated research, that every generated signal should arrive as executable code with a deterministic check.

Desk note. Formula-search lane (research automation and signal). Input: a feature panel, a library of operators, and a fixed cost model. Output: candidate expressions from a PySR-style search (Cranmer, 2023), each stored with its complexity, its search budget and its out-of-sample record. What can go wrong: the number of trials is the multiple-testing count and must be logged; expressions that look simple can hide high turnover; and formulas fit in one regime may fail silently in the next, so each should pass the regime check above before capital is assigned.

16.5 Validation: inference, attribution and testing the testers

The statistics papers in the validation stage ask how sure we can be of a number. Zhang and Shao (2026b) test hypotheses on a functional parameter with self-normalisation, and Zhang

and Shao (2026a) give a sample-splitting route to bandwidth-free inference. Both exist because classical block bootstraps and HAC estimators depend on a bandwidth whose choice can change conclusions, and serially dependent returns make that sensitivity a daily problem for Sharpe-ratio comparisons. Shalizi (2009) analyse Bayesian updating with dependent data and misspecified models; the lesson for a desk is that a posterior over strategy quality can be confidently wrong when the model is wrong. Kotłowski et al. (2024) give an online algorithm for optimising complex performance metrics such as F-measure that are functions of a confusion matrix, which matters when a signal's value is in rare, high-precision calls.

Attribution has its own pitfalls. Biparva and Materassi (2024) reweight Shapley values with a maximum-entropy approach so that information about variable ordering and causal structure enters the attribution. Méloux et al. (2025) warn, through the dead-salmon story, that interpretability analyses can find structure that is not there. A proposal: any feature-importance table in a research note should be accompanied by a null run on shuffled or random inputs.

A different question is whether a backtest suite covers enough kinds of market. Instance Space Analysis represents test instances as feature vectors and visualises where algorithms succeed. Smith-Miles and Muñoz (2023) give the methodology and tools, Muñoz et al. (2021) apply it to regression problems, and Güzel et al. (2025) provide the Python package. A natural use is to characterise each backtest window (volatility, trend, liquidity) as a feature vector, then map which strategies win where and whether the test set was diverse or merely long. Koch et al. (2026) do similar benchmarking for quantum optimisation on combinatorial problems, a niche relevant to portfolio selection only if quantum hardware becomes practical. Najarro et al. (2025) solve inverse problems in stochastic self-organising systems with invariant representations, a pattern that could recover simulator parameters for a synthetic-market model from observed statistics.

Desk note. Validation check (validation stage). Input: out-of-sample daily P&L of a candidate and of its siblings in the same search. Output: a confidence interval for the Sharpe difference from a bandwidth-free test in the style of Zhang and Shao (2026b) or Zhang and Shao (2026a), plus an Instance-Space map of the windows where the strategy wins (Smith-Miles and Muñoz, 2023). What can go wrong: asymptotic tests need long samples and daily strategies may have few independent episodes; the map is only as informative as the window features chosen; and passing the check says nothing about capacity or costs.

16.6 Automating the research loop

The last group is about agents. Novikov et al. (2025) present AlphaEvolve, an evolutionary coding agent in which LLMs propose code changes that are scored and selected, applied to open scientific problems and infrastructure optimisation. Replace the scientific objective with a backtest objective and it becomes an alpha-mining loop, with the overfitting hazards above made worse by an automated, tireless proposer. Zhang et al. (2025c) survey the combination of metaheuristics with machine learning, giving the vocabulary for hyperparameter and portfolio search. Foster et al. (2026) build preference models that predict which candidate experiments are worth their GPU cost, an idea that carries over directly to ranking trading ideas before spending compute on full backtests.

Several papers concern the scaffolding around agents. Seong et al. (2026) study building agent harnesses, Badhe et al. (2026) manage long-horizon agent skills with explicit state in place of an ever-growing transcript, Liang et al. (2026) propose creating, evaluating and connecting reusable skills, and Zhu et al. (2026b) propose a training-free multi-agent framework for recursive self-improvement through memory. For a semi-automated fund the interest is operational: repeatable research routines should be stored as versioned skills with tests, and every automated run should leave an audit trail. Terwiesch et al. (2023) find that LLMs generate product ideas that score well, yet the question of diversity remains; for idea generation, a pool of LLM-proposed alphas

may be less diverse than it looks, which would inflate apparent breadth. Chapter 15 treats these agents in the finance setting.

At the mathematical edge, Hairer (2014) gives regularity structures, an algebraic framework for describing irregular functions through local expansions, used to make sense of singular stochastic equations. We cannot claim a desk use; it is the theoretical sibling of path-signature features (see Hwang and Zohren, 2025) and of rough-volatility modelling (Chapter 11).

16.7 Bridges out of this chapter

Bridge. Change points ↔ regimes ↔ discords. What this chapter's statisticians call a "change point" (Huang, 2025; Sundararajan and Pourahmadi, 2018) is the *regime shift* of Chapter 7, and what time-series mining in Chapter 5 calls a "discord" is a subsequence that fails to match any other. A discord is a candidate change point, but a change point persists, whereas a discord can be a single event. Joint detection of both, as in Chen and Weichi (2025), is the bridge. Order-flow researchers speak of an "intensity regime", which Luo et al. (2026d) would call a change in a Poisson intensity matrix (Chapter 12).

Bridge. Label noise ↔ label design. Deep-learning researchers (Kim et al., 2021; Shu et al., 2022; van Rooyen et al., 2015) speak of corrupted labels and class-conditional noise; backtesting researchers in Chapter 14 speak of overlapping labels, triple-barrier targets and purging. These are two views of one fact, that the training target is a noisy proxy. The first group offers estimators for the noise, the second offers the temporal structure of the noise, and neither has used the other. Representation people in Chapter 6 meet the same issue as "what the encoder should be invariant to" (Zhao et al., 2019).

Bridge. Symbolic regression ↔ alpha mining ↔ multiple testing. Where the physics-inspired community says "discover the governing equation" (Basiri and Nicholson, 2026; Cranmer, 2023), the factor literature of Chapter 9 says "data-mined anomaly" and the strategy literature of Chapter 10 says "rule search". The same expression-space search sits under all three names, and the validation papers above (Zhang and Shao, 2026b; Shalizi, 2009) are the vocabulary for pricing the search. In the other direction, the LLM agents of Chapter 15 are search operators for this space, and Novikov et al. (2025) is the closest template.

Part VII

Synthesis

Chapter 17

Bridges Across Silos

17.1 Why bridges, and how to read this chapter

The topical chapters of this atlas were written one silo at a time. Their closing notes keep making the same observation: a concept that one community studies under one name is being studied by another under a different name, and, as far as the abstracts show, neither refers to the other. This chapter collects those observations and turns the strongest of them into proposals. Every bridge below has the same four parts: the communities and the mathematical object they share; what each side already holds, with citations; a research programme that exists only once the two sides are combined; and the stage of a semi-automated fund where it would land, the data it needs, and the way it most likely fails.

Two cautions apply throughout. First, the silo and pipeline-stage labels quoted in this chapter come from an automated classification of titles and abstracts, with at most three concept tags per paper. A count of zero papers carrying two tags therefore means that no paper was tagged with both, not that no such paper exists; we use such zeros as pointers, and they were the starting point for each bridge, not the end of an argument. Second, every research programme is our proposal. Where a cited paper supplies a component, we say which component; we do not claim that any cited paper has tested the combination.

Concept counts give the selection rule. We looked for concepts that are studied by three or more silos, and for concepts that are concentrated in one silo while being obviously useful to another. In the collection of 1065 papers, *motif* is tagged on 38 papers, 33 of them in time-series mining; *implied volatility* on 40, of which 30 are in the quantitative-derivatives silo; *order flow* on 63, of which 41 are in microstructure; *path signature* on 23, of which 17 are in mathematics; and *conformal* on only 5, four of them in statistics. Meanwhile *regime shift* (145 papers) and *forecast evaluation* (123) are spread over four or more silos each. The first group produces bridges that carry a tool across a wall; the second produces bridges that unify vocabulary. Table 17.1 lists all twelve, and the silo-by-stage matrix of Table 2.1 (Chapter 2) shows where the walls are.

Table 17.1: The twelve bridges. Silo names are the automated labels defined in Chapter 2 (tsml = time-series machine learning and data mining, dl = deep learning, quant-deriv = derivatives and volatility modelling, fin-econ = empirical finance and econometrics, nlp-llm = text, language models and agents).

Bridge	Silos joined	Shared object	Pipeline stage	Chapters
B1 Motifs and discords on implied-volatility paths	tsml, quant-deriv	recurring and anomalous subsequences of a surface or term-structure path	features, risk	5, 11
B2 Signatures of order-flow paths	math, microstructure	ordered iterated integrals of an event path	features, signal	6, 12
B3 Conformal intervals into sizing	stats, decision, dl	a calibrated interval or abstention rule	portfolio, risk	13, 8
B4 Change-point gates for factor sleeves	stats, tsml, fin-econ	a detected break as a regime label	regime, portfolio	7, 9
B5 Alpha-mining agents under trial accounting	nlp-llm, stats, practice	a search over formulas whose trial count matters	validation, automation	15, 14
B6 Generators as test benches for execution agents	dl, microstructure, decision	a simulated order book with stylised facts	execution, validation	3, 12
B7 Contagion measures as risk features	fin-econ, dl, stats	a time-varying dependence network	risk	7, 8
B8 Geometric stress indices with detection metrics	math, stats, fin-econ	a shape statistic whose level shift flags a break	regime, risk	6, 7
B9 Self-excitation and roughness	microstructure, quant-deriv, stats	a self-exciting intensity or power-law memory	risk, theory	12, 11
B10 Labels that carry cost and size	tsml, dl, quant-deriv, practice	the training target as a noisy proxy for P&L	signal, validation	14, 8
B11 Chart patterns as testable motifs	tsml, practice, stats	a pattern with a distance threshold and a null	signal, validation	10, 5
B12 Own-impact in decision-focused learning	decision, microstructure	endogenous uncertainty: our orders change the data	portfolio, execution	13, 12

17.2 B1. Motifs and discords on implied-volatility paths

Communities and shared object. Time-series mining and quantitative derivatives. The intraday implied-volatility surface, or a handful of its coordinates (at-the-money level at several maturities, a skew measure, a butterfly), is a multivariate series. The mining community calls its recurring stretches motifs and its never-repeating stretches discords; the derivatives community

calls them smile regimes and surface dislocations. In the automated tagging no paper carries both an implied-volatility tag and a motif or similarity-search tag, although 38 papers carry the motif tag, 50 the similarity-search tag and 40 the implied-volatility tag.

What each side has. The mining side has exact all-pairs similarity joins (Yeh et al., 2016b), variable-length motif discovery that removes the need to guess a window (Linardi et al., 2018), joint motif and discord discovery across lengths (Linardi et al., 2020a), multidimensional motifs restricted to informative dimensions (Yeh et al., 2017b), and admissible search with missing data (Zhu et al., 2021). Two papers already apply the machinery to financial series (Cartwright et al., 2021; Rodrigues et al., 2019). The derivatives side has descriptive work on smile and term-structure patterns in an emerging market (Shaikh and Padhi, 2014), evidence that implied volatility is largely explained by its own past (Wen et al., 2024) and by past returns with power-law weights (Guyon and Lekeufack, 2023), a diffusion generator of dynamic surfaces (Han et al., 2026), and a warning that the shape of at-the-money skew is hard to reconcile with rough-volatility models (Abi Jaber and Li, 2025).

Our proposal. Compute a multidimensional matrix profile over rolling five-minute vectors of surface coordinates for an index option chain. Two hypotheses follow. H1: bars whose nearest neighbour in the surface history is unusually far (discords) precede realised-volatility jumps or dealer repricing by more than the level of at-the-money volatility alone predicts. H2: the next-interval change in term-structure slope, conditional on the motif cluster of the current window, has a sign that survives costs. The control is the linear path-dependent model of Guyon and Lekeufack (2023): the question is whether shape adds anything to a linear lag model.

Landing, data, failure. Stage: features, then risk. Data: timestamped option quotes mapped to a fixed moneyness–maturity grid. Main failure: interpolation artefacts and stale quotes create motifs that are properties of the grid; searching several lengths multiplies trials, so significance must be assessed as Rodrigues et al. (2019) stress.

17.3 B2. Signatures of order-flow paths

Communities and shared object. Rough-path mathematics and market microstructure. An order-placement and cancellation stream is, to the microstructure economist, a sequence of events with signed sizes; to the rough-path theorist it is a multidimensional path whose signature collects its iterated integrals. Depth-two terms record ordered co-movement, such as whether imbalance at one level preceded a trade, which is the question microstructure asks. Of 23 path-signature papers, 17 are in the mathematics silo and one is tagged with order flow.

What each side has. The mathematics side has the feature construction for streams (Lyons et al., 2014), lead–lag embeddings for sequential learning (Fermanian, 2019) and a recent primer covering truncation choice (Chevyrev and Kormilitzin, 2026). Two papers meet microstructure directly: Gyurkó et al. (2014) extract information from the signature of order placement and cancellation streams, and Sidogi et al. (2023) add truncated signatures of limit-order-book paths to short-horizon price prediction and compare them with autoregression on prediction error. The microstructure side has deep models of price formation from the book (Sirignano and Cont, 2018) and interpretable neural models of price–trade dynamics (Naviglio and Lillo, 2026), which supply benchmarks.

Our proposal. Define the path in event time as cumulative signed volume, mid-price, depth imbalance at the first few levels and an event counter; take depth-two and depth-three signatures over a rolling window of events; predict the sign of the mid-price move over the next few seconds

net of half the spread. H: signature terms add predictive power beyond order-flow imbalance and lagged returns, *and* beat a feature set of equal dimension built from random projections, which removes the trivial explanation that more features fit better.

Landing, data, failure. Stage: features, feeding signal and execution timing. Data: level-two or level-three tick data with exchange timestamps. Main failure: dimension grows quickly with depth, timestamp jitter changes the path, and the reported gain in Sidogi et al. (2023) is a prediction-error gain, not a net P&L result.

17.4 B3. Conformal intervals feeding position sizing

Communities and shared object. Distribution-free statistics, decision theory and the deep-learning forecasters. All three produce, or consume, a number meaning how much to trust a forecast: a conformal interval, a Kelly fraction, an abstention flag. Only five papers carry the conformal tag, four in statistics, and none is also tagged with portfolio optimisation.

What each side has. Statistics offers conformalised quantile forecasts on a panel of stock characteristics with near-marginal coverage but undercoverage in high idiosyncratic-volatility and market-regime cells (Taheri Hosseinkhani, 2026), multivariate conformal ellipsoids (Zhang, 2026), a loss that trades set size against conditional coverage (Bach, 2025), and cost-sensitive conformal abstention for rare costly classes, benchmarked on high-stakes tabular decisions such as credit and fraud (Singh et al., 2026). A language-model paper abstains when evidence is sparse (Torres et al., 2026). Decision-side tools include sizing that does not assume Brownian dynamics (Barmish and Proskurnikov, 2025), robustness to corrupted features at decision time (Caunhye et al., 2026) and confidence bands for frontier quantities (Bodnar et al., 2009); Hutchison et al. (2004) give the older case for calibrating classifier probabilities before acting on them.

Our proposal. Scale each position by the lower end of a conformal interval for the signal's next-period return, set to zero when the interval straddles zero, with regime-conditional calibration cells because Taheri Hosseinkhani (2026) show that pooled calibration fails exactly in the states where risk is taken. H: at equal turnover, interval-gated sizing lowers drawdown and raises net Sharpe against sizing by the point forecast.

Landing, data, failure. Stage: portfolio and risk. Data: several years of out-of-sample forecast residuals per signal. Main failure: exchangeability is false for autocorrelated, non-stationary returns, so coverage is only approximate; wide intervals produce no trades at all.

17.5 B4. Change-point detectors as regime gates for factor sleeves

Communities and shared object. Statisticians, time-series miners and factor economists. A detected break is a regime label; the factor economist wants to know when a premium has decayed or a crowded sleeve is about to crash. No paper in the collection is tagged with both change-point and factor-model.

What each side has. Detection: online Bayesian change-point methods applied to financial series (Habibi, 2022; Huang, 2025), multivariate detection of covariance-structure breaks (Sundararajan and Pourahmadi, 2018), precision-matrix change points in heavy-tailed series (Cribben, 2019), a particle filter producing a live crisis probability (Plümer, 2026), and online matrix-profile segmentation (Gharghabi et al., 2017). Factors: risk of momentum is predictable and managing

it nearly doubles its Sharpe ratio (Barroso and Santa-Clara, 2015); data mining and crowding inflate expectations of factor premia (Arnott et al., 2019), and passive inflows compress the realised returns of long–short anomalies (Deng and Sammon, 2026); an alpha half-life shortens with AI adoption (Meng and Chen, 2026a). Regime-conditioned allocators exist (Upreti, 2026; Pan et al., 2026), and Boettiger and Hastings (2012) observe that early-warning indicators rarely report their error rates.

Our proposal. Run Bayesian online change-point detection on each sleeve's return and on the cross-factor covariance, and scale exposure by one minus the posterior probability of a recent break, with the threshold fixed by a false-alarm budget in flags per year. H: gating cuts drawdown at no more cost in net Sharpe than the volatility scaling of Barroso and Santa-Clara (2015). That benchmark is the honest one, since a break detector may only rediscover volatility.

Landing, data, failure. Stage: regime, feeding portfolio. Data: twenty or more years of daily factor returns including several crashes. Main failure: detection delay exceeds the length of the damage, a lag already noted in the factor chapter's crowding monitor.

17.6 B5. Alpha-mining agents under trial accounting

Communities and shared object. Language-model agent builders, selection-bias statisticians and backtesting practitioners. An agent that "explores a search space" and "accumulates experience" is a researcher running many trials, and the best of them is biased upward. The object is the set of candidate formulas evaluated, including those discarded. Of 28 alpha-mining papers in the language-model silo, only 6 also carry a multiple-testing or backtest-overfitting tag.

What each side has. Agents: structured search memory that records edit motifs (Yu et al., 2026e), regularised exploration against alpha decay (Tang et al., 2025), a typed operator language in a sandbox with family-aware selection (Shi et al., 2026), out-of-sample validation and economic-rationale requirements against data snooping (Huang and Fan, 2026), and closed-loop research systems (Guo et al., 2026a); Xia and Wang (2026) give a geometric account of why a search saturates when hypotheses become redundant. Statistics and practice: the deflated Sharpe ratio (Bailey and López De Prado, 2019), the probability of backtest overfitting (Bailey et al., 2013), adjustment for the number of strategies tried (Harvey and Liu, 2019), a false-discovery bound for cross-sectional predictors (Chen, 2022), a falsification audit that runs a whole workflow on zero-predictability synthetic data (Nikolopoulos, 2026), and Monte Carlo measurement of how reliable a validation protocol is (Inglese, 2026).

Our proposal. Three requirements on any mining agent in the fund. (i) Log every candidate evaluated, accepted or not, and estimate the effective number of independent trials from the correlation of candidate return series. (ii) Report the selected factor only through a deflated statistic using that number. (iii) Run the entire agent loop on placebo data in the manner of Nikolopoulos (2026) and record how often it returns a significant alpha. H: the placebo false-discovery rate of an unconstrained agent is far from the nominal level and falls when trial-aware stopping is imposed.

Landing, data, failure. Stage: validation and automation. Data: the agent's full evaluation log, plus synthetic nulls. Main failure: LLM-proposed candidates are strongly correlated, so the effective trial count is hard to estimate, and a model that has read the sample period leaks information that no deflation formula sees.

17.7 B6. Generators as test benches for execution agents

Communities and shared object. Generative-model researchers, microstructure economists and reinforcement-learning (RL) execution researchers. The shared object is a simulated order book that reproduces the stylised facts and, crucially, reacts to the agent's own orders. In the tagging, 35 papers concern synthetic data and 50 execution, yet none carries both tags.

What each side has. Generators: flow-matching generation of controllable book trajectories (Wang et al., 2026a), a state-event foundation model for counterfactual order-flow trajectories (Zhang et al., 2026e), diffusion-based book simulation (Berti et al., 2025), a learned world agent that needs no agent calibration (Coletta et al., 2022), and the open agent-based simulator (Byrd et al., 2020). Impact checks: synthetic order flow built from superposed metaorders with square-root impact, used to support a theory linking impact, order imbalance and volatility (Maitrier et al., 2025). Execution agents: liquidation with closing auctions trained on synthetic rough-Heston prices (Graf and Mastrolia, 2026), a twin-critic actor for selling large programmes (Zaznov et al., 2026), flow-matching imitation for high-frequency quoting (Li et al., 2025b), and RL trained on adversarial synthetic data for robustness (Xia et al., 2026). Cliff and Rollins (2020) warn that agents with no intelligence can beat published AI traders when methods are loose.

Our proposal. An execution gauntlet. Train each execution agent on one generator; evaluate implementation shortfall against TWAP and VWAP on two other generators and on a reactive replay of real data; admit a generator only if it passes the impact and sign-memory checks. H: agent rankings on non-reactive replay differ from rankings on reactive generators, and agents that win in their own training simulator lose their edge when transferred.

Landing, data, failure. Stage: execution and validation. Data: level-two history for calibration and held-out replay. Main failure: co-adaptation, where the agent learns generator artefacts; cross-generator transfer is the guard.

17.8 B7. Contagion measures as cross-sectional risk features

Communities and shared object. Systemic-risk economists, graph-learning researchers and statisticians. The shared object is a time-varying dependence network over assets. The economists measure spillover on it; the deep-learning silo learns it end to end to rank stocks; the statisticians detect when it breaks. No paper carries both a network-contagion tag and a portfolio-optimisation tag.

What each side has. Directional volatility spillover measures (Diebold and Yilmaz, 2012); clustering of stocks by spillover profile (Cerqueti et al., 2023); change-point analysis of cross-market network statistics (Banerjee and Guhathakurta, 2020); graphical-lasso change points in cross-asset precision matrices (Cribben, 2019). On the learning side, attention-based dynamic graphs for asset pricing (Uddin et al., 2021) and a spatio-temporal graph network for stock ranking (Yao et al., 2026); tensor-based forecasts of future stock distances as a tipping-point indicator (Cherifi et al., 2017). From microstructure, a co-crash network of high-frequency flash events (Turiel and Aste, 2022), a measured branching ratio of a liquidation cascade (Seuma, 2026), and a loop-gain matrix whose spectral radius exceeds what scalar monitoring shows (Woo, 2026).

Our proposal. For each stock and day compute in- and out-strength in a directional spillover network and its centrality in a learned graph. Use them (a) as predictors of next-month left-tail loss, controlling for beta and own volatility, and (b) as inputs to a covariance estimator, testing

whether a learned-graph covariance predicts next-quarter covariance better than a shrunk sample estimate. H: network exposure explains tail contribution beyond market beta.

Landing, data, failure. Stage: risk, then portfolio. Data: daily returns for a broad universe, plus intraday data for the high-frequency variant. Main failure: spillover estimates are window-dependent and may proxy for beta; most learned-graph papers are scored on ranking, not on risk.

17.9 B8. Geometric stress indices with detection metrics

Communities and shared object. Geometers (persistent homology, path signatures), change-point statisticians and financial economists. The object is a shape statistic of a rolling window whose level shift marks a break. Geometric papers mostly show case-study plots; the statisticians supply detection delay and false-alarm rates.

What each side has. Geometry: persistence-landscape norms rising near crashes (Gidea and Katz, 2017), persistence norms above a threshold during extreme events (Rai et al., 2024), a signature fragility index correlated with VIX (Mounir, 2025), an online signature-kernel two-sample test for regimes (Issa and Horvath, 2023), signature-based regime clustering (Bilokon et al., 2021), and persistent homology inside a Markov-switching filter reported to flag onset earlier than classical filters (Faris, 2026). Statistics: Bayesian change-point detection compared with classical tests on simulated drifts (Huang, 2025), Neyman–Pearson classification that caps false alarms (Kalan et al., 2025), and the observation that early-warning indicators seldom report error rates (Boettiger and Hastings, 2012).

Our proposal. A common bake-off. Compute the signature index, persistence norm, topological switching filter, signature-kernel test, Bayesian change-point probability and a plain realised-volatility threshold on the same dates; define the event as a drawdown beyond a stated size within twenty days; compare detection delay at a fixed alarm budget in flags per year, then compare the net result of a de-risking rule. H: at matched false-alarm rates, geometric indices lead plain volatility by days, not by zero.

Landing, data, failure. Stage: regime and risk. Data: multi-decade index and sector returns. Main failure: few independent crises, so any ranking is fragile, and thresholds tuned on the same crises used for evaluation.

17.10 B9. Self-excitation and roughness

Communities and shared object. Point-process microstructure, derivatives and statistics. A Hawkes kernel whose norm is close to one produces long memory in order flow; a rough-volatility model has a small Hurst exponent and power-law memory in variance. Theory links the two, but empirical work stays on opposite sides. Only five papers carry the Hawkes tag and none is also tagged with implied volatility.

What each side has. Hawkes-based microstructural foundations for rough Heston-type models (Wang et al., 2026f); limit theorems for nearly unstable Hawkes processes, motivated by fits that sit near the stability boundary (Jaisson and Rosenbaum, 2015); a survey by task (Bacry et al., 2015); an endogeneity ratio from Hawkes fits to index futures as a flash-crash warning (Filimonov and Sornette, 2012); and a Hawkes–order-flow-imbalance engine backtested on NIFTY50 stocks (Mishra, 2026). On the volatility side, realised volatility is rough across asset classes while the

option-implied measure identifies roughness only in some cases (Mouti, 2026), and rough models struggle with the global shape of SPX smiles (Abi Jaber and Li, 2025).

Our proposal. Per stock-day, estimate a branching ratio from trade arrivals, a roughness exponent from intraday realised variance, and, where options exist, the steepness of the short-maturity skew. H1: a branching ratio near criticality predicts next-day realised-variance spikes beyond what lagged realised variance predicts. H2: the cross-sectional ordering of branching ratios predicts the ordering of roughness estimates, which would link two literatures empirically. The change-point machinery of Luo et al. (2026d) can flag when the intensity matrix itself shifts.

Landing, data, failure. Stage: risk, with theory as a by-product. Data: trade-level timestamps and option chains for the same names. Main failure: near-critical fits can be artefacts of intraday seasonality or of clock choice, and kernel estimates are noisy for illiquid names.

17.11 B10. Labels that carry cost and size

Communities and shared object. Label engineers in time-series learning, noisy-label theorists in deep learning and statistics, and derivatives practitioners. The shared object is the training target as a noisy proxy for what the fund earns. The backtesting chapter treats triple-barrier and rank labels; the noisy-label theorists treat corrupted classes; in the papers we read, the two literatures do not refer to each other.

What each side has. Labels: information-driven bars with triple-barrier labelling (Grądzki et al., 2025); evidence that the labelling algorithm changes financial performance (Kovačević et al., 2022); a warning that a good label score need not mean good portfolio returns (Lima, 2026); separation of side and size in meta-labelling (Meyer et al., 2022); ordinal and rank targets that raise accuracy and portfolio returns in cross-sectional stock selection (Cakici and Zaremba, 2026; Guida and Coqueret, 2026); and a label built from repriced option P&L with theta, vega and delta-scaled costs (Vaghela, 2026). Noise theory: sample selection for noisy labels (Kim et al., 2021), class-aware reweighting (Shu et al., 2022) and a convex loss robust to symmetric label noise (van Rooyen et al., 2015).

Our proposal. Compare four label families on one model and one dataset: direction, triple-barrier, rank and cost-aware (realised after-cost P&L of a stated execution or hedging rule). Train each with and without noise-robust losses, with sample weights that account for label overlap. H: the ranking of label families by classification score differs from the ranking by net P&L, and the cost-aware label wins after costs on the option-trading and execution tasks.

Landing, data, failure. Stage: signal and validation. Data: any tick or minute dataset with a cost model. Main failure: a cost-aware label bakes in one execution assumption, and overlap between labels shrinks the effective sample.

17.12 B11. Chart patterns as testable motifs

Communities and shared object. Practitioners of chart analysis, time-series miners and multiple-testing statisticians. A named pattern is a template plus an implicit similarity rule; a motif is a template plus an explicit distance threshold; a rule zoo is a hypothesis family.

What each side has. Miners: image-similarity retrieval of candlestick patterns in futures (Jang et al., 2026), functional motifs of bubble surges (Cremona et al., 2023), and chart-image networks that learn price patterns largely distinct from standard trend signals (Jiang et al., 2021). Practitioners: a flag-pattern rule with filtering (Arévalo et al., 2017) and a backtest of intraday chart patterns (Montes-Gómez et al., 2024). Testers: stepwise error-rate control over tens of thousands of rules (Chuang et al., 2024), a superior-predictive-ability test over thousands of rules in dozens of markets (Rink, 2023), a pre-registered test with held-out data (Brown, 2026), and a three-gate test of significance, cost and survival (Darmanin, 2026).

Our proposal. A pattern registry. Each pattern, hand-drawn or discovered, is registered before testing with its length, normalisation, distance threshold and holding rule. All entries, including motifs found by variable-length search, enter one family-wise or false-discovery correction and one cost model. H: after correction, learned and discovered patterns retain an edge only in a small subset of markets and horizons, and discovered motifs do no better than hand patterns once the search is charged.

Landing, data, failure. Stage: signal and validation. Data: minute bars across many markets. Main failure: forking paths in normalisation and threshold choice, which are themselves a search.

17.13 B12. Own-impact in decision-focused learning

Communities and shared object. Decision-focused learning in operations research and portfolio theory, and the microstructure literature on market impact. When the fund trades size, its orders change the data it will learn from. No paper tagged with decision-focused learning is also tagged with execution or transaction costs, and only one with market impact.

What each side has. Decision side: training a model aware of decisions that bias its own observations (Cristian et al., 2025); a proof that predictive accuracy does not imply control under feedback and competition (Bajpai, 2025b); robust losses (Schutte et al., 2024); generalisation bounds for predict-then-optimise (Balghiti et al., 2023); and robustness to corrupted features (Caunhye et al., 2026). Microstructure side: synthetic order flow that reproduces square-root impact (Maitrier et al., 2025), price impact of inelastic passive demand on anomaly returns (Deng and Sammon, 2026), and learned liquidation schedules (Graf and Mastrolia, 2026).

Our proposal. Train return forecasts through an allocator whose decision layer includes a concave impact-cost term in trade size, evaluated on a reactive simulator of the kind in B6. H: at small size the impact-aware and mean-squared-error-trained forecasts perform alike, but beyond a size threshold the impact-aware version has lower turnover and higher net Sharpe; the crossing size is an empirical estimate of capacity.

Landing, data, failure. Stage: portfolio and execution. Data: signal history and a calibrated impact model. Main failure: impact parameters are noisy and a mis-specified curve is trained into the forecasts.

17.14 Bridges the collection cannot yet build

Table 17.2 lists cells where the literature in this collection is empty or nearly so. Each row is a place where a bridge would be useful but no source exists on one side. Counts come from the automated tagging and stage labels and mean only that the collection, not the literature, is silent. In the silo-stage matrix of Table 2.1, the quantitative-derivatives silo has no paper in

the features, portfolio, execution or automation stages; the language-model silo has none in the regime or execution stages; financial economics has none in execution; and time-series mining has one. Only 76 of 1065 papers carry an execution stage, and 39 of those are from microstructure.

Table 17.2: Empty or near-empty cells and what would fill them (our reading).

Cell with no tagged paper	Why it matters	What is missing
Implied volatility × change-point, surface × change-point	a surface regime break is what a vega book fears, and B1 and B4 depend on it	online detectors evaluated on option surfaces, not on spot or factor returns
Conformal × execution, order flow, order book	execution cost forecasts are uncertain, and sizing slices needs intervals	any conformal study of impact or slippage forecasts
Implied volatility × path signature, surface × topology	the surface is a path and a shape, the tools of B2 and B8 apply	a signature or persistence study of option surfaces
Topology, signatures × limit order book (apart from the signature study in B2)	the book is the richest high-frequency object	stability of topological features under tick noise
Network contagion × execution	cascades hit liquidity first, and slicing should react	a bridge from cascade measures to schedule adaptation
Options × execution and portfolio stages	a vol desk needs cost and capacity for multi-leg trades	transaction-cost and impact models for option baskets
Language-model agents × regime and execution stages	agents need circuit breakers and cost-aware actions	agent evaluation across regimes with fills, not just returns
Hawkes × implied volatility	B9 would link intensity to option-implied skew	a joint estimation study

These gaps are partly an artefact of how the collection was assembled, and partly the usual one: communities do not read each other. The next two chapters take the bridges in different directions. Chapter 18 is organised by the ten stages of a working fund rather than by bridge, but several bridges have natural homes there: B10 (cost-aware labels) in signal generation, B3 and B4 (calibrated sizing, regime gates) in the risk stage, B6 and B12 (simulators, own impact) in execution and portfolio construction, and B5 (trial accounting) in validation and research automation. Chapter 19 restates B3, B4, B5, B6, B10 and B12 as open problems for mathematicians and machine-learning researchers, and uses the thin cells of Table 2.1 to steer future additions to the collection. The empty cells of Table 17.2 remain open; neither later chapter fills them.

Chapter 18

Blueprint of a Semi-Automated Systematic Fund

The fourteen topical chapters of this atlas are organised by the vocabulary of the research communities that wrote the papers. A fund is organised differently: it is a pipeline in which data become signals, signals become positions, positions become orders, and every stage is watched. This chapter re-reads the collection along that pipeline. It describes the architecture of a realistic small-to-mid semi-automated systematic fund, one with a handful of researchers, one or two engineers, a person who owns risk and operations, and a portfolio manager who signs off on capital. Its ten operational stages are cut differently from the eleven classification stages of Chapter 2: data and features map to Stages 1 and 2, signal and forecast to Stage 3, portfolio to Stage 4, regime and risk to Stage 5, execution to Stage 6, validation to Stages 7 and 8, and automation to Stage 9; governance (Stage 10) has no classification counterpart, and theory has no operational stage. For each stage we say which chapters and papers supply components, where a human must stay in the loop, and, most importantly, what a real fund needs that the collected literature does not provide.

Two conventions apply throughout. First, statements attributed to papers are limited to what their abstracts or the earlier chapters report. Second, everything labelled *practitioner guidance* or *our synthesis* is ours: it is standard industry practice or our own proposal, it is not drawn from the collection, and it should be read as a starting point for a fund's own design and not as a tested recipe. The silo and stage labels used when we count papers come from an automated classification of titles and abstracts, and are approximate.

18.1 The architecture at a glance

Table 18.1 gives the whole chapter in one place: for each stage, its main components in the atlas (with chapters), the human gate, and the gap. The stages are numbered as in the sections that follow. The vertical arrows of a real system (a monitoring alert that halts execution, a validation failure that sends a signal back to research) are described in the text of the stages concerned.

Table 18.1: Stage-to-component map of the fund blueprint. Chapter numbers refer to the atlas; the final column lists what the collection does not supply.

Stage	Components from the atlas (chapters)	Human gate	Gap (practitioner guidance in this chapter)
1. Data and reference data	Trade signing, liquidity proxies (12); outlier gates (16); regulatory calendars (4); synthetic data (3)	Vendor onboarding; quarantined records	Corporate actions, identifiers, as-of storage, feed reliability

Stage	Components	Human gate	Gap
2. Research environment and feature store	Platforms (14); feature banks and representations (5, 6)	Feature definitions and lineage review	Versioning, point-in-time joins, compute budgets
3. Signal generation	Factors (9); forecasters (8); rules and patterns (10); volatility and options (11)	Hypothesis approval before testing	Cost-aware labels, signal half-life, per-signal capacity
4. Combination and portfolio	Convex and decision-focused allocation (13)	Constraint and objective sign-off	Sleeve capital allocation, turnover budgets
5. Risk	Regime gates (7); volatility targets (11); uncertainty-aware sizing (13)	Limit setting; override of gates	Margin and gap stress, liquidity-adjusted limits
6. Execution	Impact, schedulers, TCA (12)	Strategy-level cost review; broker choice	Order state machine, pre-trade checks, multi-leg options
7. Validation and promotion	Overfitting statistics, baselines, nulls (14)	Promotion committee	Paper-trading protocol, staged capital, reconciliation
8. Monitoring and decay	Process control, break detection (14, 7); crowding evidence (9)	Review of alerts; reinstatement	Decay-versus-drawdown tests, kill-switch ladder
9. LLM research automation	Agentic mining, audits, security (15, 16)	Approval of every production artefact	Trial ledger, permissions, sandboxing
10. Governance	Control and delegation papers (4, 13)	Committee, change control	Model inventory, incident process

18.2 Stage 1: data and reference data

Components

The microstructure chapter supplies the building blocks of the tape-level layer. Trade-sign classification is not a solved problem: Lee and Ready (1991) document timing problems with quote-based classification, Chakrabarty et al. (2012) find the conventional rules more accurate than bulk volume classification, and Bessembinder (2003) show that the trade–quote time adjustment changes estimated execution costs. We read these together as a requirement that signed-flow series carry a data-quality metric. Intraday seasonality in volume and spreads on the NSE (Krishnan and Mishra, 2013) means that features must be conditioned on the time of day before they enter a model. For outlier screening, the general-methods chapter offers the copula-based detector of Li et al. (2020), and the time-series chapter offers discord scores in the matrix-profile family (Yeh et al., 2024), which can flag a window that is unlike anything seen before.

Reference data of a different kind come from the market-structure chapter. Exchange rule changes (lot sizes, expiry days, auction sessions) are structural breaks in the data. Gunnam et al. (2024) review regulatory changes in derivatives trading, Khatwani (2026) evaluate the closing auction session, and Hussain and Olekar (2026) use a regulator-mandated expiry change as a falsification test of a volatility claim. A regulatory calendar belongs in the reference-data layer next to the holiday calendar.

Point-in-time discipline has surprisingly few papers behind it. Cai et al. (2026) build an auditable point-in-time evaluation for portfolio agents, and Darmanin (2026) use point-in-time index membership and delisting corrections in cross-sectional tests; Castellanos Macias (2026) state survivorship as an explicit ceiling on their evidence. Synthetic data enter at this stage as augmentation and as a sandbox: the controlled environments of Sun et al. (2026), the simulation harness of Galas (2014) and the order-flow generator of Berti et al. (2025), together with the large-tick simulator of Noble et al. (2026), are the nearest things to a standard test bench (Chapter 3).

Human in the loop

A person approves each new vendor feed, reviews the quarantine queue, and signs off on every change to the exchange-calendar table. Automated outlier gates should quarantine and never silently delete, because the extreme observations that a gate flags are sometimes the stress episodes a risk model most needs. A person decides which they are.

Gaps

Practitioner guidance. The collection says almost nothing about the following, all of which cause real losses. (i) *Corporate actions and identifiers.* Splits, dividends, mergers, symbol changes and contract rolls must be applied from a reference table that is versioned, so that a price series can be reconstructed as it looked on any past date. (ii) *As-of storage.* Store both the event time and the time the fund learned of it, so that restated fundamentals and late vendor corrections do not leak into past backtests. (iii) *Cross-vendor reconciliation.* Compare two sources daily and alert on disagreement above a tolerance. (iv) *Feed reliability.* Treat the market-data connection as a production service: heartbeat checks, sequence-gap detection, a documented fallback, and a rule for what the trading system does when data are stale (the safe default is to stop opening new risk). (v) *Clock discipline.* Timestamp every record at ingestion with a synchronised clock, because the latency and leakage problems that the microstructure chapter describes reappear in the fund's own plumbing. In the atlas, the data stage holds 122 records by our automated classification, and a reading of the data-stage chapters suggests that most concern similarity search, series generation or cleaning, not the operation of a data pipeline.

18.3 Stage 2: research environment and feature store

Components

Yang et al. (2020a) describe Qlib, an AI-oriented platform that organises data, models and backtests, and Yang et al. (2026c) describe FinRL-X, in which backtest and broker execution share a weight-centric interface; the second is the closest the collection comes to the principle that research and live code should be the same code (Chapter 14). For feature construction, the time-series chapter offers feature banks such as the descriptor library of Fulcher and Jones (2017) and fractional differencing (Hosking, 1981), and the representation chapter offers signatures, topological summaries, learned embeddings and pretrained forecasters as candidate features (Chapter 6). The recurring warning in those chapters is that any transform fitted on the full history, whether a decomposition, an encoder or a clustering, leaks the future, so each must be refitted inside each fold.

Human in the loop

A reviewer approves every feature definition before it can be used in production, with a short note on its economic rationale and its information timestamp. The review is about meaning and timing, not about performance; a feature is rejected if nobody can say when its value becomes known.

Gaps

Practitioner guidance. A feature store for a small fund needs: immutable, versioned feature definitions; point-in-time joins that can reproduce the feature matrix as of any past date; lineage from raw data to model input; a rule that research and production read features through the same code path; and a cost accounting of compute, since retraining frequency is often limited by compute and not by accuracy. None of the platform papers in the collection addresses lineage,

versioning or the cost of recomputation at scale. A second gap is the *shared feature registry* across sleeves: two researchers who each build a volatility feature produce two numbers that differ in a decimal place, and the portfolio then trades a disagreement.

18.4 Stage 3: signal and alpha generation

Components

Three families of signal feed the fund, and the atlas supplies different amounts of evidence for each.

Classical factors and rules are the best-documented. Time-series momentum has a known failure mode, and Barroso and Santa-Clara (2015) show that the risk of momentum is predictable and that managing it nearly doubles the Sharpe ratio while virtually eliminating crashes. Kurth et al. (2026) document the demise of short-term trend following and its dependence on signal speed and asset class. The collection of formulaic alphas of Kakushadze (2016) is a standard library of cross-sectional price–volume expressions, and Arnott et al. (2019) list three blunders of factor investing, starting with expectations inflated by data mining, crowding and optimistic cost assumptions (Chapter 9). Volatility and option-derived signals are covered in Chapter 11: implied volatility is largely past-dependent (Wen et al., 2024), so a feature built from option data may repackage what returns already contain, and Duarte and Jones (2008) estimate a volatility risk premium that rises with market volatility.

ML forecasters are the largest group (126 records in the return-forecasting chapter alone). The most useful of them for a fund are about the target and the benchmark. Guida and Coqueret (2026) show that the label alone changes long–short decile performance and explains more out-of-sample variation than the model; Cakici and Zaremba (2026) reach a compatible result across 35 equity markets; and graph overlays in the style of Liu et al. (2026a) or Yao et al. (2026) are candidate residual corrections to a tabular baseline. On the sceptical side, Pandhe (2026) find a gradient-boosted ranker not distinguishable from plain 12-1 momentum on the Nifty 500 after costs, and Bandi et al. (2026) argue that the zero forecast is too weak a benchmark in stock-return forecasting.

Pattern and regime features come from Chapters 5, 6 and 7. Online change-point detectors (Habibi, 2022; Huang, 2025), particle-filter regime detection (Plümer, 2026), and discord scores (Yeh et al., 2024) are used as conditioning variables or filters, seldom as stand-alone alphas. Microstructure signals (order-flow imbalance, participant mix) enter as short-horizon filters on slower strategies (Chapter 12).

Human in the loop

A person approves the *hypothesis* before a test is run: what the signal is, why it should exist, its intended horizon, and which market participant is on the other side. This is a defence against the multiple-testing problem rather than a bureaucratic step, because a hypothesis registered in advance fixes the number of trials. Humans also decide when a signal sits in the wrong family: a regime detector promoted to an alpha, or a risk filter promoted to a return source.

Gaps

Practitioner guidance. (i) *Cost-aware labels.* Chapter 14 documents that classification gains may not survive conversion to a portfolio (Lima, 2026); Vaghela (2026) label option bars by repriced profit and loss with theta, vega and costs inside the target. These are exceptions. We recommend that every label be built from the intended instrument and execution assumption: label by net-of-cost outcome of the trade that would actually be placed, with the fund's own cost curve, and report the gross label as a diagnostic. (ii) *Signal half-life and horizon matching.* Each

signal needs a measured information-coefficient decay profile by horizon, so that holding period, rebalance frequency and execution urgency are chosen from it. (iii) *Per-signal capacity.* See the next stage and the monitoring stage; capacity should be a field in the signal's registry entry from the first day. (iv) *Signal correlation.* A candidate signal should be judged by its marginal contribution to the existing book, not stand-alone, which the synergy-oriented mining papers argue (Yu et al., 2023).

18.5 Stage 4: combination and portfolio construction

Components

Chapter 13 supplies most of the layer. Covariance hygiene starts from the classical random-matrix finding that empirical correlation matrices are mostly noise (Laloux et al., 1999; Plerou et al., 2002); OWL shrinkage (Hiraki and Sun, 2022) gives similar weights to correlated stocks. Boyd et al. (2024) describe a Markowitz extension with transaction costs, leverage limits and parameter uncertainty that remains a convex problem, and Boyd et al. (2017) give the multi-period version that executes only the first planned trade. Sizing rules are reviewed in the same chapter, with Kelly-type rules revisited by Stiffelman (2026). For training forecasts through the allocator, decision-focused learning is surveyed and benchmarked by Mandi et al. (2024), and Bajpai (2025b) prove, in a stylised model, that high predictive power need not imply profit once convex execution costs and feedback are present. Signal gating by predictability state (Farmer et al., 2018; Lin et al., 2021) and regime-conditional allocators (Upreti, 2026) sit between forecasting and sizing.

Human in the loop

The portfolio manager owns the objective and the constraint set: the risk-aversion and cost-aversion parameters, the exposure and concentration limits, the turnover budget, and the allowed instruments. The optimiser proposes, and a person reviews the proposed trade list against a few sanity rules (gross and net change, largest single order, share of daily volume) before the first live rebalance of any new configuration. After that review the process can run unattended inside limits.

Gaps

Practitioner guidance. (i) *Combining signals.* The collection has little on how to weight many weak signals with unstable correlations. A sound default is a simple, shrunk combination (equal-risk weights over signals orthogonalised to the existing book), with weights updated slowly. (ii) *Capital allocation across sleeves.* Allocating between a daily equity sleeve and an intraday options sleeve depends on capacity, drawdown tolerance and correlation in stress, none of which the portfolio papers address at the fund level. (iii) *Turnover budgets.* Make turnover an explicit constraint tied to the cost curve of the execution stage, not a penalty tuned on the backtest. (iv) *Financing, borrow and margin.* Short-borrow availability and margin requirements are constraints of the optimiser and rarely appear in the research papers; for options they dominate.

18.6 Stage 5: risk

Components

The risk layer is the stage where the atlas is most helpful, because many silos converge on it. *Regime gates*: Musaev et al. (2024), Habibi (2022) and Huang (2025) offer change-point detectors for financial series, and Plümer (2026) a real-time regime detector calibrated to the distributional

shape of stress. *Volatility targeting*: the risk-managed momentum result of Barroso and Santa-Clara (2015) is the template, with a volatility forecast maintained as a shared service as described in Chapter 11, in which GARCH-type baselines are challenged under a superior-predictive-ability test (Reinhard Hansen and Lunde, 2003). *Drawdown control*: Mendes and Lavrado (2017) implement and test Maximum Drawdown at Risk, and Harding et al. (2003) discuss the strengths and weaknesses of drawdown as a risk statistic. *Uncertainty-aware sizing*: Sanderink (2026) separate a slow decision about whether to trade from a fast sizing rule and report that naive inverse-uncertainty sizing hurts because uncertainty and signal strength are correlated; Taheri Hosseinkhani (2026) find conformal intervals undercover in high idiosyncratic-volatility and regime cells; Hutchison et al. (2004) give the case for calibrating a classifier before turning probabilities into sizes. *Tail and structural risk*: the flash-crash taxonomy of Ng (2026) and the dealer-gamma caveat of Sahu (2026) mark how far vendor-quoted levels can be trusted.

Human in the loop

Limits are set by people and enforced by code. The risk owner sets gross, net, per-name, per-sleeve, loss and liquidity limits, and decides the response to a regime flag (scale down, flatten, escalate). Regime gates should be able to reduce risk automatically but should require a person to restore it, because the cost of a false alarm is asymmetric to the cost of a false all-clear.

Gaps

Practitioner guidance. (i) *Margin and gap stress.* Run stress scenarios against the exchange margin schedule and overnight gaps (the first-bar gap effect documented in Zielkowski, 2026 is a reminder that stops do not protect against gaps). (ii) *Liquidity-adjusted limits.* Express positions in days-to-liquidate at a stated participation rate. (iii) *Concentration of gates.* If several sleeves share one regime detector, one false alarm de-risks the whole fund at once, and the de-risking itself costs spread and impact. (iv) *Short-option risk.* A short-volatility book needs limits on short-gamma exposure and on the loss from a strike-level gap at expiry, with settlement mechanics modelled explicitly.

18.7 Stage 6: execution

Components

Chapter 12 is the supply. Impact theory explains why large orders are costly: order splitting and liquidity replenishment are jointly needed to reproduce the square-root exponent (Zhou et al., 2026a). For schedules, Rastegar and Eghbalreihani (2019) fit an I-star impact function to split a large order, and Lehalle and Neuman (2019) add a Markovian signal to the transient-impact framework so that alpha decay and impact decay can be traded off. Barzykin et al. (2026) treat optimal execution with passive impact, and Maciejewski (2026) propose a model-free passive execution method. After the trade, Markov (2019) compare broker algorithms on implementation shortfall with hierarchical pooling, and Zovko (2026) point out that real-time slippage detection needs hundreds of fills to separate impact from volatility. Learned executors such as Zhang et al. (2023b) are evaluated in simulators, which is why the simulator of Noble et al. (2026) and the closing-auction work of Graf and Mastrolia (2026) matter for a desk that trades the close.

Human in the loop

A trader or risk owner reviews the cost model each quarter against realised slippage, picks brokers and algorithms from the TCA ranking, and holds the manual override: the ability to pause, cancel or flatten with one action, which must work even when the research stack is down. Large

or unusual orders (above a threshold share of volume, in a thin contract, near an expiry) require approval before release.

Gaps

Practitioner guidance. (i) *Order state machine and reconciliation.* Orders, acknowledgements, partial fills and cancels must be tracked in a state machine, and positions reconciled against broker and clearing records at least daily and after every restart. (ii) *Pre-trade checks.* Fat-finger, price-collar, maximum-order-size, rate-limit and duplicate-order checks must sit between the strategy and the venue. (iii) *Multi-leg option execution.* The collection treats execution mostly for equities and futures; spreads and multi-leg options require leg-risk handling and expiry-day rules. (iv) *Cost-model feedback.* The impact curve fitted from the fund's own fills feeds the portfolio stage and the capacity estimate, closing the loop that the TCA papers leave open. Capacity itself is not estimated by any paper that we found; Kurth et al. (2026) test capacity constraints as an explanation of trend-following's decline, which is a use of the word and not an estimation method.

18.8 Stage 7: validation and promotion gates

Components

This is the stage where the collection is strongest, and where Chapter 14 should be read in full. The gate has five layers. *Splits*: walk-forward evaluation with purging and embargo around overlapping labels (standard practice; the effective-sample-size problem is shown vividly by Diao, 2026). *Selection correction*: the number of trials must be counted (Lopez De Prado, 2013); the probability of backtest overfitting (Bailey et al., 2013), the deflated Sharpe ratio (Bailey and López De Prado, 2019) and the probabilistic Sharpe ratio (Bailey and López De Prado, 2011) turn the count into a number; the test for superior predictive ability (Hansen, 2005) compares a family of forecasts with a benchmark. *Placebo and null environments*: Nikolopoulos (2026) falsify workflows that produce significant walk-forward results from nothing, and Inglese (2026) show that stacking redundant corrections costs power. *Baselines*: Pandhe (2026) and Cliff and Rollins (2020) are models of the comparison to run. *Cost realism*: Mesfin (2026b) test fourteen intraday signal families against a fixed round-trip cost and find none surviving, and Wiecki et al. (2016) report that reported backtest Sharpe ratios predicted out-of-sample results poorly across 888 strategies.

Human in the loop

A promotion committee (at least the portfolio manager, the risk owner and one researcher who did not build the signal) reads a fixed one-page promotion pack and may only promote, return or reject. The committee exists because the researcher who built a signal is the person least able to count the alternatives that were tried and dropped.

Gaps

Practitioner guidance. The collection stops at the backtest; the following steps belong to a fund. (i) *Paper trading with a protocol.* Run the production code path against live data with simulated fills for a pre-set number of weeks, with exit criteria written before it starts. (ii) *Staged capital.* Promote in steps, for example 10, 25 and 50 percent of target size, each step requiring that realised cost and slippage match the model. (iii) *Live-versus-backtest reconciliation.* Re-run the backtest on the live period with the live data and the realised fills, and decompose the difference into data, signal, sizing and execution components.

Desk note. Promotion pack and reconciliation (validation stage; our protocol). *Inputs:* the hypothesis note; the trial ledger with the number of labels, features, windows and parameters tried; walk-forward results net of the fund's own cost curve; the deflated Sharpe ratio and the probability of backtest overfitting (Bailey and López De Prado, 2019; Bailey et al., 2013); the placebo result in the manner of Nikolopoulos (2026); and the baseline comparison of Pandhe (2026). *Output:* promote, return or reject, plus a stage-one capital figure. *After promotion,* each week compute a four-way reconciliation: (a) *data*, the difference between the research and live feature values for the same timestamps; (b) *signal*, the difference between backtest and live scores on identical inputs; (c) *sizing*, the difference between intended and actual target positions; (d) *execution*, realised minus modelled cost. Any component outside its tolerance halts further capital steps. *What can go wrong:* the ledger is incomplete, the placebo does not match the real data's dependence, and the tolerances are set after seeing a disappointing week.

18.9 Stage 8: monitoring and decay detection

Components

Monitoring is thinly covered but not empty. Hassan et al. (2010) propose statistical process control for trading systems, treating them as machines with variable outputs, which supplies the control-chart vocabulary. Atluri (2026) motivate monitoring from the failure side: a meta-labelling filter showed a structural inversion of the direction-to-profit relation during a drawdown. On decay, Meng and Chen (2026a) formalise an alpha half-life that shortens as AI adoption rises, Deng and Sammon (2026) and Unterberg (2026) link passive flows to weaker anomaly returns and active-fund alpha, and Arnott et al. (2019) list crowding among the reasons for inflated factor expectations. Break detectors (Habibi, 2022; Kalan et al., 2025) can monitor a signal's own error stream, and Ozimek (2026) compare strategies through regime-conditional distributions of fold-level results, a useful reference distribution for live performance. Zovko (2026) give the statistical lesson for in-flight cost monitoring: individual fills are too few to decide anything.

Human in the loop

A daily monitoring page is read by a person who has the authority to pause a sleeve. Alerts are tiered, with the highest tier reserved for events that stop trading automatically. Reinstatement after a halt always needs a human decision and a written reason.

Gaps

The central gap is that a decaying signal and an ordinary drawdown look the same over short windows, and the collection offers no test that separates them. *Practitioner guidance.* Compare live information coefficients, hit rates and cost-adjusted returns with the walk-forward distribution from the promotion pack, using a sequential test (a CUSUM on standardised live residuals is a simple choice) with a false-alarm budget set in advance. Track the signal's crowding proxies separately. Review capacity by comparing the fund's share of volume in each name with the cost curve of Stage 6, re-estimated as the book grows.

Desk note. Kill-switch ladder (monitoring and risk; our design). Level 0, *observe*: an alert is logged. Level 1, *throttle*: a sleeve's risk multiplier falls (for example to half) when its live drawdown, cost overrun or rejection rate leaves the promotion-pack range; this is automatic. Level 2, *stop opening*: no new positions, exits only, triggered by sleeve loss limits, stale data, failed reconciliation or a regime flag from the detectors of Stage 5; this is automatic. Level 3, *flatten*: a human, or a fund-level loss limit, flattens the book through execution with pre-approved algorithms. Level 4, *disconnect*: the broker or exchange kill facility, tested on a schedule. *Design rules:* the switch lives outside the research stack; each level has a named owner; restoring risk costs more approvals than removing it; every trigger is tested in the paper environment. *What can go wrong:* thresholds too tight turn noise into turnover, de-risking by several sleeves at once adds to impact, and an untested switch fails when needed. Among the papers, only a small set discuss halts or circuit breakers, mainly at the exchange level (Abad et al., 2018; Jain et al., 2016).

18.10 Stage 9: research automation with LLM agents

Components

Chapter 15 describes agents that mine formulaic alphas, read text, and in some cases trade. For a fund the useful pattern is the narrow one. Shi et al. (2026) combine an LLM with a domain-specific operator language and an AST execution sandbox, so that generated programs cannot do arbitrary things; Tang et al. (2025) add regularised exploration to counteract alpha decay in mining; Li et al. (2026c) replace free-form agent messages by structured, replayable transport so that a hand-off is verifiable; Wang et al. (2026d) test whether generated code behaves as the described strategy. The evaluation papers supply the audit layer: Roy and Roy (2026) filter memorisation-contaminated signals by cross-model disagreement, Zhu et al. (2026c) propose a memory-controlled benchmark with masking, and Ye et al. (2026a) argue headline Sharpe ratios of LLM agents are not deployment evidence. Security is addressed by Na et al. (2026), Sua et al. (2026) and Yan et al. (2025), and the measurement of agents in production by Barton et al. (2026).

Human in the loop

Agents propose; people dispose. Our rule is that an agent may read data, write code in a sandbox and submit candidates with a stated hypothesis, but may not modify the production feature store, change a limit, or send an order. A person approves every artefact that enters production, and an LLM used inside the live loop is an analyst whose output is a veto or a size-down flag at most, never a size or an order.

Gaps

Practitioner guidance. (i) *Trial ledger.* Every candidate an agent generates is a trial; the ledger must be written by the harness, not by the agent, so that memory modules that discard failures cannot hide them from the deflation step. (ii) *Permissions and secrets.* Agents get read-only credentials to research data and none for brokers. (iii) *Reproducibility.* Pin model versions and record prompts, because an agent-built signal that cannot be regenerated cannot be audited. (iv) *Cost and latency.* Inference cost and API availability are operating risks that the benchmark papers do not price.

18.11 Stage 10: governance

Components

Governance receives the least direct attention. Raschner (2021) examine, from a legal perspective, the testing and simulation requirements for trading algorithms under MiFID II, and Hansen (2025), drawing on 182 practitioner interviews, argue for attending to the whole technical stack of automated trading systems rather than to algorithms alone. Jain et al. (2016) supply a framework for assessing whether high-frequency quoting increases systemic risk, and Dixon (2026a) formulate how much decision authority an AI agent should receive as evidence quality changes, which is a governance policy in the literal sense. Pre-registration practices (Castellanos Macias, 2026) and axioms stated before testing (Li, 2026) show how a fund can bind itself before it sees results.

Human in the loop

The governance body (in a small fund, the principals plus an independent reviewer) owns the model inventory, approves exceptions, and reviews incidents. It is the only body that can raise a limit.

Gaps

Practitioner guidance. (i) *Model inventory and change control:* a register of every live model with owner, version, data dependencies, validation record, limits and retirement criteria; every change goes through the promotion gate or a documented fast path. (ii) *Incident process:* a blameless post-mortem for every breach or near miss, and a log that feeds back into the monitoring thresholds. (iii) *Segregation:* the person who builds a signal does not promote it. (iv) *Regulatory compliance:* venue and regulator rules on algorithmic trading (in India, those of SEBI and the exchanges) must be checked by the fund's compliance function; Gunnam et al. (2024) review regulatory changes in derivatives trading, but none of the atlas papers is a compliance manual, and nothing here is legal advice.

18.12 Worked examples

The two traces below follow a hypothetical strategy through every stage. They are illustrative designs, not results, and every performance question in them is open until the gates above have been passed.

Desk note. Intraday index-options strategy (hypothetical). *Idea:* trade short-dated Nifty option structures in the direction of an intraday forecast, with a volatility filter. *1 Data:* option chains, India VIX and spot bars, timestamped at ingestion; expiry calendar and rule changes (Chhimwal et al., 2025; Gunnam et al., 2024); stale-strike checks as in Gupta (2018); a simulated stress set from Noble et al. (2026) and Sun et al. (2026). *2 Features:* model-free implied volatility, smile level and slope, put–call ratio, residualised against a past-return baseline (Wen et al., 2024; Guyon and Lekeufack, 2023); time-to-expiry flags, with intraday seasonality removed (Krishnan and Mishra, 2013). *3 Signal:* a direction forecast gated by regime state in the manner of Sharma et al. (2026), labelled by repriced option profit and loss (Vaghela, 2026), with a net-of-cost label in our recommended form. *4 Portfolio:* fixed premium-at-risk per structure, no optimiser; a cap on net vega. *5 Risk:* short-gamma and expiry-gap limits informed by Duarte and Jones (2008) and Sahu (2026); a volatility forecast challenged under Reinhard Hansen and Lunde (2003); a change-point flag (Habibi, 2022). *6 Execution:* limit-order placement for legs with leg-risk rules (our guidance); cost model from realised fills and TCA (Markov, 2019). *7 Validation:* placebo and trial count (Nikolopoulos, 2026; Bailey and López De Prado, 2019); a simulated-null test like that of Pillai (2026). *8 Monitoring:* control charts (Hassan et al., 2010); the kill-switch ladder. *9 Automation:* an agent may propose feature variants through a sandbox (Shi et al., 2026); a person approves. *Gaps that bite here:* capacity in far strikes, margin, leg execution, and the fact that the cited Indian studies are mostly descriptive.

Desk note. Daily cross-sectional equity strategy (hypothetical). *Idea:* a long–short book on a liquid universe ranked by a blend of price and fundamental features. *1 Data:* adjusted prices with point-in-time index membership and delistings as in Darmanin (2026); corporate-action table (our guidance). *2 Features:* window-safe transforms (Hosking, 1981), a graph-relation feature in the style of Liu et al. (2026a), features logged in the platform of Yang et al. (2020a). *3 Signal:* labels chosen by ablation (Guida and Coqueret, 2026; Cakici and Zaremba, 2026); a gradient-boosted ranker that must beat 12-1 momentum (Pandhe, 2026) and the mean benchmark of Bandi et al. (2026); marginal contribution to the alpha pool (Yu et al., 2023). *4 Portfolio:* multi-period convex rebalancing (Boyd et al., 2017, 2024) with a cleaned covariance (Laloux et al., 1999; Hiraki and Sun, 2022). *5 Risk:* volatility scaling of the momentum sleeve (Barroso and Santa-Clara, 2015), a two-level trade-or-abstain gate (Sanderink, 2026), drawdown limit (Mendes and Lavrado, 2017). *6 Execution:* participation schedule from a fitted impact curve (Rastegar and Eghbalreihani, 2019; Zhou et al., 2026a); broker ranking (Markov, 2019). *7 Validation:* deflated Sharpe, effective sample size (Diao, 2026), superior-predictive-ability test (Hansen, 2005), then staged capital. *8 Monitoring:* live information coefficient against the walk-forward range, crowding proxies (Deng and Sammon, 2026), break detector on errors (Habibi, 2022). *9 Automation:* synergy-oriented formula search (Yu et al., 2023) screened with a multi-dimensional, backtest-free evaluation (Ding et al., 2026), confined to a quarantined window with a trial ledger (our guidance). *Gaps that bite here:* borrow cost on the short leg, capacity of small-cap names, and the live-versus-backtest reconciliation of Stage 7.

18.13 What the blueprint does not claim

Three limits deserve plain statement. The pipeline above is a synthesis, and no paper in the collection tests an end-to-end fund. Several stage assignments are ours: papers were written for one community and we have placed them where a fund might use them. And the gaps are gaps of this collection, which reflects what the Zotero library holds; some of the practices we call missing (corporate-action handling, reconciliation, incident management) are well known to practitioners and simply do not appear in research papers. The main contribution of the blueprint is to show where a mathematician, a machine-learning researcher or a trader can pick up a part of

the pipeline and know what it connects to (Chapter 17), and where the open problems that Chapter 19 lists begin.

Chapter 19

Gaps, Open Problems and Conclusion

The atlas was built to let people who do not read one another find one another. This closing chapter does three things. It states a set of open problems, framed so that a mathematician or machine-learning researcher can see the well-posed question inside the market vocabulary. It measures what the collection over- and under-represents, using the records themselves. And it says how the atlas should be maintained as the library grows. A short conclusion restates the contribution.

19.1 Open problems at the silo boundaries

Each problem below sits where two or more silos meet, and each is stated as a question the markets pose, with the part that is mathematically clean separated from the part that is empirical. They are our selection, drawn from the gaps of Chapter 18 and the bridges of Chapter 17; the papers cited show that the pieces exist, not that the problems are solved.

1. Labels that close the gap between classification and profit. *Market side:* a model with a better F1 score on a trade label can lose money after costs (Lima, 2026), and the correlation between classification metrics and strategy performance is low (Kovačević et al., 2022). *Question:* given a cost function and a decision rule, characterise the family of labels whose Bayes-optimal classifier maximises expected net P&L, and bound the loss when the label is only an approximation. The decision-focused learning literature (Mandi et al., 2024) is the natural toolkit; the triple-barrier labels of the time-series community are special cases to be analysed.

2. Inference with few independent observations. Overlapping labels leave about twenty independent observations in one audited setting (Diao, 2026), and a handful of regime episodes leaves fewer. *Question:* give finite-sample, assumption-light tests for the difference of two strategies' Sharpe ratios under overlapping labels, heavy tails and regime switching, with power curves that practitioners can read. The deflation and probability-of-overfitting tools (Bailey and López De Prado, 2019; Bailey et al., 2013) are asymptotic or simulation-based; sharper small-sample theory would be a direct contribution.

3. Detection delay against the cost of acting. Change-point detectors (Habibi, 2022; Huang, 2025) and the Neyman–Pearson framing of Kalan et al. (2025) give false-alarm budgets, yet a trading action costs turnover and impact. *Question:* formulate regime response as an optimal-stopping or sequential-decision problem with an explicit cost of switching, and find the detector threshold that is optimal for net P&L, not for detection accuracy. The stopping-problem papers in Chapter 13 (for example Colaneri and De Angelis, 2022) suggest the continuous-time setting.

4. Learning with feedback from one's own trades. Cristian et al. (2025) study learning when decisions bias the data they later generate, and Bajpai (2025b) show that predictability need not imply profit under feedback and convex costs. *Question:* what is the minimal experimental design (randomised participation, small probes) that identifies the fund's own impact curve

without paying for the identification in cost, and how should a forecaster be trained on data it has itself moved? Impact theory (Zhou et al., 2026a) supplies the structural model.

5. Scoring generators by the decisions they support. The authors of Berti et al. (2025) and Kwon and Lee (2024) both note the absence of agreed metrics for market generators. *Question:* define a proper scoring rule, or a family of them, for synthetic paths and order flow so that a lower score provably bounds the error in a downstream quantity (execution cost, drawdown, or fill rate). The controlled benchmark of Sun et al. (2026) provides ground truth against which candidates can be tested.

6. Separating decay from drawdown. Meng and Chen (2026a) derive an alpha half-life; Hassan et al. (2010) propose control charts. *Question:* design a sequential test that, given a walk-forward distribution of a signal's performance, detects a drop in its mean as early as possible at a fixed false-alarm rate when returns are autocorrelated and volatility is time varying, and quantify the expected delay.

7. Calibrated confidence without exchangeability. Conformal intervals under-cover precisely in high-volatility and regime cells (Taheri Hosseinkhani, 2026), and uncertainty-based sizing can cancel an edge when uncertainty and signal strength are correlated (Sanderink, 2026). *Question:* give coverage guarantees conditional on a volatility or regime state, under distribution shift, that remain useful for position sizing. Online and adaptive conformal results (Noarov and Roth, 2026; Zhang, 2026) are the starting point.

8. Trial-counting for automated search. Agentic miners generate and discard many candidates (Tang et al., 2025; Shi et al., 2026). *Question:* define the effective number of trials of an adaptive search with memory, so that the deflated Sharpe ratio of Bailey and López De Prado (2019) can be applied to its output, and detect contamination of signals by a language model's training data (Roy and Roy, 2026).

19.2 What the collection over- and under-represents

All counts in this section are computed from the 1065 records of the atlas, and the silo, stage and relevance labels come from an automated classification of titles and abstracts. Market counts come from keyword matches in titles and abstracts and are approximate, since a paper may mention several markets or none.

By silo, year and relevance

The two largest silos, time-series machine learning (180 records, 16.9 percent) and deep learning (176, 16.5 percent), together account for a third of the collection. Empirical finance and econometrics holds 136 records (12.8 percent), microstructure 115 (10.8 percent), decision theory and control 104 (9.8 percent), statistics 95 (8.9 percent), language models and agents 88 (8.3 percent), practitioner writing 55 (5.2 percent), mathematics 49 (4.6 percent), quantitative derivatives 43 (4.0 percent) and other domains 24 (2.3 percent). The mathematicians and the derivatives specialists, whose tools Chapters 6, 13 and 11 rely on most, are among the smallest groups.

The collection is young. Records dated 2026 number 330 (31.0 percent), and 500 (46.9 percent) are from 2025 or later, against 145 (13.6 percent) from 2015 or earlier and only 15 from before 2000. The language-model silo is almost entirely recent: 83 of its 88 records date from 2025 or later, while microstructure and financial economics lean older, with 35 of 115 and 37 of 136 records respectively dated 2015 or earlier. A reader should therefore expect the fashionable methods to be over-sampled relative to the slow-moving foundations, and should treat the many 2026 preprints as claims awaiting replication, as Chapter 14 urges.

On relevance, 753 records (70.7 percent) were rated 3, 261 (24.5 percent) rated 2 and 51 (4.8 percent) rated 1, the peripheral class. Peripheral records concentrate in time-series methods (23)

and general methods (15) by chapter. By silo, the language-model silo has the most in absolute terms (13 of 88), but the other-domain silo has by far the highest share (11 of 24 records), which is to be expected of imports from outside finance.

By pipeline stage and market

Table 2.1 in Chapter 2 shows how silos distribute over the pipeline stages. A record can carry more than one stage label. The stage counts across the collection are: signal 288, forecast 206, validation 167, risk 131, data 122, theory 115, features 79, execution 76, portfolio 68, regime 59 and automation 46. The pipeline is thus top-heavy: signal and forecast papers outnumber execution and portfolio papers by roughly three to one. Execution is thin in the deep-learning and time-series columns (one record each), where most of the new modelling talent sits, while risk is led by financial economics (38) and microstructure (24), and validation by statistics (39) and practice (30). Automation has only 46 records, 14 of them from the language-model silo.

By market, a keyword scan of titles and abstracts finds United States references in 167 records (15.7 percent), India in 96 (9.0 percent), China in 70 (6.6 percent), cryptocurrencies in 41 (3.8 percent), futures in 49 (4.6 percent), options in 78 (7.3 percent), commodities in 27 (2.5 percent), foreign exchange in 23 (2.2 percent) and bonds or rates in 13 (1.2 percent). Only 415 records match any of these keywords, so for most records the market is not stated in the abstract. The Indian share is high for a global literature, reflecting the origin of the library, and the Indian options and expiry studies of Chapter 11 are largely descriptive. Fixed income and foreign exchange are nearly absent, as are practices that a fund needs but papers rarely discuss: in the same scan only 7 records mention point-in-time data, 2 mention survivorship, and none that we found estimates a strategy's capacity. These are the gaps catalogued in Chapter 18.

19.3 Maintaining the atlas

The atlas is a snapshot of a Zotero collection, and the collection will keep growing. We propose the following routine, all of it implementable with the scripts used to build this volume. (i) *Incremental classification:* new items receive silo, stage, concept and relevance labels from the same automated classifier, and a sample is checked by hand to estimate the label error rate that this edition has not measured. (ii) *Concept index refresh:* counts per concept and per chapter are recomputed, and a concept that crosses a size threshold or changes its silo mix triggers a review of the relevant chapter. (iii) *Bridge audit:* each new item is checked against the bridge statements of the chapter it joins, and an item that contradicts one is flagged for the author, since the sceptical papers are the most informative. (iv) *Gap tracking:* the blueprint's gap list is kept as a table, and each addition that fills a gap (a capacity paper, a reconciliation protocol) is recorded against it. (v) *Retire and merge:* duplicate records, such as the two versions of a walk-forward study reported in Chapter 14, are merged, and preprints are updated when a published version appears. (vi) *Rebalance the intake:* since the collection over-represents recent model papers, searches should aim at the thin cells of Table 2.1: execution, portfolio and automation outside the language-model silo, and the older econometric foundations.

19.4 Conclusion

First, a restatement of what the atlas can and cannot claim. Every statement about a paper rests on its title and abstract, not on its full text, so methods, costs, universes and test periods are often less certain than a reader of the full paper would know them to be. The silo, stage, concept and relevance labels come from an automated classification of those abstracts and were not audited by hand; every count in this volume is an estimate on a convenience sample. The reference list carries authors' surnames only, as recorded in the library. And the bridges, desk notes, blueprint

and open problems are our proposals: where a cited paper supplies a component, it supplies only what its abstract reports, and no cited paper has tested the combinations proposed here.

Research that a systematic fund can use exists in shards. A time-series-classification group and an implied-volatility group can work for years on the same decision problem without sharing a word. This atlas is a contribution in bridge-building: it reads 1065 papers into one vocabulary of silos, pipeline stages and concepts; it shows chapter by chapter where a term in one community is a term in another; and it re-reads the whole as the architecture of a semi-automated fund, with humans where judgement is needed and with an explicit account of what the literature does not yet supply.

Three practical messages follow. For traders, the largest gains are likely to come from borrowing validated components across silos (a label design from one chapter, a deflation test from another) and not from a new architecture. For mathematicians and machine-learning researchers, the markets offer well-posed problems with real data, real costs and unforgiving evaluation, several of them listed above. For everyone, the evidence of Chapter 14 is that a result is worth only what its protocol is worth, so the shared checklist (label, horizon, baseline, trial count, cost) is the cheapest common language we can offer. The atlas is a map, and it is meant to be redrawn as the territory is explored.

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